

Direct Cardiac Compression: Device Development, Simulations and New Concepts on Compliance

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Sárosi, Kristóf

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DIRECT CARDIAC COMPRESSION: DEVICE
DEVELOPMENT, SIMULATIONS AND NEW CONCEPTS
ON COMPLIANCE

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presented by

KRISTÓF ÁDÁM SÁROSI

born on August 4, 1993

accepted on the recommendation of

Prof. Dr. Patrick Jenny, examiner
Prof. Dr. Med. Stefanos Demertzis, co-examiner
Dr. Stijn Vandenberghe, co-examiner
Prof. Dr. Dominik Obrist, co-examiner

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Direct Cardiac Compression: Device Development, Simulations and New Concepts on Compliance

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To the memory of my grandfathers and Sun

Abstract

Patients living with heart failure or suffering cardiac arrest require mechanical circulatory support (MCS) to maintain blood circulation. Despite recent advancements in the field of MCS, there is still a need for devices to support patients in emergencies. Most MCS devices are either designed for long-term support or are in contact with the bloodstream. This leaves patients requiring short-term support or those who have no access to anticoagulation therapy without a suitable device.

We aimed to develop a novel MCS device to support the circulation of the patient in the short term without direct contact with the bloodstream. We also aimed to design a device that can be deployed quickly to suit emergency applications. The device is a direct cardiac compression (DCC) device, which compresses the ventricles of the heart directly to support circulation. The device is designed to be used in open-chest emergency scenarios, such as when opening the chest for access or during open-chest surgery.

In the development of the device, we used the concept of Kummer et al., who designed a DCC device with two contact patches driven by a Bowden wire cable. We modified the design to reduce the degrees of freedom during actuation, reducing the complications during DCC device support. The device is actuated by a single Bowden cable. The patches are connected on one side with an adjustable axis, allowing for quick size adjustment to fit hearts of different sizes. The two patches have distinct designs to allow control over the induced pressure in the left and right ventricles. The suction cups in the middle of the patches provide a stable attachment to the heart surface. We developed a motorised and a manual actuator for the device.

The device was tested in *in vitro* and *ex vivo* experiments. The *in vitro* experiments were conducted on a silicone heart sample, while the *ex vivo* experiments were conducted on porcine and ovine hearts. We built an improved version of Kummer's mock circulation loop (MCL) to improve the *in vitro* experimental setup. The *ex vivo* experiments were conducted in the facilities of the LifeTec Group. The results show that the device can achieve comparable performance to open-chest cardiac compression, with a cardiac output of 1.5 L/min and a mean arterial pressure of 50 mmHg. Importantly, in the *ex vivo* experiments, the device showed different pressures in the left

and right ventricles, which is a unique feature of our device.

To optimise the design, we created a simulation framework based on the model of Kummer et al. The proposed lumped parameter model (LPM) simulates the heart and the device, allowing for the optimisation of the device parameters. The model was built on the results we obtained from Kummer’s finite element method (FEM) simulation framework. To simulate the interaction of the heart and the device, a novel compliance-based approach was developed to replace the computationally expensive FEM model. In the process of building this framework, we ran a thorough analysis of the compliant nature of the heart.

The results of this thesis show promise for the development of the device. Our short-term support application target is unique among similar devices, as it focuses on providing immediate assistance in emergency situations. This relieves surgeons from performing open-chest cardiac compression, allowing them to support patients with other life-saving operations. By developing a simulation framework for the device, we have shown that we can integrate DCC devices into LPMs, while also presenting new approaches to modelling heart compliance.

Zusammenfassung

Patienten mit Herzinsuffizienz oder Herzstillstand benötigen eine mechanische Kreislaufunterstützung (MCS), um die Blutzirkulation aufrechtzuerhalten. Trotz jüngster Fortschritte auf dem Gebiet der MCS besteht weiterhin Bedarf an Geräten zur Unterstützung von Patienten in Notfällen. Die meisten MCS-Geräte sind für die Langzeitunterstützung ausgelegt oder stehen in direktem Kontakt mit dem Blutkreislauf. Für Patienten, die eine kurzfristige Unterstützung benötigen oder keinen Zugang zu einer Antikoagulationstherapie haben, steht somit kein geeignetes Gerät zur Verfügung.

Unser Ziel war die Entwicklung eines neuartigen MCS-Geräts, das den Kreislauf des Patienten kurzfristig ohne direkten Kontakt mit dem Blutkreislauf unterstützt. Zudem sollte das Gerät für den schnellen Einsatz in Notfallsituationen geeignet sein. Bei dem Gerät handelt es sich um ein direktes Herzkompressionsgerät (DCC), das die Herzkammern zur Unterstützung des Kreislaufs direkt komprimiert. Das Gerät ist für den Einsatz in Notfallszenarien am offenen Brustkorb konzipiert, beispielsweise bei der Öffnung des Brustkorbs für einen Zugang oder während einer Operation am offenen Brustkorb.

Bei der Entwicklung des Geräts haben wir das Konzept von Kummer et al. verwendet, die ein DCC-Gerät mit zwei Kontaktflächen entwarfen, die über einen Bowdenzug angetrieben werden. Wir haben das Design modifiziert, um die Freiheitsgrade bei der Betätigung und damit die Komplikationen während der Unterstützung durch das DCC-Gerät zu reduzieren. Das Gerät wird über einen einzigen Bowdenzug betätigt. Die Kontaktflächen sind auf einer Seite mit einer verstellbaren Achse verbunden, was eine schnelle Grössenanpassung an Herzen unterschiedlicher Grösse ermöglicht. Die beiden Kontaktflächen sind unterschiedlich gestaltet, um den im linken und rechten Ventrikel erzeugten Druck zu steuern. Saugnäpfe in der Mitte der Kontaktflächen gewährleisten eine stabile Befestigung an der Herzoberfläche. Für das Gerät haben wir einen motorisierten und einen manuellen Aktor entwickelt.

Das Gerät wurde in *in vitro*- und *ex vivo*-Experimenten getestet. Die *in vitro*-Experimente wurden an einem Silikonherzmodell durchgeführt, die *ex vivo*-Experimente an Schweine- und Schafherzen. Zur Verbesserung des *in vitro*-Versuchsaufbaus haben wir eine weiterentwickelte Version von Kummers

Mock Circulation Loop (MCL) gebaut. Die *ex vivo*-Experimente wurden in den Einrichtungen der LifeTec Group durchgeführt. Die Ergebnisse zeigen, dass das Gerät eine mit der offenen Herzdruckmassage vergleichbare Leistung erzielen kann, mit einem Herzzeitvolumen von 1,5 l/min und einem mittleren arteriellen Druck von 50 mmHg. In den *ex vivo*-Experimenten erzeugte das Gerät zudem unterschiedliche Drücke im linken und rechten Ventrikel, was ein einzigartiges Merkmal unseres Geräts ist.

Zur Optimierung des Designs haben wir ein Simulationsframework auf Basis des Modells von Kummer et al. erstellt. Das vorgeschlagene Lumped-Parameter-Modell (LPM) simuliert das Herz und das Gerät und ermöglicht so die Optimierung der Geräteparameter. Das Modell basiert auf den Ergebnissen, die wir mit dem Finite-Elemente-Methode (FEM) Simulationsframework von Kummer erzielt haben. Um die Interaktion zwischen Herz und Gerät zu simulieren, wurde ein neuartiger, auf Compliance basierender Ansatz entwickelt, der das rechenintensive FEM-Modell ersetzt. Im Zuge der Entwicklung dieses Frameworks haben wir eine gründliche Analyse der nachgiebigen Eigenschaften des Herzens durchgeführt.

Die Ergebnisse dieser Arbeit sind vielversprechend für die Weiterentwicklung des Geräts. Unser Anwendungsziel der kurzfristigen Unterstützung ist unter ähnlichen Geräten einzigartig, da es sich auf die sofortige Hilfe in Notfallsituationen konzentriert. Dies entlastet Chirurgen von der Durchführung einer offenen Herzdruckmassage und ermöglicht es ihnen, Patienten mit anderen lebensrettenden Massnahmen zu versorgen. Mit der Entwicklung eines Simulationsframeworks für das Gerät haben wir gezeigt, dass sich DCC-Geräte in LPMs integrieren lassen, und gleichzeitig neue Ansätze zur Modellierung der Herz-Compliance vorgestellt.

Resumaziun

Pazients cun in'insuffizienz u fermada dal cor dovran in sustegn mecanic da la circulaziun (MCS) per mantegnair la circulaziun dal sang. Malgrà ils ultims progress en il champ dal MCS basegni anc adina d'apparats per sustegnair pazients en cas d'urgenza. La gronda part dals apparats da MCS èn concepids per il sustegn a lunga vista u stattan en contact direct cun la circulaziun dal sang. Per pazients che dovran in sustegn temporar u che n'han nagin access ad ina terapia d'anticoagulaziun, na datti damai nagin apparat adattà. Nossa finamira è stada da sviluppar in nov apparat da MCS che sustegna la circulaziun dal pazient temporarmain senza contact direct cun la circulaziun dal sang. Ultra da quai duai l'apparat esser adattà per l'utilisaziun pronta en situaziuns d'urgenza. I sa tracta d'in apparat da cumpressiun directa dal cor (DCC) che cumprima directamain las chombras dal cor per sustegnair la circulaziun. L'apparat è concepì per l'utilisaziun en scenaris d'urgenza al pèz avert, per exempel tar l'avertura dal pèz per in access u durant in'operaziun al cor.

Per sviluppar l'apparat avain nus utilisà il concept da Kummer et al. che han concepì in apparat da DCC cun duas surfatschas da contact impulsadas da cabels da Bowden. Nus avain modifitgà il design per reducir ils grads da libertad tar l'activaziun e cunquai las cumplacaziuns durant il sustegn tras l'apparat da DCC. L'apparat vegn activà sur in sulet cabel da Bowden. Las surfatschas da contact èn colliadas d'ina vart cun in'axa adattabla, quai che pussibillitescha d'adattar svelt la grondezza tut tenor dimensiun dal cor. Las duas surfatschas da contact èn concepidas differentamain per manipular la pressiun generada en il ventricul sanester e dretg. Ventosas en il center da las surfatschas da contact garanteschon ina fixaziun stabila al cor. Per l'apparat avain nus sviluppà in actuatur motorisà ed in manual.

L'apparat è vegnì testà en experiments *in vitro* ed *ex vivo*. Ils experiments *in vitro* èn vegnids exequids sin in model da cor da silicon, ils experiments *ex vivo* sin cors da portgs e nursas. Per meglierar la structura da l'experiment *in vitro* avain nus costruì ina versiun sviluppada vinavant da Kummer's Mock Circulation Loop (MCL). Ils experiments *ex vivo* èn vegnids exequids en las instituziuns da la LifeTec Group. Ils resultats mussan che l'apparat po cuntanscher ina prestaziun cumparegliabla ad ina massascha dal cor,

cun in volumen cardiac dad 1,5 l/min ed ina pressiu arteriala media da 50 mmHg. En ils experiments *ex vivo* ha l'apparat era generà mintgamai differentas pressions en il ventricul sanester ed il ventricul dretg, quai ch'è ina caratteristica unica da noss apparat.

Per optimar il design avain nus creà in framework da simulaziun a basa dal model da Kummer et al. Il model da lumped parameters (LPM) proponì simulescha il cor e l'apparat e pussibilitescha uschia d'optimar ils parameters da l'apparat. Il model sa basa sin ils resultats che nus avain cuntanschì cun il framework da simulaziun da la metoda dals elements finits (FEM) da Kummer. Per simular l'interacziun tranter il cor e l'apparat è vegnì sviluppà ina nova metoda a basa da la compliance, che remplazza il model FEM intensiv areguard la performanza. En il decurs dal svilup da quest framework avain nus exequì in'analisa profunda da las proprietads da s'extender dal cor.

Ils resultats da questa lavur èn empermettents per l'ulteriur svilup da l'apparat. Nossa finamira d'utilisaziun en furma d'in sustegn temporar è unica en cumparegliaziun cun apparats sumegliants, perquai ch'ella sa concentrescha sin l'agid immediat en situaziuns d'urgenza. Quai distgargia ils chirurgs da l'execuziun d'ina massascha dal cor averta ed als pussibilitescha da provider pazients cun autras mesiras da salvament. Cun il svilup d'in framework da simulaziun per l'apparat avain nus mussà ch'ils apparats DCC sa laschan integrar en LPMs, e nus avain era preschentà novas metodos per modellar la capaciad dal cor da s'extender.

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Nomenclature

Upper-case Roman

A	Matrix of resistances and diodes
A	Amplitude of DCC compression
B	Parameter for systolic fraction in DCC compression
C	Matrix of compliances
C	Compliance
D	State of a diode (valve)
E	Elastance
H	Heaviside step function
K	Constant coefficient for valve resistance
L	Inductance
Q_L	Vector of flows through inductors
R	Resistance
S	Source term vector
T	Period of cardiac cycle / DCC compression
V	Volume

Lower-case Roman

c	Compliance
e	Heart-device interaction coefficient
k	Scaling factor for elastance

m	Rate constants for elastance model
p	Pressure
q	Flow rate
t	Time

Lower-case Greek

ω	DCC patch displacement
ϕ	Phase shift for DCC compression
τ	Time constants for elastance model

Subscripts

ao	Aortic valve
la	Left Atrium
L	Left
lv	Left Ventricle
mi	Mitral valve
pa	Pulmonary Arteries
pp	Pulmonary Capillaries
pu	Pulmonary valve
pv	Pulmonary Veins
ra	Right Atrium
R	Right
rv	Right Ventricle
sa	Systemic Arteries
sp	Systemic Capillaries
sv	Systemic Veins

tr Tricuspid valve

Abbreviations

ADC	Analog-to-Digital Converter
ANOVA	Analysis of Variance
BiVAD	Biventricular Assist Device
CA	Cardiac Arrest
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CO	Cardiac Output
CPR	Cardiopulmonary Resuscitation
CT	Computed Tomography
DCC	Direct Cardiac Compression
ECMO	Extracorporeal Membrane Oxygenation
EDPVR	End-Diastolic Pressure-Volume Relationship
ESPVR	End-Systolic Pressure-Volume Relationship
FEM	Finite Element Method
HF	Heart Failure
I ² C	Inter-Integrated Circuit
LPM	Lumped Parameter Model
LV	Left Ventricle
LVAD	Left Ventricular Assist Device
MAP	Mean Arterial Pressure
MCL	Mock Circulation Loop
MCS	Mechanical Circulatory Support

MPAP	Mean Pulmonary Arterial Pressure
OCCC	Open-Chest Cardiac Compression
OR	Operating Room
PLA	Polylactic acid
RV	Right Ventricle
SV	Stroke Volume
TAH	Total Artificial Heart
UART	Universal Asynchronous Receiver-Transmitter
VAD	Ventricular Assist Device

Part I

Preface

A PhD is a marathon, not a sprint, a phrase we often hear as PhD students. Our peers and supervisors usually remind us of this even before we start or at the beginning of our PhD journey. Finding a purpose in this marathon to never surrender is essential. For me, this purpose was to develop a new device to advance mechanical circulatory support. Essentially, the results of my suffering might help people who suffer from heart diseases. Did I achieve this? I hope so, but I will let you be the judge of that.

Part II introduces the reader to human physiology, including the anatomy of the heart and the circulatory system, cardiovascular diseases, and the need for mechanical circulatory support (MCS) devices. It also provides an overview of the current state of MCS devices, including their limitations and challenges. This part also highlights the research gap that the work of this thesis was intended to fill.

Part III guides us through the history of MCS devices, currently available devices, and introduces direct cardiac compression (DCC) devices. It describes the history, design, and development of our DCC device, which is the main focus of this thesis. The part discusses the design features, actuation mechanisms, and the unique aspects of our device compared to other DCC devices.

Part IV presents our results from the *in vitro* experiments, which were conducted on a silicone heart sample. This part introduces the mock circulation loop (MCL), a standard testing platform for cardiovascular devices. The part describes the experimental setup, methods, and results of the *in vitro* experiments, showing that our device can achieve comparable performance to open-chest cardiac compression.

Part V presents our results from the *ex vivo* experiments, which were conducted on porcine and ovine hearts in Eindhoven, at the facilities of the LifeTec Group. The part describes the experimental setup, methods, and results of the *ex vivo* experiments, showing that our device can achieve comparable performance to open-chest cardiac compression on an ovine cardiac arrest model, supporting the findings of Part IV.

Part VI introduces the reader to computer simulation techniques used in cardiovascular research. It presents our lumped parameter model (LPM), which we developed to simulate the heart and the DCC device. The part describes the methods used to build the LPM, including the compliance-based approach to simulate the interaction between the heart and the device. It also discusses the validation of the model against the results an FEM simulation framework. The LPM allows for the optimisation of the device parameters to support future *ex vivo* studies.

Part VII discusses the challenges we faced during the development of the LPM, particularly in modelling the compliance of the heart. It presents our analysis of the compliant nature of the heart and how it affects the performance of the DCC device. The part also discusses the implications of our findings for future research in simulating heart cycles and MCS devices.

Part VIII concludes the thesis with a summary of the key findings, addressing unresolved questions and providing perspectives for future research.

Part IX includes supplementary results and elaborates on some of the aspects discussed in the main body of the thesis.

But before the heavy lifting, let us warm up with the Introduction.

This thesis is based on the following manuscripts:

K. Sarosi et al. “Demonstration of a Mechanical External Biventricular Assist Device for Resuscitative Thoracotomy”. In: *Cardiovascular Engineering and Technology* (July 2025). ISSN: 1869-408X. DOI: <https://doi.org/10.1007/s13239-025-00793-y>

K. Sarosi et al. “Direct Cardiac Compression Device for Short-Term Mechanical Circulatory Support”. In: *ASAIO Journal* (June 2025). ISSN: 1058-2916. DOI: <https://doi.org/10.1097/MAT.0000000000002493>

In writing this thesis, I used AI assistants to check the grammar and improve the overall quality of the writing.

Part II

Introduction

Chapter 1

Physiology

This thesis focuses on understanding and tackling issues related to heart diseases, with the goal of demonstrating the development of a device supporting surgeons in special cases. I also discuss the development of a simulation environment to optimise the device, and hence understand the interaction between similar devices and the heart. Before we head into these topics, it is important to give a quick overview of heart diseases and anatomy (as a mere engineer, I do not dare to go further than scratching the surface) to familiarise ourselves with the different regions of the heart and their functions. In writing these sections of this chapter, the knowledge base of the Mayo Clinic website, Arnold M. Katz's Physiology of the Heart, and N. P. Rhodes' Clinical Engineering helped me greatly. After the initial sections, we move to the boundary of medicine and engineering, talking about the mechanical properties of the heart and engineering analogues of this complex biological system. In the later sections of this part, I introduce the reader to the history of device solutions tackling specific heart diseases, and the development of simulation environments. I must highlight that this part is not describing my work but rather helping the reader getting into the topic.

The work presented in the later chapters of this thesis builds heavily on the knowledge presented in this part. In this part, I try to abstain from typical jargon and abbreviations when possible while building the necessary foundation for the following chapters. Those later chapters build on previous publications, unpublished and ongoing research, where jargon and abbreviations are necessary and unavoidable. You will find the related nomenclature at the beginning of this thesis.

1.1 Cardiovascular Diseases

Cardiovascular diseases are the leading cause of death in Switzerland, the European Union, and the world [3, 4, 5], amounting to almost one-third of all deaths. The leading cause among cardiovascular diseases is ischaemic heart disease, something that is more commonly known as coronary heart disease,

a condition where the blood and thus the oxygen supply to the heart itself is congested. Ischaemic heart disease is the cause of many other conditions such as myocardial infarction, commonly known as a heart attack; ischaemic cardiomyopathy, the weakening of heart muscles due to the lack of oxygen supply or a heart attack; and may eventually lead to heart failure, a clinical condition caused by a structural and/or functional cardiac abnormality [6].

1.1.1 Heart Failure

Heart failure is a broader term as its cause is not necessarily ischaemic heart disease but any disease (e.g. valve disease) that causes abnormal cardiac function fulfilling the clinical criteria for diagnosis. Heart failure is categorised into different stages, depending on the severity of the condition. While living with heart failure is possible, chronic or end-stage heart failure is life-threatening, requiring device therapy or a transplant.

Heart failure is a global epidemic. Our ageing and growing society is only making this epidemic worse as total numbers and prevalence are both increasing. Estimation shows that more than 64 million people are living with heart failure [7]. The prevalence for developed countries is higher than globally, between 1-2%, and 11.8% among the population aged 65 years and over [8]. There is a lack of data from developing countries, despite this part of the world carrying an estimated 80% of the global cardiovascular disease burden [9]. Moreover, prevalence is expected to rise in low- and middle-income countries [10], resulting in a serious economic burden on individuals and healthcare systems [11, 12]. Given the unrecognised cases and lack of access to treatments due to unavailability or costs, better screening and accessible treatments are needed for heart failure patients.

1.1.2 Cardiac Arrest

Cardiac arrest is another cardiovascular event that poses a great threat to our ageing population. It is a condition when heart function is lost due to an irregular heart rhythm. Given the surge of atrial fibrillation in the United States [13], cardiac arrest cases are only expected to rise. In contrast to heart failure, there is no blood flow during cardiac arrest, leading to life-threatening emergency cases. Sudden cardiac arrest is the condition that most of us have in mind when we think of cardiovascular diseases. The heart stops or, on the contrary, beats so fast (fibrillates) that heart functions fail, the person collapses, and cardiopulmonary resuscitation (CPR) and shocks

by a device called an automated external defibrillator (AED) are carried out on the scene [14]. Rapid medical intervention is key as blood circulation must be restored within minutes. Cardiac arrest can be triggered by heart failure, stress, anaesthesia in surgery, and many other factors. Cardiac arrest can be treated by CPR, defibrillation, or antiarrhythmics; however, in certain cases, these techniques cannot be used. In the case of cardiac arrest following a penetrating event to the chest, cardiac arrest after recent cardiac surgery, decompression of pericardial tamponade (release of pressure on the heart following accumulated liquid in the pericardial sac surrounding the heart), and cardiac arrest during open-chest surgery, open-chest cardiac compression is recommended [15].

1.2 Anatomy

This section uses the works of Arnold M. Katz, *Physiology of the Heart* [16], and Nicholas P. Rhodes's *Anatomy and Physiology* [17] extensively.

1.2.1 Human Circulation

The human circulation consists of the network of blood vessels and organs such as the heart and the lungs. The main purpose of the vasculature is to transport oxygen and nutrients to the tissues. The arteries deliver blood from the heart, and veins carry blood into the heart. The circulation is driven by the beating heart. Arteries and veins are similar in structure, but the muscle layer of arteries is thicker to sustain the higher pressures they are exposed to. Veins in the legs have valves to prevent backflow. The circulation is often divided into two regions, the systemic and the pulmonary circulation. Figure 1.1 shows the simplified structure of the human cardiovascular system.

It is important to note that due to the pulsatile nature of the heartbeat and blood flow, the compliant veins and arteries, the losses, and the complex network of the vasculature, different parts of the body are exposed to different pressures, and pressure waves travel along the network as the compliant vessels carry blood in the circulation. This temporal and spatial heterogeneity in flow properties must be considered when designing a device interacting with the heart or the cardiovascular system.

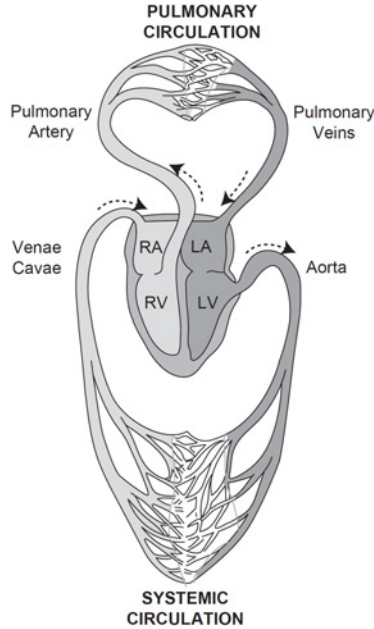


Figure 1.1: Human circulation. The figure shows the heart and the two main parts of the vasculature, the systemic and the pulmonary circulation. Important to note that this is not an anatomically correct image, it only represents the functionalities and the different regions of the circulation. Light shading: deoxygenated blood; dark shading: oxygenated blood. RA, right atrium; LA, left atrium; RV, right ventricle; LV, left ventricle [16].

1.2.2 Lungs

Oxygen is vital for the organs and thus for life. The oxygen exchange happens in the lungs; therefore, we must have a look at its structure. The lungs have a tree-like structure, branching into smaller divisions as going down the airways. The bigger tubes (bronchi) break into bronchioles, which branch off to millions of alveoli, on the surface of which small blood vessels, the capillaries, are responsible for oxygen exchange in the circulation (see Figure 1.2). The small vessels are sensitive to pressure changes as with high pressure, these thin structures can break. This phenomenon is called pulmonary hypertension, a dangerous condition where the lungs can be damaged. When designing a circulatory support device, one must keep in mind the effects of the device on other organs and the vasculature. Clinicians

and engineers these days are very conscious about the interconnectedness of different organs through the vasculature. They call this interaction organ crosstalk.

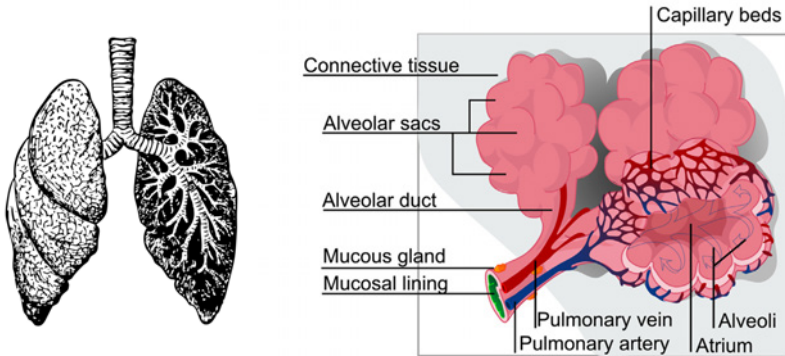


Figure 1.2: The branching structure of the lungs and the alveoli, where the oxygen exchange happens. The alveolar sacs are filled with air after respiration, and the oxygen enters the blood stream in the capillary beds. The red color indicates capillaries filled with oxygenated blood, while the blue capillaries are filled with oxygen-poor blood [17].

1.2.3 Heart

The performance of the heart is affected by two factors in the circulation (apart from the valve functions). These are the pressure before (preload) and after (afterload) the ventricle. These are pressures in the atria and the outflow arteries. They are important for two aspects: these pressures define whether the valve between these regions and the ventricle opens or closes, and they also affect the pressure inside the ventricles. Preload and afterload can also affect the outcome of circulatory support device performance.

Chambers

The heart consists of four main chambers, acting as individual volumetric pumps in a complex system (more on that later). These four chambers are the left and right atria, and the left and right ventricles, depicted in Figure 1.3. The left side of the heart is responsible for delivering oxygenated blood from the lungs to the body (systemic circulation) and the heart itself, and the right side of the heart collects oxygen-poor blood and transports it

to the lungs (pulmonary circulation) to oxygenate the blood. The structure of the two ventricles is very different. The left ventricle, creating pressures about three times that of the right ventricle, has much thicker walls and is therefore more rigid than the right ventricle. Applying the same force on both ventricles can create an issue for the right ventricle as the induced pressure in the right ventricle can be much higher than the physiological pressure, leading to pulmonary hypertension. The two ventricles are separated by the interventricular septum, a thick wall of heart muscle. This wall helps the left ventricle with ejection as it moves towards the wall of the left ventricle during contraction (systole). The relaxation of the heart muscles and the filling of the ventricles happen during the phase called diastole. The pulsatile contraction of the heart is dictated by the sinus node, sending electric signals to the heart.

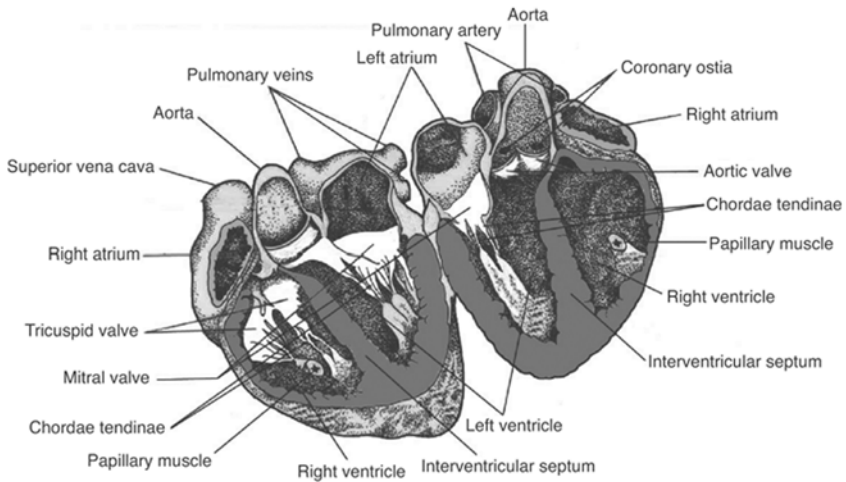


Figure 1.3: The structure of the human heart [16].

The contraction of the heart is an interesting topic on its own. The contraction of the four chambers happens with a small delay as the electric signal travels in the tissue. The contraction or the stiffening of the heart

muscles changes the mechanical properties of the heart, making it less compliant. Modelling the heart action in simulations usually happens by describing the elasticity of the heart. However, the contraction is more complex than just describing it with a global parameter as the shape of the heart deforms; the helical structure of the heart muscles leads to twisting the apex (the tip) of the heart while the ventricles deform and shrink, ejecting blood into the circulation. The same happens with the atria as they contract just before the ventricles do to fill up the ventricles before ejection.

Valves

The filling and ejection is controlled by the heart valves as they open and close depending on the pressure on the two sides of the valves. A valve with healthy function also acts as a regulator as it only lets the blood flow in one direction. The four valves are the aortic valve, tricuspid valve, pulmonary valve, and mitral valve, in this order following the direction of blood flow from the left ventricle through the right heart and back to the left ventricle (see Figure 1.4). The two atria and two ventricles are separated by the atrioventricular valves (tricuspid, mitral). These valves prevent backflow from the ventricles towards the atria. The semilunar valves (aortic, pulmonary), shaped like half-moons from the side, separate the ventricles from their respective main blood vessels, the aorta and the pulmonary artery. The free margins of the semilunar aortic and pulmonary valve cusps are supported by thick tendinous edges. The larger cusps of the mitral and tricuspid valves are tethered at their free margins by fibrous chordae tendinae that attach to “fingers” of myocardium called papillary muscles that project into the right and left ventricular cavities [16].

1.3 Heart as a Pump

Contrary to skeletal muscles, the reason behind the contraction of the heart muscles is not to move parts of the body around a joint through the shortening of the muscles, but to drive the circulation. Therefore, the heart is best characterised as a pump, relating pressure to ejected volume, and not only by strain and stresses. Talking about units of measure, medicine is an ancient practice, using units that feel almost prehistoric. In the later parts of this thesis, pressure is represented by the unit *mmHg*, that is millimetre of mercury, a manometric unit of pressure, the pressure generated by a one-millimetre-high column of mercury. If you can recall, the normal blood

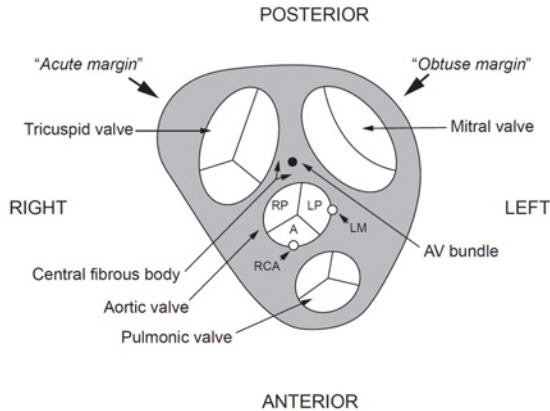


Figure 1.4: The valves of the heart, viewed from above, showing the four valves. [16].

pressure is indicated as 120/80, that is 120 mmHg at systole and 80 mmHg at diastole in the aorta.

As mentioned previously, the heart works by contracting and relaxing periodically. Combining this periodicity and the important relation between pressure and ejected volume, a standard depiction of cardiac performance is the pressure-volume loop. With consistent pumping, these graphs form a loop for every period of heart action, similar to the cycles of an internal combustion engine. Figure 1.5 shows this representation of the heart's performance. It is a very useful representation, as any abnormalities or changes regarding ejection or pressure can be seen immediately.

Before systole, the atrium contracts and fills the ventricle with blood. Then, the onset of systole closes the atrioventricular valve as the pressure in the ventricle reaches the preload levels, leading to isovolumetric contraction, meaning that the heart contracts without ejecting any blood to the artery or receiving any from the atrium, thus rapidly increasing the pressure in the ventricle. The semilunar valve opens when the pressure in the ventricle is higher than that of the afterload, leading to the ejection phase of systole. When the pressure in the ventricle drops below the pressure in the artery (afterload), the semilunar valves close, while the heart muscles relax, entering diastole. In the initial phase of diastole, the two valves are closed, leading to isovolumetric relaxation. The pressure in the ventricle rapidly declines until it reaches the level of the atrial pressure (preload), when the atrioventricular

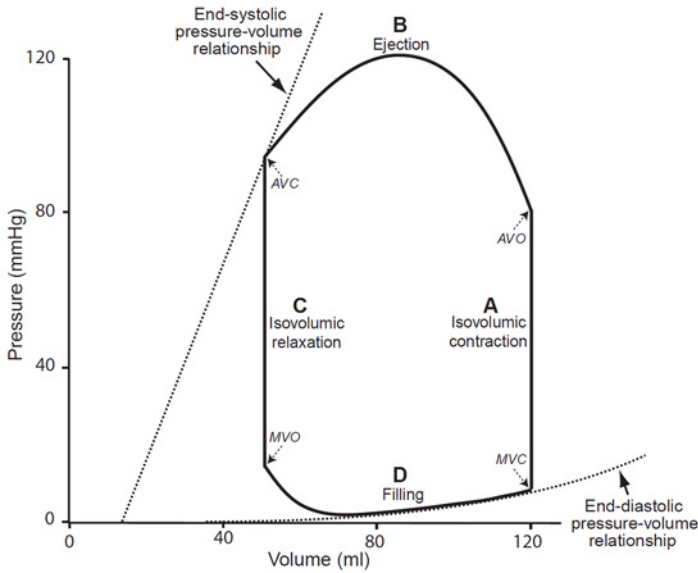


Figure 1.5: 'Pressure-volume loop generated by a normal left ventricle. The loop is constrained by the end-diastolic pressure-volume relationship and the end-systolic pressure-volume relationship. Systole begins at a point along the end-diastolic pressure-volume relationship that represents the preload, after which the mitral valve closes (MVC) and pressure increases rapidly. Because both the aortic and mitral valves are closed, this phase is isovolumic (A). Ejection (B) begins when the aortic valve opens (AVO) and the ventricle meets its afterload, the aortic pressure. Systole ends when ventricular pressure and volume reach the end-systolic pressure-volume relationship, which describes the inotropic state of the ventricle. After aortic valve closure (AVC) separates the afterload (aortic pressure) and the ventricular chamber, blood can neither enter nor leave the ventricle; as a result, relaxation begins under isovolumic conditions (C). When left ventricular pressure falls below that in the left atrium, the mitral valve opens (MVO) and blood flows from the atrium into the ventricle during the phase of filling (D). The cycle ends when ventricular pressure and volume reach the end-diastolic pressure-volume relationship.' — Arnold M. Katz "Physiology of the Heart" p. 304 [16]

valve opens again, while the atria contract, closing the cycle. The ratio of the time of the systolic to the diastolic phase is called the systolic fraction.

While the cyclic pumping action of the heart is best represented in a pressure-volume loop, the electric signals of this pumping action and the effects of pumping on the connected parts of the circulation are usually represented in a Wiggers diagram (see Figure 1.6). The diagram shows the pressure, volume, and electrical events developing in time during the cardiac

cycles.

When a ventricle dilates, the mechanical strain and stress increase in the ventricular walls, and therefore its ability to perform contraction increases. This self-regulating mechanism is described in the Frank–Starling mechanism. However, this is only true for a healthy heart. The failing heart can stretch over time while losing the ability to work. The increased cost of ventricular contraction is not compensated for by the benefits described in the Frank–Starling law. The mass balance of ejecting as much blood as the ventricle received breaks due to heart failure, and it builds up volume over time.

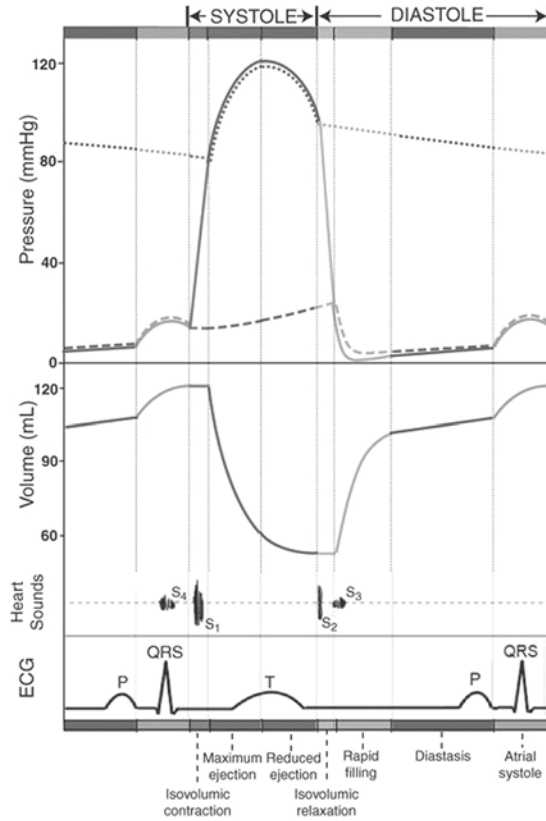


Figure 1.6: 'The cardiac cycle (Wiggers diagram) showing seven phases of left ventricular systole. By convention, the cycle begins with the onset of ventricular systole. The top three curves represent aortic pressure (upper dotted line), left ventricular pressure (solid line), and left atrial pressure (lower dashed line). The solid line below these pressure curves is left ventricular volume, below which are the heart sounds: S4, fourth (or atrial) sound; S1, first heart sound; S2, second heart sound; S3, third heart sound. The bottom line shows the timing of the electrocardiogram that records the electrical events during the cardiac cycle.' — Arnold M. Katz "Physiology of the Heart" p. 306 [16]

Chapter 2

Mechanical Circulatory Support

Currently available treatments for heart failure and cardiac arrest include surgical intervention, pharmacologic and device solutions. Lam et al. [6] provide an extensive review of recent developments in the treatment of heart failure, mostly from the pharmacologic aspect, while Callan and Dimarakis [18] share an overview of available mechanical circulatory support (MCS) devices.

2.1 Open-Chest Cardiac Compression

During open-chest cardiac compression, cardiac surgeons manually massage the heart, supporting blood circulation. It is an invasive technique, requiring a chest incision or an open chest to access the heart (see Figure 2.1). D’Souza provides a modern view on open-chest cardiac compression and emergency thoracotomy (a surgical procedure in which a cut is made between the ribs to access the heart or the lungs) [15]. As there are no standards for open-chest cardiac compression, only recommended practices, there is great variability in outcomes depending on the surgeons and the patients [19, 20]. It requires surgeons to maintain circulatory support until recovery or device support is established. In many cases, open-chest cardiac compression is only a temporary solution before the patient is supported by an extracorporeal membrane oxygenation (ECMO) device. Before covering devices and device support, it is important to learn the basic anatomy of the heart and the human circulation to understand the limitations of these devices.

2.2 Ventricular Assist Devices

One of the initial goals of ventricular assist device (VAD) design was to replace open-chest cardiac compression with a machine capable of maintaining long-term blood flow. In the 1950s and 60s, this led to the concept of direct mechanical ventricular assistance [20, 19, 22, 23]. Later, closed-chest CPR



Figure 2.1: Open chest cardiac compression, illustrated by Netter [21].

replaced open-chest cardiac compression for most cases [24], and modern technology enabled implantable, small-scale turbomachines for MCS. While highly effective in *bridge-to-transplant* scenarios [25], modern MCS devices have a significant drawback: they require invasive heart surgery and operate in direct contact with the patient's blood [26, 27, 28]. This increases the risk of complications, limiting their suitability for many patients. Furthermore, most devices support only the left ventricle (LVADs), causing pressure imbalances within the heart that can lead to aortic insufficiency [29], right heart failure [30], and septal shift [31]. Direct contact with the blood not only can lead to infections [32], but also requires medication to tackle haemorrhagic and thromboembolic complications [22], further restricting patient eligibility. Available devices require invasive techniques, special medication, and target *bridge-to-transplant* and long-term solutions. This leaves many without a safe short- or mid-term solution.

2.2.1 Left Ventricular Assist Devices (LVADs)

LVADs are the most common MCS devices for heart failure patients, and in acute cases, biventricular assist devices (BiVADs) are used. There are many studies evaluating the performance of such devices in clinical practice [33, 34]. Continuous flow LVADs traditionally suffer from bleeding and thrombosis issues [28]. However, *HeartMate 3* (HM3), a recent LVAD (see Figure 2.2), using a magnetically levitated centrifugal continuous-flow pump [35] managed to tackle this issue better than its previous iterations. The magnetically levitated rotor technology could reduce the risk of thrombosis [36], and reach a 1-year survival rate of over 80% [37, 38, 36]. With the current level of devices, LVADs could provide a safe destination therapy for patients with left ventricular diseases. Unfortunately, in the long term, LVAD therapy can lead to right ventricular failure, requiring circulatory support in the right ventricle as well [39, 40, 41].



Figure 2.2: The HeartMate 3 (HM3) [18] and Impella 5.5 with SmartAssist [42] LVAD devices.

2.2.2 Bi-Ventricular Assist Devices (BiVADs)

Biventricular support can be provided with two LVADs or a specific BiVAD. In recent years, the use of BiVADs is declining [43], despite new devices being available for *bridge-to-device*, *bridge-to-decision*, *bridge-to-recovery*, and *bridge-to-transplant* therapy [44, 45, 46, 47]. The Berlin EXCOR BiVAD (see Figure 2.3) showed superior performance to using two LVADs, indicating

that biventricular support should be provided by pulsatile flow BiVADs [45]. Even in these studies with people living with advanced biventricular heart failure, 50% of the cohort INTERMACS 1 patients (with critical cardiogenic shock) die due to end-organ failure [48, 49]. Studies using two LVADs show varying outcomes in clinical practice. Biventricular support is usually less effective [50] as biventricular heart failure is a more advanced stage of heart failure than left ventricular or right ventricular heart failure. Some studies with two LVADs show similar performance to the Berlin EXCOR [51, 52], some even improved with the novel HM3 device [53]. Although comparing the performance of such devices is difficult as patient cohorts are varying, Packer et al. aimed at comparing the performance of LVADs and BiVADs [54].

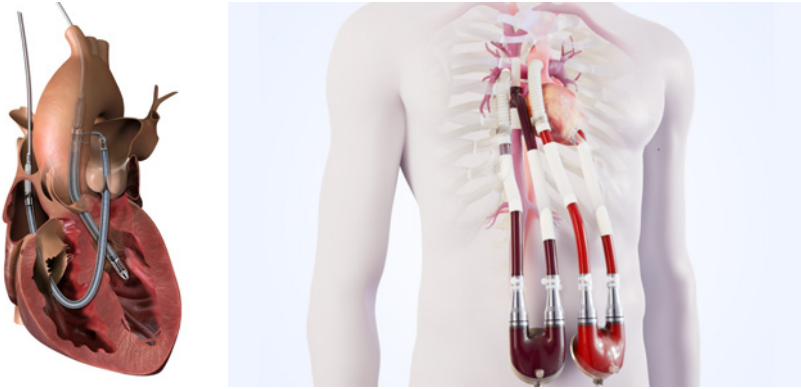


Figure 2.3: The Bipella, using two Impella devices for BiVAD support [42], the Berlin Heart EXCOR BiVAD [55].

2.3 Total Artificial Hearts

All previous devices are implanted or in interaction with the ventricles, but there are other MCS solutions, such as circulatory support through cannulation or the replacement of the heart with a device for *bridge-to-transplant*. Despite the latter sounding like far-fetched science fiction from a steampunk novel, devices replacing the heart do exist. They are called total artificial hearts, or TAH for short. The history of these devices is not as rich as for VADs, but applications do exist, and their development is necessary for *bridge-to-transplant* and potentially for destination therapy.

Current devices include the BiVACOR [56], a total artificial heart rotary blood pump, the SynCardia [57] artificial ventricles, and the total artificial heart Carmat [58], a pulsatile volumetric pump in-lined with animal tissue for blood safety. The former two can be seen in Figure 2.4.

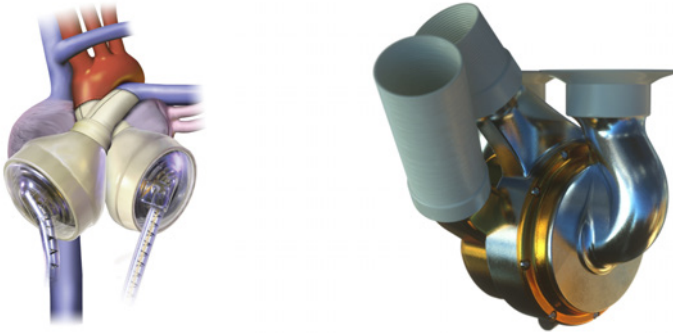


Figure 2.4: The SynCardia total artificial heart [18] and the BiVACOR total artificial heart [59].

2.4 Extracorporeal Membrane Oxygenation

MCS devices such as ECMO provide support in the operating room, supporting patients' circulation and oxygenation in emergencies. An example of an ECMO setup is shown in Figure 2.5. ECMO provides biventricular assistance in heart failure, cardiogenic shock, and cardiac arrest cases [60, 61, 62, 63, 47, 64]. ECMO is a volumetric blood pump, being in direct contact with the bloodstream through two cannulas inserted in the vasculature, hence requiring targeted anticoagulant drug use. Similar to LVAD complications due to anticoagulation, ECMO support suffers from an increased risk of bleeding. The prevalence of bleeding events is 40%, and the rate of infections is 31% due to cannulation [40]. It also suffers from renal failure, pneumonia, and stroke [65, 61]. Using ECMO devices is an expensive treatment. It requires the ECMO machine, a specialist (operating the machine), drugs, and cannulation. The reported costs can reach \$70,000 [66, 67, 68, 69, 70, 65]. Considering this, it is fair to say that when ECMO use could be avoided, it should be, and non-blood-contacting alternatives should be developed.

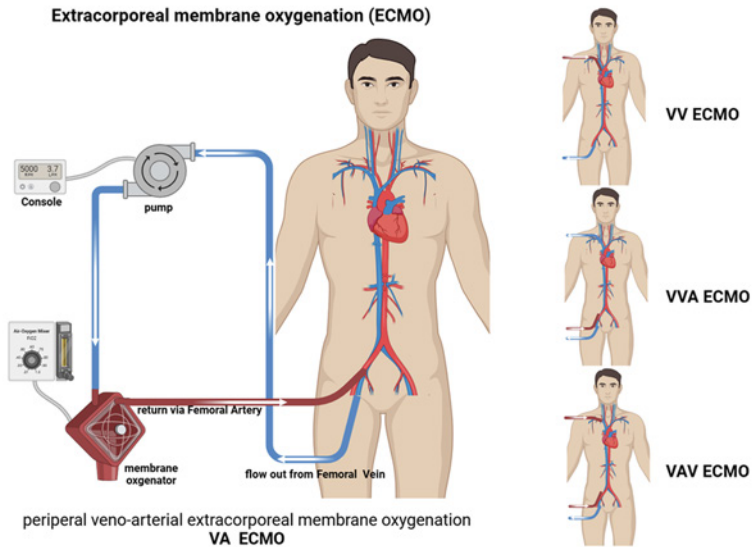


Figure 2.5: Peripheral fémoro-femoral veno-arterial ECMO. The blood is circulating in the opposite direction as the natural blood flow to get the oxygenated blood back to the systemic circulation [71].

2.5 Complications of MCS Devices

Current VADs are in direct contact with the bloodstream, requiring anticoagulation therapy. Anticoagulation therapy can be contraindicated for some patients, leaving many without a circulatory support solution. Those who can receive anticoagulation drugs face complex anticoagulation strategies [72], and an increased risk of bleeding [28, 40].

Infections are a major factor in device complications with 30% prevalence after 6 months [32], potentially affecting all components of the device. Despite reducing infection rates with continuous flow VADs, still close to 15% of VAD patients die due to infections [73] and 28% die after infections [32]. Driveline infections are not only the deadliest of all infections but also the most prevalent [32]. The driveline is the connection between the external power source or controller and the internal device (see Figure 2.6). Using two LVADs for BiVAD support with two drivelines therefore exposes patients

to an elevated risk of infections. Even in cases with no effect from infections on survival, it can deteriorate the quality of life of survivors [32]. Driveline infections are responsible for 47% of all device-related callbacks [40], leading to hospital readmission and surgical interventions. Removing drivelines could be a major improvement in the life of VAD patients, but trials with transcutaneous power induction show no improvement in infection rates [51]. Given the novelty of these devices, surgeons follow treatments from other devices (e.g., pacemakers) as there are no standard procedures for VAD-related infections [32, 40].

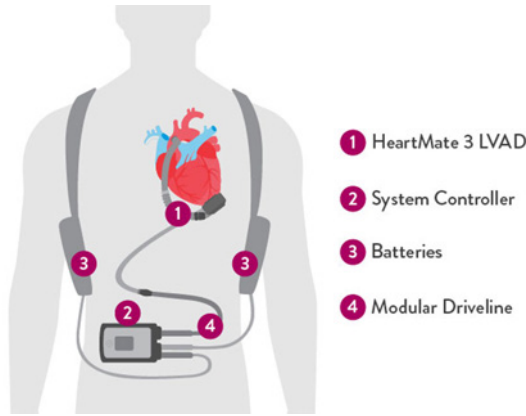


Figure 2.6: Heartmate 3 LVAD system. The driveline connects the internal pump to the external controller that is powered by two batteries [74].

The total cost of VAD therapy including the cost of the device, the operation, anticoagulation therapy, follow-up treatments, and with other costs reaches up to \$300,000 by the end of the first year [37]. Accessibility to these technologies, the high costs, and complexity even in short-term MCS scenarios such as in cardiac arrest cases limits the use of these devices.

The development of novel approaches for blood-contacting and non-blood-contacting devices are an active area of research. Despite advancements in VAD technology, challenges remain in optimising blood flow while minimising device-related complications. Research directions include a pulsating pump using a hemocompatible membrane [75], an invasive soft robotic device moving the ventricular walls [76], further improvements of small-scale turbo machines [77], and new approaches in direct cardiac compression (DCC) [78].

2.6 Direct Cardiac Compression Devices

With the high costs and invasive nature of previously mentioned techniques, alternative MCS solutions are still sought after. DCC devices have been under investigation since before the time of LVAD and ECMO devices. They are similar to open-chest cardiac compression as there is no direct contact with the bloodstream and the device attaches directly to the surface of the heart (epicardium). A sketch of a DCC device is shown in Figure 2.7. The breakthrough in soft robotics fuels a renaissance of DCC devices [79, 40, 80, 81, 76]. One of the earliest DCC prototypes was designed by Vineberg in 1957, driven by pneumatics [20]. In 1965, Anstadt had a major breakthrough with a novel DCC device using a glass cup and inflatable cushioning for short-term support and resuscitation [82]. DCC devices are also used in circulatory support [83, 84], *bridge-to-decision*, and *bridge-to-device* therapies [40]. Even early DCC prototypes could outperform open-chest cardiac compression and closed-chest cardiac compression (modern CPR) in organ perfusion (how well-saturated the organ is with blood) and cardiac output (the volume of blood ejected in a unit of time) [23, 19]. The AbioBooster could reach 1.5 L/min in fibrillation, and could fully recover stroke volume (ejected blood volume in one heartbeat) in cardiac arrest [85]. Patient outcomes between adults and paediatric groups differ, showing lower survival for paediatric patients with complications in anticoagulation therapy [78, 38]. The CorInnova device [86] aims to provide circulatory support for paediatric patients [78], replacing PC-ECMO.

2.6.1 Complications of DCC Devices

So far, no DCC device can provide positive results on human subjects. A potential challenge with these devices lies in precisely controlling pressure within both ventricles. Non-physiological pressure levels or imbalances between the left and right ventricles can lead to complications. The right ventricle is particularly vulnerable to excessive pressure due to equal force application on both ventricles by direct compression. Reducing diastolic filling pressure for better filling of the ventricles is an issue for most DCC devices as they do not provide active suction [27, 76]. As DCC devices compress the heart, and with that deform the muscle tissue and the support structures (chordae, annulus) of the valve apparatus, they can cause valve regurgitation [27]. Mechanical depolarisation of the heart due to direct mechanical compression could hinder cardiac recovery [26]. Increased filling

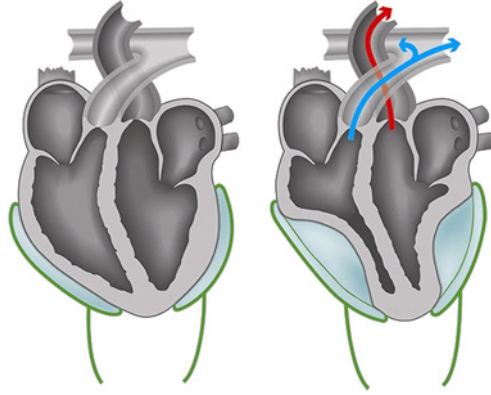


Figure 2.7: Sketch of *HeartPatch* direct cardiac compression device. Each patch was designed to conform to either the left ventricular or right ventricular contour and consisted of a thin inner and thicker outer wall. [87]. Most of current devices follow a similar design.

pressure [26, 78, 27] and epicardial surface adhesion [85] could also lead to complications with DCC devices. To optimise DCC devices, using *in vivo* models (testing on a living animal) would be important, but unfortunately, the device function depends highly on the subject, making animal models a poor substitute [27]. Computer models are therefore essential for optimising and understanding the working mechanisms of DCC devices. There are many complex computer models of heart tissues with possibilities for adding myocardial infarction regions, and modifying heart geometry and material properties [88, 89].

2.6.2 Our DCC Device

Most of the previously mentioned devices aimed for short-term support at first, achieving great results, and later investigated long-term support, but never made it to medical practice. There is no single MCS or DCC solution solving every problem of patients with heart failure or cardiac arrest. We created an accessible DCC device concentrating on emergency scenarios and short-term circulatory support. Our research is based on the work of Kummer et al [90].

Chapter 3

Simulating Human Circulation

DCC devices managed to prove excellent in preclinical studies, but they failed to deliver in the clinical environment. Unlike LVADs and BiVADs, where the devices are in direct contact with the bloodstream, operating in a well-controlled environment, DCC devices provide MCS through a very complex mechanical interaction with the heart tissue. To understand the interaction of MCS devices with the heart and the circulation, computer simulations supported by experiments are required.

Computer simulations are essential in understanding the mechanics and the behaviour of the heart and the circulation. Experiments require a lot of preparation, expensive equipment, and ethics approvals when using animals, and are therefore very expensive. Simulations can provide an environment where ideas and devices can be tested [91]. Scientists and engineers have come up with solutions where detailed models can predict patient outcomes even with different pathologies. Thus, computer simulations can be used as a tool to help device development and build a solid understanding of the cardiovascular system. Some even think that computer simulations could be sufficient for device approval by the governing bodies, but these solutions definitely speed up the process and can reduce costs by reducing the number of experiments required for approval.

Early adaptations of this approach appeared already in the 1960s with analogue computer models. McLeod was one of the pioneers in the field of simulating human circulation with computers [92]. With the tunability of computer models, adjusting resistance, inductance, capacitance, and diodes (i.e. pressure loss due to wall friction, inertia of blood in large vessels, vascular compliance, and heart valves respectively) parameters in the simulation creates the opportunity to investigate individual cardiovascular conditions. Niederer and Smith argued that to improve clinical translation, these computer simulations are essential [93].

The complexity of human physiology is well beyond just the mechanical properties or the visible events. The complex cell-level, biochemistry processes with the biophysics pose a level of complexity that is hard to model

and even harder to compute even with the fastest computers. Therefore, simplifications are unavoidable, and most of the models concentrate on one part of this complex system. The most common approach is to capture macro-scale dynamics while building efficient models.

3.1 Lumped Parameter Models

Lumped-parameter models (LPMs) are the only currently available tools where local and global parameters such as flow rate and pressure in the circulation can be simulated [94]. They are essentially an electric 0D analogue of the human circulation, moving the analogue computer idea into the digital world. They contain resistance, capacitance, inductance and diode elements. An example of an LPM is depicted in Figure 3.1. As LPMs are computationally cheap and can include patient data, the idea of predicting their outcome with medical or device intervention was investigated by many [94, 95]. Human physiology can be integrated not only through the vasculature but also by modelling the functions of the heart [96]. Patient-specific conditions can include pathophysiologic conditions such as atrial fibrillation, aortic stenosis, heart failure, and cardiac arrest. What still remains unsolved is the interaction of the human cardiovascular system, be it in standard or non-standard conditions, with MCS devices. The general understanding of how the heart's control mechanism aligns with the devices [97], and how tissue interacts with MCS devices, such as DCC devices, is missing from the literature [98]. Understanding the underlying mechanisms of these diseases and their consequences has been of interest in the field [80, 99, 88].

3.2 Higher Order Models

More detailed models, concentrating on micro-scale behaviour, such as finite element simulations of tissue dynamics can provide more insight with increased computational costs. A very common approach is to couple LPMs (0D model) with higher-order models such as 1D or 3D models simulating great vessels or the heart [100]. At the initial phase of this project, Thomas Kummer focused on developing a similar framework, coupling an LPM to a finite element model, with simulated heart infarction and DCC device support [88].

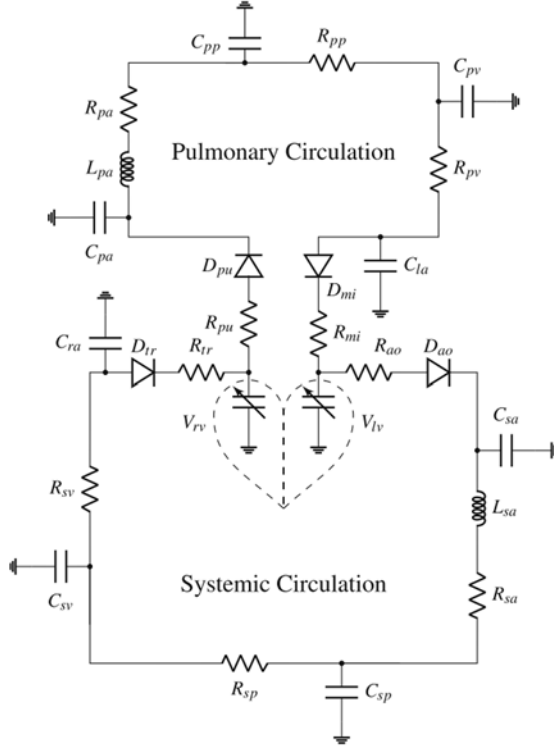


Figure 3.1: An example of the lumped parameter model. The electrical elements of the circuit resemble the resistance, compliance, inertia and flow obstructions in the different parts of the human circulation [88].

3.3 Simulating Pathologies

Simulating pathologies such as heart infarction is important as these conditions are hard to recreate even in *in vivo* models, eventually killing the animal in the process. Therefore, simulating heart, valve and other cardiopulmonary diseases is very important for device development and patient support in general. Heart valve diseases can introduce complications for

MCS, lead to heart failure, and thus understanding the systemic effects through computer simulations is very important [101, 99]. As most of MCS is applied during heart failure, simulating these conditions is essential for the outcome of MCS patients [102, 103]. However, understanding the basic physiology in heart failure is only the first step. The more important part is understanding device interaction with the circulation or the heart in heart failure. These complexities usually require more than a simple LPM, leading to more complex models [104, 105]. Often overlooked but understanding cardiac arrest and thus atrial fibrillation is equally important [106].

Chapter 4

Research Gap

On one hand, we pointed out the need for short-term non-blood-contacting support to help surgeons and the medical team in the operating room. Current devices are either not aiming for or cannot fill this gap. Patients requiring temporary support for recovery, decision or other device support could benefit from a direct cardiac compression device.

Additionally, improving the current understanding of the compliance of the heart, and thus developing efficient computer codes to simulate heart failure, cardiac arrest, mechanical circulatory support, and especially a virtual testing framework for novel direct cardiac compression devices is needed. Most of the computer codes use simple heart models which fail to simulate the interaction of mechanical circulatory support devices and the heart or are computationally very expensive, making optimisation impossible.

In this thesis, we aim to address these three topics: developing a novel direct cardiac compression device for non-blood-contacting short-term support, creating the foundations for a novel compliance model for lumped parameter models, and integrating mechanical circulatory support functionalities into lumped parameter models, especially for optimising our direct cardiac compression device in a virtual environment.

Part III

Direct Cardiac Compression Device Development

Chapter 5

Introduction

Although available solutions for patients living with heart failure and suffering from cardiac arrest are very successful, alternatives are under development. All current mechanical circulatory support (MCS) devices are in contact with the bloodstream. This necessitates anticoagulation therapy for patients; however, some cannot take these therapies, leaving them with no available device treatment. The only solution for these patients is a heart transplant, but the number of available donor hearts unfortunately does not meet the demand.

Advanced heart failure and cardiac arrest require immediate medical intervention. The former, being a disease, requires long-term solutions, while cardiac arrest requires treatments to stabilise the patient and restart blood flow. Thus, designing device solutions for patients with heart failure poses a challenge in fulfilling long-term requirements, while cardiac arrest solutions target short-term interventions for recovery. Heart failure is generally treated with MCS devices such as left ventricular assist devices (LVADs) and biventricular assist devices (BiVADs). Cardiac arrest is treated with CPR, defibrillation, direct cardiac compression by the heart surgeon, or using the ECMO (see Figure 2.5) device.

5.1 Current MCS Solutions

Current VAD solutions provide an outstanding performance and survival rate for patients in the first few years. Initially starting out as a *bridge-to-transplant* solution, these devices proved excellent, now even used as destination therapy due to the lack of donors. However, not only availability of treatment but also risk factors of VAD therapies could be reduced by avoiding anticoagulation therapy. Taking blood-thinning medication combined with the continuous flow rate of most MCS solutions can lead to severe gastrointestinal bleeding and an increased risk of stroke. Many patients have to be rehospitalised or die because of these events.

Cardiac arrest in the operating room is a rare but critical event, often treated by initial cardiac compression by the surgeon, then transferring the patient to an ECMO device. The latter is an expensive intervention as it requires the machine, a specialist to operate the device, cannulation, and anticoagulation drugs. If the surgeon could continue with cardiac compression, the complications and costs of ECMO could be avoided. Unfortunately, the surgeon is tied to the patient while carrying out the compression and hence cannot perform any other life-saving tasks. The need for a device that can perform cardiac compression to relieve the surgeon from this task was evident and led to the development of the first MCS devices.

However, current MCS devices do not include solutions to replace open-chest cardiac compression or non-blood-contacting support for patients with heart failure. It might be surprising that the initial goals of MCS devices have never been fulfilled, nor does it seem to be the goal of current devices. But how did we get here?

5.1.1 History of MCS devices

As mentioned earlier, the first MCS devices were developed to replace open-chest cardiac compression. One of the first devices was the *Mechanical Heart Massager* by Arthur Vineberg in 1957, initially using a metal shell around the heart and later switching to a flexible sac inlaid with inflatable diaphragms. The compression was driven by a piston pump, delivering a 1:2 systolic fraction. The construction is depicted in Figure 5.1 along with the *Direct Mechanical Ventricular Actuation* device by Mark P. Anstadt in 1989, which used a similar approach but with a rigid glass shell for the diaphragm, and constant suction at the apex of the heart to keep the device attached to the heart. Open-chest cardiac compression was later replaced by modern-day CPR, and the need for these devices declined. However, the idea of a device that can compress the heart without being in contact with the bloodstream remained. Initial pulsatile MCS devices were also replaced by small-scale turbomachines providing continuous flow, which are more efficient in maintaining blood flow.

5.1.2 Modern MCS devices

Modern rotores had been struggling with thrombosis and haemolysis until the introduction of magnetically levitated impellers, which reduced the shear forces on the blood. These devices are now even used as destination

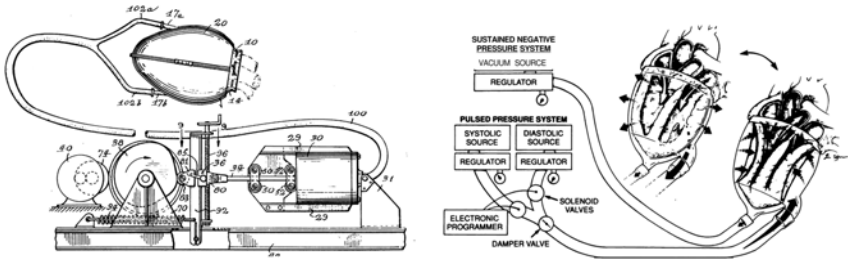


Figure 5.1: The first MCS devices: the Mechanical Heart Massager by Arthur Vineberg (left) [20] and the Direct Mechanical Ventricular Actuation device by Mark P. Anstadt (right) [107].

therapy but still require anticoagulation therapy. The size of the devices has also decreased over the decades, allowing solutions for people with a small body surface area (i. e. children, women). The most successful devices are the *HeartMate 3* and the *Impella* devices, which offer high flow rates with reduced shear rates. The *HeartMate 3* is a continuous flow device with adjustable pulsatility for *bridge-to-transplant* and destination therapy, while the *Impella* is a temporary device for cardiac support during surgery or in the event of cardiogenic shock. The *HeartMate 3* is implanted in the left ventricle at the apex, connecting to the ascending aorta, while the *Impella* is inserted through the femoral artery and placed in the left ventricle through the aorta. Both devices are shown in Figure 2.2. To further reduce blood-related complications, a new technology, using a membrane instead of a rotor, is under development. The *CorWave* device is similar to the *HeartMate 3* but uses a flexible membrane to create pulsatile flow, similar to how a speaker generates sound, which is expected to reduce shear forces and thrombosis (see Figure 5.2).

5.2 Direct Cardiac Compression Devices

However, with all previous solutions being in contact with the bloodstream, the need for anticoagulation therapy remains. The direct cardiac compression (DCC) device, which is the focus of this thesis, is a non-blood-contacting device that can compress the heart and provide cardiac support without the need for anticoagulation therapy. There are many DCC devices under

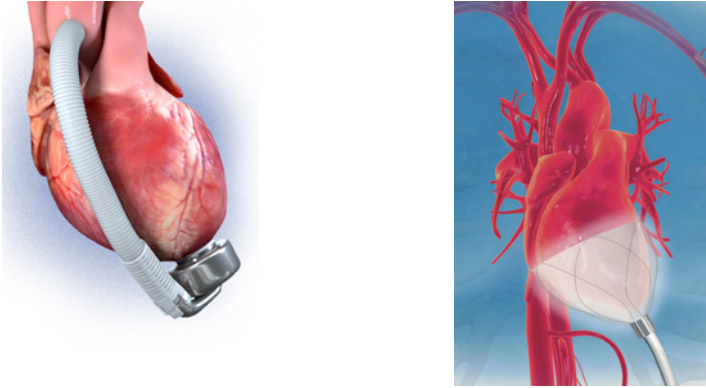


Figure 5.2: Modern devices: the CorWave LVAD (left) [108] and the CorInnova DCC device (right) [105].

development even today; Bonnemain et al. provide an excellent review [79] on the topic. One of the most prominent DCC devices is the CorInnova device, which uses a similar design to the Anstadt device, but it uses a flexible frame around the diaphragm to allow for minimally invasive implantation. The device is placed around the heart and supports the failing heart during systole, providing cardiac support without being in contact with the bloodstream. The device is shown in Figure 5.2. Similar to previous approaches, the CorInnova device targets long-term support for heart failure patients. We believe that LVAD solutions are already proving excellent for heart failure patients, even targeting destination therapy, but many remain without device support apart from the ECMO device in the event of cardiac arrest. To unlock the potential of DCC devices, we follow the guidelines of the literature. These devices could be used in prolonged total circulatory support, resuscitation in cardiac arrest, MCS following ischaemia and infarction, *in vivo* organ preservation, MCS in hypothermia, and in drug overdose cases [23]. Bonnemain et al. defined requirements for such devices: systolic enhancement, low filling pressure, avoiding blood exposure, and adapting to varying conditions [79]. Oz described additional criteria for ideal DCC devices: left and right ventricle stroke volume balance, safe control of right ventricle pressure, simple to apply, adjustable, MCS in cardiac arrest, and easy removal [26].

5.3 Our Approach

Based on these requirements, the development of our device started in 2017. The initial target of Thomas Kummer was to provide a non-blood-contacting MCS device for heart failure patients, similar to the targets of other DCC devices. However, we must be aware of the almost century-long history of DCC devices, which have not been successful in the past for these applications. On the other hand, these devices have proved extremely successful in replacing open-chest cardiac compression, sometimes providing MCS for days. Therefore, our target with the device is to provide a solution for surgeons to assist them with cardiac compression in the event of cardiac arrest without the immediate need to support the patient with ECMO. With this solution, we can buy some time for the medical team in the operating room to decide on the next steps and treat any other complications, while MCS is provided by a non-blood-contacting device, avoiding unnecessary interventions and anticoagulation therapy. Our device can also support medical teams where ECMO support is not possible. We aim to develop a device that is compact, quick to deploy in an open-chest environment, low-cost, and can maintain MCS for a few hours until the patient can be stabilised or transferred to ECMO.

Chapter 6

Previous Designs

As my thesis and project are built on the work of Thomas Kummer [90], I will provide an overview of the previous designs that have influenced our current approach.

6.1 Heart Pincher

The original name of our project was the *Heart Pincher*, which was the name of the first prototype of the design. The first iterations and the final design of the first prototype were built by Michael Liem [109]. The prototype, depicted in Figure 6.1, squeezes the heart with three finger-like levers, hence the name. The device was actuated through a Bowden cable by an actuation device. The early prototypes were tested with active suction, but they caused significant tissue damage, so they were equipped with suction cups instead.



Figure 6.1: The first prototype of the device, the Heart Pincher. The second iteration was built with a more robust structure, using suction cups to attach to the surface of the heart. The development and results are discussed by Michael Liem in his master's thesis [109].

6.2 Contact Patches

During his project, Liem et al. noticed that the suction cups were easily moving along the surface of the *ex vivo* heart during their experiments in the Cardiac Biosimulator at the LifeTec Group in Eindhoven. Initially, they wanted to stay with the design but then added a support at the apex of the heart. Later, they moved to a new design using patches. The idea had already appeared in the work of Liem, but the first prototypes were built by Jonas Conrad [110]. The patches were actuated by a Bowden cable, using a splitter to pull on the patches on the two sides, designed by Patrick Amstad [111]. Figure 6.2 shows the first iterations of the patch design with the suction cups already in the middle. The two patches were then identical, not shaped differently for the left and right ventricles. The first prototypes with different patches for the left and right ventricles were designed by Tobias Spiegelburg [112]. Their design is depicted in Figure 6.2. They had already experimented with connecting the two patches on one side and having different suction locations on the patches.



Figure 6.2: The first iterations of the patch design. These devices used two bowden cables for actuation. The patches moved closer to each other as the cables were pulled. The design on the left was the first iteration of the patch design by Jonas Conrad [110], and the prototype on the right was designed by Tobias Spiegelburg [112].

6.2.1 Kummer's Design

The final design of Thomas Kummer [90] is depicted in Figure 6.3. It used two Bowden cables for actuation, a vacuum pump for suction in the centre of the patches, and a motorised actuator to control the patches. The design features two distinct patches for the left and right ventricles to control the

pressure in the ventricles during compression. Kummer et al. also designed a new actuator device with servo motors to replace the previous one, which used a spindle motor.



Figure 6.3: The last prototype of Thomas Kummer. It used two Bowden cables for actuation and already had the vacuum cups in the centre. The image on the right is taken from a CAD software [90].

6.3 Results and Limitations

The device was tested on a silicone heart [113] in a mock circulation loop [114], both designed by Raphael Holdener, *ex vivo* in the same mock circulation loop, *ex vivo* in the Cardiac Biosimulator at the LifeTec Group in Eindhoven, and in a cadaver study at the University Hospital Zürich. The results were promising; however, they never managed to match the performance of open-chest cardiac compression. The device was able to perform regular compression on the heart with low but stable cardiac output and mean arterial pressure. The reader can find more information on the results in the thesis of Thomas Kummer [90].

The unsolved issues of the device at the time of taking over the project were, on one hand, the performance, of course, but other design flaws also needed correction. The most important one was the buckling of the patches during contraction. As the patches were moving towards each other, the lower parts could move faster, essentially pinching the apex of the heart. This resulted in the ballooning of the apex, potentially stressing and damaging the tissue. This also indicated that the device could potentially have issues removing the blood from the apex during contraction. Removing the degrees of freedom was necessary to avoid this phenomenon, and improving the

Chapter 6 Previous Designs

general performance of the device was essential for successful applications. There were also issues with tissue-driveline contact, damaging the heart during actuation.

Chapter 7

Current Design

The current design was inspired by the work in the previous chapter. Kummer et al. already discussed the option to connect the two patches to reduce the degrees of freedom during actuation. This idea was taken up in this project by removing the cable attachments from one side and connecting the patches with an adjustable axis. With the patches connected on one side, the driveline on the same side was also removed. The extension at the magnetic lock had not been discussed; the input for this came after the final porcine cadaver experiments in 2019.

7.1 Design process

Designing the patches was made possible by the initial design of Thomas Kummer. I used the CAD software SpaceClaim [115] to modify the design and prepare it for 3D printing. The three main components added to the previous design were the adjustable axis, the single driveline, and the extension at the magnetic lock.

The adjustable axis was added by removing the features on one side of the patches and adding arms to that side. During the design phase, the movement of the axis and the patches, and the placement of the patches on the heart had to be considered. Making sure that the arms did not damage the heart during compression and that there was enough play and tolerance for the axis to move properly. The axis is a simple M3 screw, and the size adjustment is made by multiple holes in the arms. The axis is placed through these holes to define the centre of rotation of the patches.

This allows the patches to be actuated with a single Bowden cable, rotating around the axis. The original lateral motion of the patches was thus replaced by a rotational motion, similar to the motion of a pair of pliers or a clam.

The extension on the other side of the patches was added to decrease the chance of tissue damage during compression due to contact with the wire cable. The features of the patches were removed on that side, and an extrusion was added to the side of both patches. This 50 mm extension makes

the patches bigger and could cause potential issues with device placement in a compact chest, but at the time of writing this thesis, we did not consider this a limitation. The removed features for connecting the driveline were then placed on the ends of the extrusions.

7.2 Manufacturing

Initial prototypes were 3D printed in the lab using an Elegoo Neptune 3 Pro 3D printer with PLA filaments. The final design was printed at the Wyss Center, Zürich, using HP PA12 material in a Jet Fusion process. The material is not biocompatible, and at the time of writing this thesis, we did not consider this necessary.

7.3 Current design

The final design is depicted in Figure 7.1. The device kept the initial differences between the left and right patches to have a smaller contact surface with the left ventricle, hence producing higher pressure in the left ventricle in theory. The suction remained in the middle of the patches, using a neoprene cushion to seal the area under vacuum. The suction is provided by vacuum lines connected to the back of the patches (see Figure 7.1). The edges of the additional features are not rounded or chamfered to simplify the printing process, as in the current phase, we were only considering the functionality of the patches.



Figure 7.1: The current design with the axis on one side and the single Bowden cable on the other side of the patches. The picture on the right shows the inlet location for the vacuum line and the connected Bowden cable.

7.4 Improvements over the previous design

The device is shown in Figure 7.2 with the intended placement on a silicone heart, manufactured by Elia Morello [116]. The magnetic lock head is flushed in the extrusion on the right ventricle patch. The locking mechanism uses neodymium magnets to hold the head in the socket during operation. The other connector uses two screws and friction to keep the Bowden cable in place on the left ventricle patch. The standard wire cable fittings were fixed by pressing them onto the cable.



Figure 7.2: The current design with the intended placement on a silicone heart sample.

7.4 Improvements over the previous design

With the new design and the additional design features, we were able to challenge the issues of the previous design. The adjustable axis allows for adjusting the patches to different sizes of hearts, and the rotational motion avoids the buckling present with the lateral compression of the previous design. The single Bowden cable reduces the complexity on the actuation side and also helps with placement inside the chest. The extension on the side of the driveline decreases the chance of tissue damage during compression due to contact with the moving wire cable.

Chapter 8

Actuator and Other Parts

The direct cardiac compression device consists not only of the patches but also of an actuator and a vacuum pump. The actuator, be it motorised or manual, is used to actuate the patches, and the vacuum pump helps to keep the patches in place. We used the motorised actuation of Thomas Kummer [90] and built on his concepts, essentially replacing almost every element in the housing. I also designed a manual actuator as a backup solution in case the motors fail during operation. I must mention that the development of the actuators was not part of my thesis, but it was necessary to run tests with the device before the manual actuation was designed. Also, motorised operation provides more control over the device and allows the operators to work on other parts of the experiments while the device is running.

We took the servo motor design of Thomas Kummer, and as the performance was not sufficient, we had to build an actuator with stronger servo motors, also fitting the new single driveline design. The new motorised actuator was designed and manufactured during the master's thesis project of Jan Vincent Spitzer [117].

The following sections are based on the following manuscripts:

J. V. Spitzer. “Testing and developing and extraventricular assist device”. Master's Thesis. ETH Zurich, 2022

8.1 Actuator

8.1.1 Motorised actuator

The actuator is shown in Figure 8.1. As we took over the housing of the actuator device, constraints were imposed by the dimensions of the box. Another important aspect was that the previous design suffered from reliability issues along with insufficient performance. The new design uses new servo motors, connected by a mechanical fork, and the I^2C protocol

was replaced by *UART* communication. The issue with *I²C* was that it is known to be prone to electromagnetic interference, and when one line breaks in the communication chain, the whole system shuts down. The number of Arduino boards was reduced from three to two to further simplify the system. The Arduinos were combined in a single package to be easily replaceable in case of control failure. The actuator can be controlled via the physical knobs on the control interface or via WiFi using a web interface. The adjustable settings are the amplitude, frequency, and systolic fraction of the compression. The new actuator is able to provide feedback to the user regarding limitations in the settings. As the speed of the servo motors is limited, the user can potentially set amplitude, frequency, and systolic fraction combinations that cannot be achieved by the device.



Figure 8.1: The contact patches with a silicone heart, connected to the motorised actuator. The actuator allows setting amplitude, frequency, systolic fraction of the compression.

8.1.2 Manual actuator

The manual actuator was designed to replace motorised actuation if the actuator failed during the experiments. Also, using a manual actuator, one could test whether the motorised actuator is a limiting factor in the performance of the DCC device. After the initial design phase, aiming at using non-standard 3D-printed parts to assemble the manual actuator, I decided to use standard parts to simplify the manufacturing process and limit potential reliability issues. As the device uses Bowden cables for actuation, it was only natural to use standard bike parts to build the manual actuator.

The actuator consists of a handlebar and a brake lever. The Bowden cable is attached to the brake lever, and the other end is connected to the patches. The actuator is shown in Figure 8.2. The manual actuator allows the user to control the patches manually. To pace the actuation, the user can use a simple audio signal at the desired rate of compression. The actuator has a very simple and rigid design, allowing the user to use it with their hands or feet.



Figure 8.2: The manual actuator, using standard bike parts.

8.2 Vacuum pump

The vacuum pump is a commercial, off-the-shelf device, an ATMOS C 161 respirator. It is capable of providing suction up to 750 mmHg and is connected to the patches via two vacuum lines. The patches have a vacuum channel from the connection point to the central circle of the patches.

Chapter 9

Conclusions and outlook

Our device is different from other DCC devices in the literature. Most DCC devices follow the design of the Anstadt cup, or modern devices use soft robotics to provide MCS support. Our device, however, uses only two rigid patches to compress the ventricles, and the actuation is completely mechanical, as the vacuum is only used to keep the device in place. This is a significant difference, as it allows for a simpler design that can be deployed quickly in an emergency scenario. The device is also designed to be used in short-term MCS, unlike other devices that are designed for *bridge-to-transplant* or long-term support. Our device is the first DCC device that can show different pressures in the left and right ventricles in cardiac arrest experiments, potentially avoiding the risk of pulmonary hypertension during DCC support.

Our design features two patches with design differences to control the pressure in the left and right ventricles during compression. This is a unique feature among DCC devices, as most other devices report similar pressure increments with DCC support on both the left and right ventricles [78, 86]. Our device includes suction on the surface of the ventricles, another unique feature that could reduce the filling pressure on the ventricles. High filling pressure is one of the obstacles in DCC support, as it can lead to decreased cardiac output. Thanks to the magnetic locking, our device can be deployed in a matter of a minute in an open-chest emergency scenario.

Our device has the potential to impact the field of mechanical circulatory support. We noticed during the *ex vivo* experiments that the curvature of the right ventricle is too high for bigger hearts, damaging the tissue significantly. The current size adjustment is not feasible for emergency scenarios in an operating room, so a solution for quick adjustment is needed. The patches are not biocompatible, and a plan for providing a sterile device for applications is needed. As we did not focus on the ergonomics of the device, it is relatively bulky and requires optimisation for a more compact design.

We noticed that vacuum support is not necessary, even in the *ex vivo*

experiments. Following this line, and trying to reduce the size of the device (the vacuum channel in the patches limits the minimal thickness of the patches), we have plans for removing the vacuum support and using a net to keep the device on the heart. The design would be similar to the ACORN device [118], combined with a thin version of the patches. The net would be used to keep the device in place, and the patches would provide the compression. This design would allow for a more compact device that could be deployed quickly in an emergency scenario. A prototype of this new design is shown in Figure 9.1.

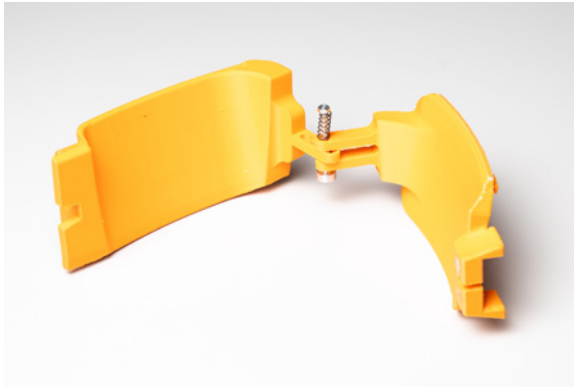


Figure 9.1: The latest prototype, designed by Annika Hämmer during her Semester Project. It features a quick adjustment mechanism, flatter patches, and no vacuum support.

The first version of this device was developed by Annika Hämmer during her Semester Project. The new design was based on the feedback we got from the *ex vivo* experiments. We wanted to reduce the curvature of the RV patch, come up with a new axis design for quick adjustments, and remove the vacuum lines from the patches while providing the possibility for net support. The new patches are flatter, and the quick adjustment mechanism allows for a quick change of the size of the device. The current design does not include a concrete net support, but it has hooks on the outside of the patches to allow the net to be pulled over the heart and the patches to keep them attached.

To improve the manual actuator, we are currently working on a new foot pedal design, as actuating the device in the long term with the hands can be painful or tiresome.

Part IV

In Vitro Testing

Chapter 10

Introduction

In the development of the novel direct cardiac compression (DCC) device, the stages of testing are as follows: first, *in vitro* testing on silicone heart models in a mock circulation loop (MCL); then, *ex vivo* testing on animal heart samples; then, cadaver testing on a diseased animal; and finally, *in vivo* testing in a live animal model with induced heart failure and cardiac arrest. This part focuses on the first stage of testing, which is the *in vitro* testing of the DCC BiVAD system in our in-house testing facility.

In Chapter 7 and Chapter 8, the DCC patches, the actuation devices, and the vacuum pump were already introduced. To conduct *in vitro* experiments, an MCL as a test bench is required. The design of the MCL and the data acquisition system is equally important for the purpose of this thesis and requires a detailed description. The original MCL was built by Thomas Kummer [90], but the measurement instrumentation was completely redesigned to improve the accuracy of the data acquisition system. It is important to note that there is no off-the-shelf solution for MCLs, meaning that every biomedical lab has its own design with unique components. To be transparent regarding the limitations, details beyond the usual specifications in publications are required. The MCL, the silicone heart, and the data acquisition system are described in Chapter 11.

This part introduces an early prototype of a short-term emergency direct cardiac compression (DCC) device designed to replace open-chest cardiac compression (OCCC) with biventricular support. Although the original Anstadt cup was meant to replace open-chest cardiac compression, current DCC devices aim to provide long-term solutions. While using a similar actuation concept, our device addresses the unmet need for rapid intervention in resuscitative thoracotomy. DCC requires no blood-thinning medication as the device is not in direct contact with the patient's blood. As the success rate of open-chest cardiac compression in resuscitative thoracotomy is currently low, we aim to improve on that by introducing little to no new complication factors. The goal of our research is to develop a DCC concept that simultaneously maintains physiological pressure levels for both

ventricles while ensuring sufficient cardiac output.

Here, we present a 3D-printed epicardial patch system actuated by the motorised actuator via wire cables. Figure 10.1 demonstrates our DCC patches on a silicone heart model. The design allows quick application in emergency cases, freeing medical staff from open-chest cardiac compression. Additionally, thanks to its straightforward design, the DCC device requires minor training for application or operation. This DCC device aims to offer control over pressure and flow rates, facilitating the exploration of optimal resuscitation strategies.



Figure 10.1: Our DCC patch device on a silicone heart model.

To experimentally validate the DCC device concept, we built upon the simulation framework by Kummer et al. [88], which explored the interaction of a failing heart’s digital twin with a DCC patch system. The potential of a DCC patch system was already demonstrated in their study, albeit only in a simulation framework. Their lumped parameter model of human circulation is replicated on a test bench using components such as ball valves, compliance chambers, and check valves. Pressure and flow rates were measured simultaneously for both ventricles. Comparative studies of open-chest cardiac compression and mechanical actuation with the patch system were conducted on silicone hearts. Initial experiments with porcine heart samples were inconclusive, leading to the choice of silicone models. However, these preliminary *ex vivo* experiments were valuable for assessing the tissue damage caused by the device. The digital twin of Kummer et al. [88], the silicone heart, and the geometry of the patches were based on

the same generic human heart CAD model [119]. This proof-of-concept study aimed to demonstrate the DCC patch system as a viable alternative to open-chest cardiac compression.

This study demonstrated the feasibility of our DCC device using a silicone heart model. At a fixed rate of 60 bpm, we compared our system against open-chest cardiac compression, focusing on the critical 1.5 L/min cardiac output target. The results include raw and statistically evaluated pressure and flow data for both ventricles. Importantly, our DCC device managed to operate in the range of the target cardiac output while significantly reducing peak ventricular pressures compared to open-chest cardiac compression. Additionally, the DCC device reliably maintained the desired actuation rate, cardiac output, and peak pressure performance.

This part of the thesis builds on the previously published work on the *in vitro* testing of the DCC device, the master’s thesis project of Jan Vincent Spitzer, and the bachelor’s thesis of Elia Morello:

K. Sarosi et al. “Demonstration of a Mechanical External Biventricular Assist Device for Resuscitative Thoracotomy”. In: *Cardiovascular Engineering and Technology* (July 2025). ISSN: 1869-408X. DOI: <https://doi.org/10.1007/s13239-025-00793-y>

J. V. Spitzer. “Testing and developing and extraventricular assist device”. Master’s Thesis. ETH Zurich, 2022

E. Morello. “Parametric studies on heart models”. Bachelor’s Thesis. Institute of Fluid Dynamics, ETH Zurich, 2023

Chapter 11

In vitro setup

Building an MCL with completely redesigned instrumentation was a major part of this project. The MCL built by Kummer et al. [90] was a good starting point, taking over features from MCLs in the literature. However, the instrumentation and thus the accuracy of the data acquisition system were not sufficient for testing our device. Some parts of the MCL were redesigned to include the new flow meters and to provide the standard measures for instrument placements. During our experiments, we noticed other design issues with the original MCL, but these did not affect the results of our experiments. However, these issues must be addressed in the future to improve the MCL and simplify the measurement process. The rebuild of the MCL would have been impossible without the expert support of Prof. Thomas Rösgen and the hard work of Jan Vincent Spitzer during his thesis.

11.1 Mock Circulation Loop

This study utilizes a hydraulic analogue of human circulation for testing cardiac assist device performance. Figure 11.1 presents the schematics and the actual setup, highlighting the most important parts. The MCL consists of a systemic and a pulmonary circuit, two water pumps, and the sensors. Each circulation has a compliance chamber modeling the compliance of the blood vessels, a ball valve modeling the flow resistance of the system, and a water pump to fill the compliance chamber. Compliances and resistances are adjustable with adjusting the water level or the volume of air in the compliance chamber, and adjusting the valve in the circulation. We used distilled water in the setup as our flow sensors were calibrated for that.

The compliance chambers model the compliance as they are closed to ambient pressure, and the initial volume of air in the chamber defines the compliance of the system. Kummer provides a detailed description of the compliance chambers and their target water levels in his thesis [90]. Their initial design included flow metering based on the pressure of air in the compliance chambers. However, this is not a standard method for flow

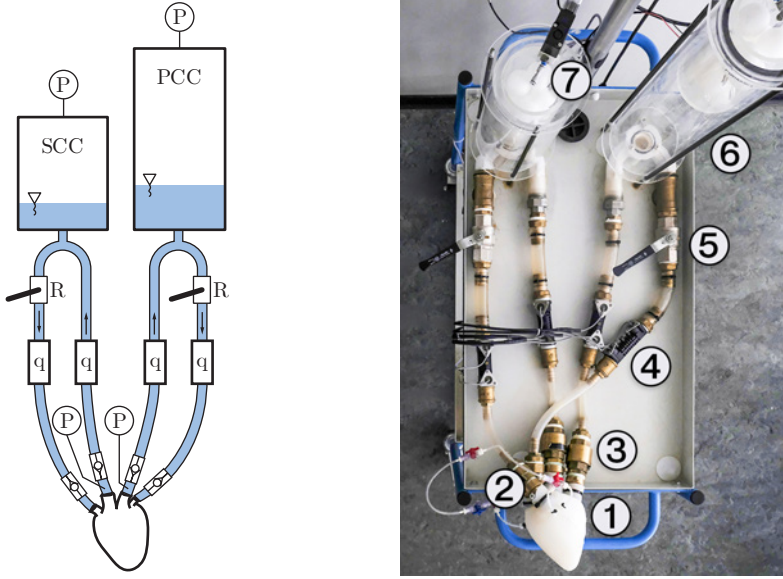


Figure 11.1: The Mock Circulation Loop, including a silicone heart. **Left:** The schematics of the Mock Circulation Loop. *SCC* stands for Systemic Compliance Chamber, *PCC* for Pulmonary Compliance Chamber, *P* for static pressure taps, *q* for ultrasonic flow meters, and *R* for adjustable resistances. **Right:** The Mock Circulation Loop consists of the heart (1), the pressure taps for the ventricles (2), the check valves (3), the Audiowell flow meters (4), the ball valves (5), the compliance chambers (6), and the pressure taps of the compliance chambers (7).

metering, especially not in the highly complex case of a pulsatile, bidirectional flow in the range of 0-40 L/min. After creating a model pointing out the flaws of the original flow metering, we proved that using specific flow meters provides superior results.

The resistors in the system are modelled by industrial ball valves. The resistance can be set by opening or closing the valve. There is no precise way to measure the resistance, but the valve can be set according to the desired flow rate and pressure values.

The heart valves are represented by industrial check valves. These valves are placed directly at the heart's inlet and outlet, and they are designed to allow flow in one direction only. The valves open at a given pressure difference on both sides. More details regarding the selection of the valves can be found in [90].

11.2 Data acquisition

Accurate data acquisition is paramount in medical device testing. The new MCL employs four Audiowell HS0016 ultrasonic flow meters with four TI EVM430-FR6047 data acquisition boards to measure rapid flow changes without altering the flows through obstructions. We received an EmTec BioPro TT clamp-on ultrasonic flow meter with an EmTec DIGIFLOW EXT-1 data acquisition board from *Cardiocentro Ticino* for testing and calibrating our system.

The Audiowell flow meters were placed on the two systemic and two pulmonary lines representing the aorta, vena cava, pulmonary artery, and pulmonary vein. The flow meters are placed in their respective positive flow directions. The Audiowell flow meters were not designed to serve in such a setup; therefore, their settings had to be precisely adjusted, and the connection to the data acquisition system had to be modified to work in our setup. We used the more precise Emtec BioPro TT clamp-on sensor to calibrate our setup with the Audiowell flow meters, and we also used it at various locations in the MCL. To include the EmTec flow meters in our setup, we used Saint-Gobain Tygon S3 E-3603 tubing, as it was individually calibrated with this tubing with water. More details regarding the flow meter setup can be found in Appendix A.2.

The digital pressure sensors of the initial MCL were replaced by analogue ones to improve their reliability. The analogue pressure sensors are connected to the left ventricle, the right ventricle, and the two compliance chambers through static pressure tapping.

The data from the sensors is collected by two Arduino Portenta H7 boards simultaneously capturing flow rates and pressures at 100 Hz / ADC 16bit. One board is connected only to the EmTec flow meter's data acquisition system, while the other one is connected to the four Audiowell flow meters and the pressure sensors, and receives the data from the other board via WiFi. The two separate systems are depicted in Figure 11.2. The data acquisition systems are designed to transmit data through WiFi at 100 Hz. More details regarding the data acquisition system can be found in [117].

For further information on data acquisition components, see Table A.2.

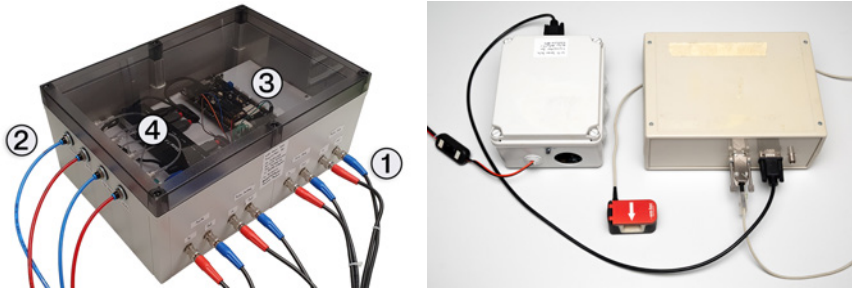


Figure 11.2: The data acquisition system for the flow meters and the pressure sensors. **Left:** The data acquisition box for the four Audiowell flow sensors and the pressure sensors. **Right:** The data acquisition system of the EmTec BioPro TT flow sensor.

11.3 Silicone Heart

Initial experiments and compliance measurements showed that the silicone heart of Kummer et al. [90] was not suitable for our experiments. The silicone hearts were too stiff compared to porcine samples; thus, the compliance of the heart was not sufficient to simulate the human heart. Therefore, Irma Burazorović and Elia Morello redesigned the silicone heart to better match the compliance of a porcine heart.



Figure 11.3: The silicone heart used in the experiments. The unit was cast using *Dragon Skin™* silicone. The pressure taps can be found on the brass connectors. The heart valves are represented by check valves.

For the silicone heart, we used the material *Dragon SkinTM* based on the research of Irma Burazorović. This silicone provides a soft material while providing flexibility and durability for our pressure ranges and our applications. The silicone hearts are cast and closed using silicone adhesive. The process was carried out by Elia Morello [116] and followed the guidelines of Raphael Holdener [114]. The shape of the heart was given by a generic 3D model, provided by Thomas Kummer [119]. The silicone heart used in the experiments can be seen in Figure 11.3.

Chapter 12

Methods

This chapter provides details regarding the DCC patch system and the methodology in our *in vitro* experiments.

Our goal was to design a DCC device that provides sufficient blood flow in the circulation while keeping pressure levels safe in both ventricles without damaging the epicardium. When imposing external actuation through the epicardium, it is important that not only the flow rates but also the pressures remain in sustainable ranges throughout the actuation, while keeping the coronary and capillary flow unobstructed. In extreme cases, the heart can deform in such a way that the tissue or the blood vessels get damaged. The deformation may cause severe valve leakage or even rupture the chordae tendineae. We used the initial design of Kummer et al. [90] and modified it to address the issues they noticed in their studies.

12.1 DCC Device

The DCC device includes patches that directly compress the heart, an actuation device to control the applied force, and a vacuum pump to ensure the patches remain securely attached to the epicardium. This system is presented in Figure 12.1. The patches are in direct contact with the heart and are responsible for compressing the heart and controlling the compression in different regions. The actuation device provides the total compression force for the patches which can be adjusted to remain in the desired range. The vacuum pump is connected to the patches, which each have a vacuum channel to create suction inside the central circular cavity of the patches. Without this suction, the device may easily slip off the wet epicardium.

More information about the DCC device can be found in Chapter 7.

12.1.1 Actuation

The actuator includes two digital servo motors. These motors are connected to a single Bowden cable that transmits the force to the patches. The

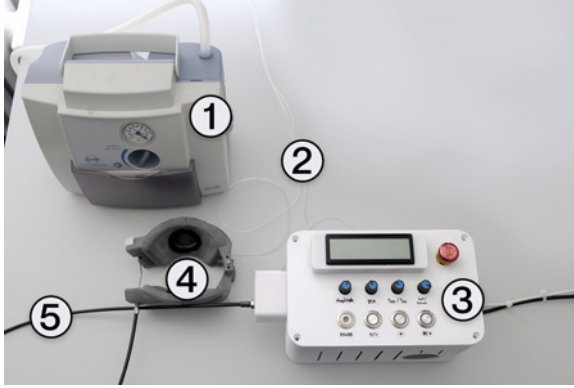


Figure 12.1: The DCC BiVAD system, including the vacuum pump (1), connecting to the force patches (4) via vacuum lines (2). The patches are in direct contact with the heart, and actuated by the actuation device (3) through a single Bowden cable (5).

two servo motors are controlled by two Arduino devices, which adjust the amplitude, frequency, and the systolic and diastolic periods of the actuation.

More details about the actuation and its components can be found in Chapter 8 and Appendix A.1 respectively.

12.2 Experiments

The experiments were conducted in an MCL setup, which is described in Section 11.1. The MCL was designed to simulate human circulation, and it was used to test the DCC device. The MCL includes a silicone heart, which is used to simulate the left and right ventricles. We measure the pressure in the aorta and the pulmonary artery, and the flow rate in the aorta, pulmonary artery, pulmonary vein, and the vena cava. The sampling rate of the data acquisition system is 100 Hz, which is sufficient to capture the rapid changes in flow and pressure.

This study compared three actuation modes: manual patch compression, open-chest cardiac compression (OCCC), and actuation via a specialised device (Figure 12.2). Manual patch compression demonstrates the system’s capabilities without the actuation device. Amplitude, frequency, and $T_{systolic}/T_{diastolic}$ are adjustable using the actuation device. In both manual patch compression and OCCC, the only variable is the rate of actuation. Given the use of a silicone heart model, synchronisation with a heartbeat



Figure 12.2: The three modes of actuation. **Left:** manual compression of the patches, **Centre:** open-chest cardiac compression, **Right:** compression of the patches by the actuation device

was not required; also, cardiac output and pressure are supplied entirely by the mode of actuation. During patch device actuation, the patches were secured to the heart's surface with the locking mechanism.

While our study on silicone heart has been published [1], we also conducted experiments using *ex vivo* porcine hearts. These experiments were not successful in assessing the performance of the DCC device, but they provided valuable information about the tissue damage caused by the patches and the suction.

12.2.1 Silicone Heart

The results shown in Chapter 13 were acquired using a silicone heart. To find more details about the silicone heart, please refer to Section 11.3.

12.2.2 Porcine Heart

We also conducted experiments with *ex vivo* porcine hearts, which were obtained from *Stadt Zürich, Umwelt- und Gesundheitsschutz, Veterinaerdienste*, but they were unsuccessful for different reasons. However, we could use those experiments to assess the tissue damage caused by the patches and the suction.

12.3 Data Processing

To ensure robust statistical analysis, the collected data was segmented into individual heart cycles. Each presented measurement lasted one minute, yielding over 50 heart cycles per configuration. To evaluate the expected performance (see Figure 13.3) of the three modes, the individual cycles were resampled to their median length to neglect the inconsistency of the rate of actuation (see Figure 13.4). Understanding the actuation mode's influence on actuation frequency, cardiac output, and peak pressure variability is crucial. Due to differences in consistency between the three modes, the data did not meet homoscedasticity assumptions, precluding the use of ANOVA. Therefore, Welch's t-test was used to compare expected values, and Levene's test was employed to assess variance.

Chapter 13

Results and discussion

In demonstrating the performance of a VAD, the most important factors are cardiac output, pressure values, tissue damage, consistency and reliability. The selected results have matching frequency and orientation of the device to provide a fair comparison for all three modes of actuation (see Figure 12.2). The orientation of the patches was an important design aspect, having a significant effect on the outcomes. The intended use was with the smaller patch compressing the LV, with the two patches being parallel to the interventricular septum. All of the following cases were recorded at 60 bpm, with the patches oriented in the aforementioned configuration.

The following values were recorded during the experiments: flow rate (aorta, pulmonary artery, pulmonary vein, vena cava) and pressure (aorta, pulmonary artery). Regarding cardiac output, we found that actuating the heart at 60 bpm was the ideal rate of actuation. The orientation of the device is such that the heart sits perfectly between the patches (the intended orientation during the design phase). It is important to note that the results are shown for a non-self-actuating silicone heart, meaning that the flow and pressure output are caused entirely by the mode of actuation.

For a proof-of-concept study of the DCC device, the results are compared to those of open-chest cardiac compression. It was also important to see if manually actuating the patches could improve on the results of open-chest cardiac compression. Figure 13.1 shows results comparing the DCC device to open-chest cardiac compression, and Figure 13.2 shows raw data for all three modes of actuation for comparison. Open-chest cardiac compression operates at higher flow rates and peak pressures than using the device. The same is true for actuating the patches by hand over using the actuation device. On the other hand, using the device results in lower peak pressure levels in both ventricles over the other two modes of actuation.

From Figure 13.2, one cannot draw conclusions on the expected performance and the consistency of the output. This requires statistical studies on the collected data as discussed in Section 12.3. Figure 13.3 shows results using means and variation of the data for each mode of actuation. While

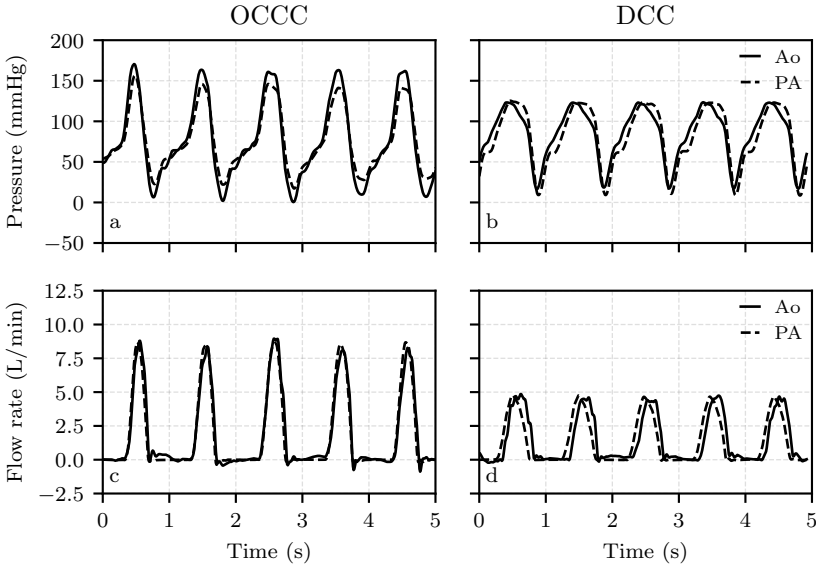


Figure 13.1: Raw data of five heart cycles applying OCCC and using the DCC device with motorised actuation on a silicone heart sample.

the expected cardiac cycles show similar results as in Figure 13.2, we can now also see how consistent the different modes are. Regarding consistency, the actuation device produces the least variance among the three modes.

Essentially, this study also compares human operation to machine actuation. Section 12.3 mentions differences in the length of the average cycles between the three modes of actuation. This is the result of disparity in the consistency of actuation. In Figure 13.4, the box plots of the stroke volume and the actuation frequency for all three modes are shown. It is visible that the difference between the modes in terms of the means of frequency of actuation is not highly significant, while the stroke volume shows more variation. The variation of the actuation frequency and the stroke volume are significantly smaller using the device compared to open-chest cardiac compression.

From the averaged values (Figure 13.3), we can derive the expected

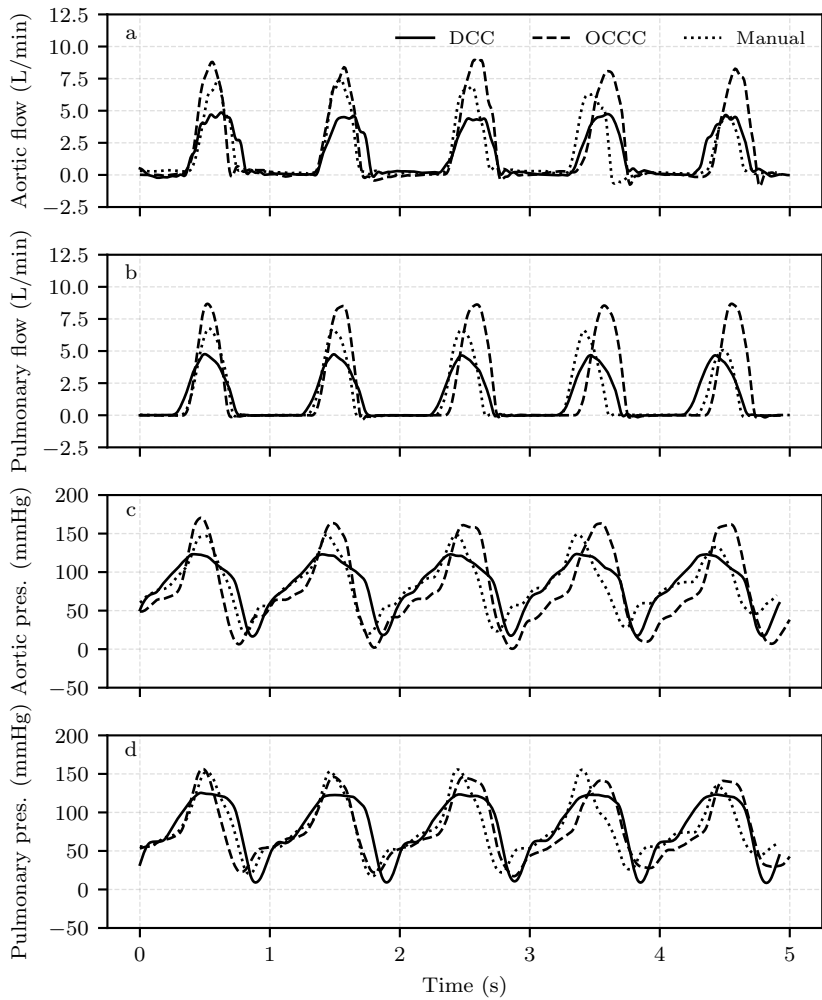


Figure 13.2: Raw data of the three actuating modes for five heart cycles. Nomenclature: OCCC - open-chest cardiac compression, DCC - using the device with the actuator, Manual - manual actuation of the patches.

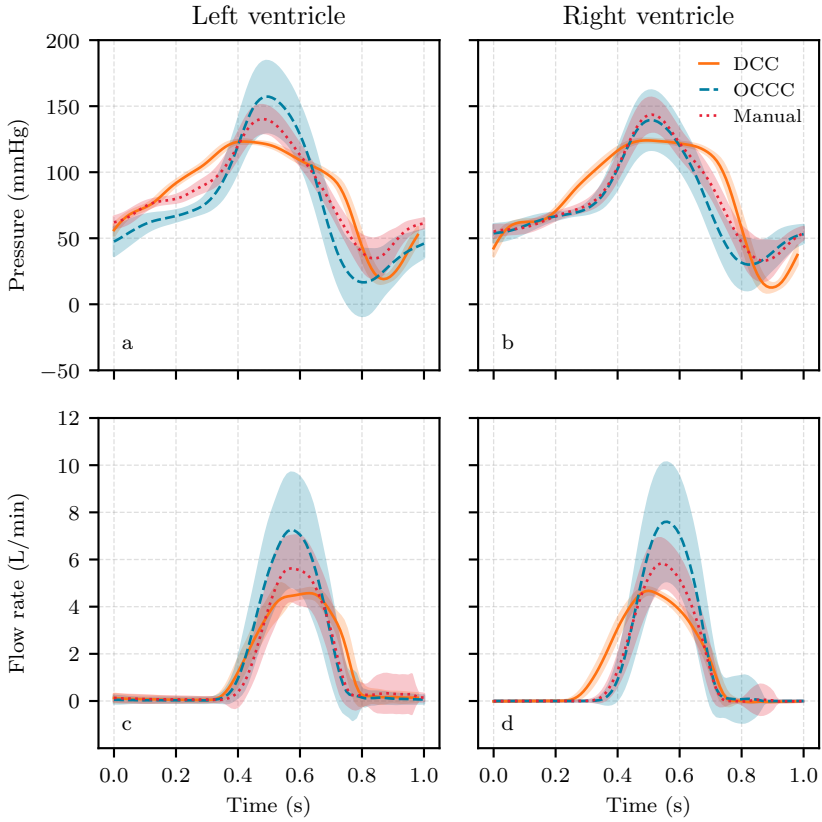


Figure 13.3: The mean values and the standard deviation bands were obtained by sampling individual heart cycles in a measurement. Data was recorded at 60 bpm in all cases. Nomenclature: OCCC - open-chest cardiac compression, DCC - using the device with the actuator, Manual - manual actuation of the patches.

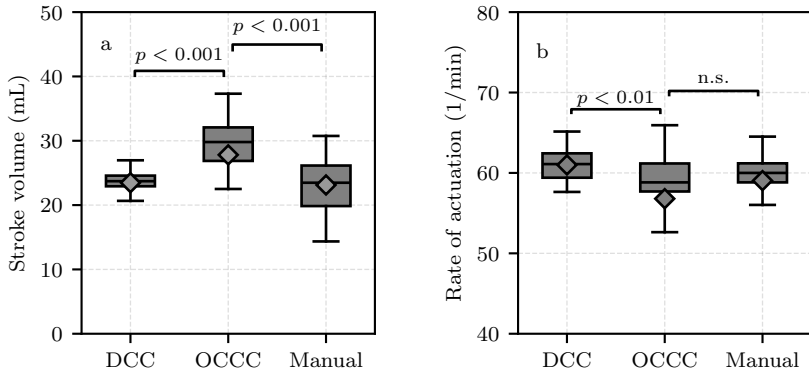


Figure 13.4: Box plots of the stroke volume of the left ventricle and the rate of actuation. The means are represented by the rhomboids and the horizontal line represents the median. Nomenclature: OCCC - open-chest cardiac compression, DCC - using the device with the actuator, Manual - manual actuation of the patches.

pressure-volume diagram for both ventricles. Figure 13.5 shows these results, from which we notice that the ejected volume is approximately a third of the physiological 95 mL [120] for both ventricles and all modes, matching the findings in Figure 13.4. The device produces lower pressure values than the other two modes while having a similar output as with manual patch actuation. As seen in the previous figures, open-chest cardiac compression produces higher pressure and higher cardiac output compared to the rest.

For the concluding assessment of the actuation modes, the cardiac output, the peak pressures in the left and right ventricles, and their consistency must be thoroughly examined. Figure 13.6 shows the three properties for all modes of actuation. As seen before, open-chest cardiac compression produces higher cardiac output, but using the patches actually increases the mean pressure while reducing the peak pressures (see Figure 13.3). All three modes are close to or exceed the 1.5 L/min cardiac output required to sustain the life of a patient. Using the device, all measures show significantly lower variance compared to open-chest cardiac compression.

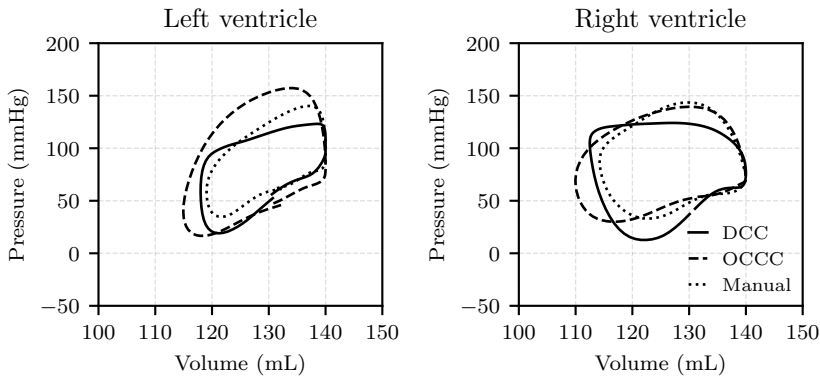


Figure 13.5: Pressure-volume diagrams of the three actuation modes for both ventricles. The results represent the mean values of pressure and volume based on the entire measurement (see Figure 13.3). Nomenclature: OCCC - open-chest cardiac compression, DCC - using the device with the actuator, Manual - manual actuation of the patches.

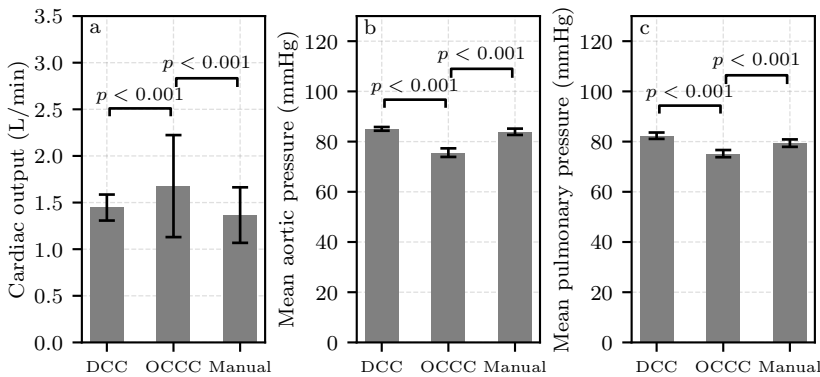


Figure 13.6: Plots of the three main performance indicators. Nomenclature: OCCC - open-chest cardiac compression, DCC - using the device with the actuator, Manual - manual actuation of the patches.

13.1 Discussion

Data (Figures 13.2 and 13.3) indicate that all modes of actuation were relatively consistent. Being machine-operated, consistency is an obvious consequence of using the DCC device, but surprisingly, manual operation and open-chest cardiac compression also show consistent trends. With the latter two modes, we used a metronome to signal 60 bpm pacing for actuation. Prior to the final measurements, we conducted experiments at 40 and 80 bpm, but 60 bpm was chosen based on cardiac output and consistency of the output. Using audio pacing during resuscitative thoracotomy might improve the consistency of the output.

Even though the DCC device shows a significantly lower peak flow rate, it produces higher cardiac output compared to manual patch actuation. The device provides flow over a longer period, a feature that can be adjusted by the ratio of $T_{systolic}/T_{diastolic}$ (we used a ratio of 2:1).

The DCC device managed to keep peak pressure in the LV in the physiological range; however, it failed to do so for the RV. As mentioned in Chapter 10, controlling pressure in both ventricles is crucial for these devices. Other publications usually do not report on the pressure levels in the pulmonary circulation [86], or when they do, the additional pressure is similar for the systemic and pulmonary circulations [78]. We wanted to be transparent about our findings and limitations regarding the pulmonary circulation (related to the RV).

Something worth noting regarding the ejected volume from the ventricles: Figure 13.5 shows different behaviour for the LV and the RV for all modes of actuation. The figure shows a conservative system, excluding the implications of measurement errors. As the wall of the LV is thicker than that of the RV, Figure 13.5 shows a lower end-systolic volume for the LV. Usually, the RV has a lower stroke volume than the LV [120].

There are indications regarding the quality of the recorded data on the box plot diagrams (Figure 13.4). As the mean and median values are almost identical in terms of stroke volume, we can assume negligible skewness in the measured properties. This means that the cardiac output and peak pressures provide distributions similar to a normal distribution.

As shown in Figure 13.6, the device was able to operate in the critical 1.5 L/min cardiac output range to sustain the life of a patient. Not only could it produce sufficient cardiac output, but it also managed to achieve this with significantly lower peak pressures, and it produced much more consistent heart cycles than open-chest cardiac compression. This implies that using

the DCC device provides safer, more consistent but physiologically still sufficient support for patients than open-chest cardiac compression. Given its unique design, our DCC patch system can open new directions in direct mechanical ventricular actuation.

Regarding damage to the tissue and safety of operation, our preliminary experiments on porcine heart samples showed that the vacuum cups leave only a temporary mark on the surface of the heart. Figure 13.7 shows this temporary mark caused by the vacuum in the sealing rings. However, these are results from *ex vivo* experiments. To further assess the effect of the device on the heart, *in vivo* experiments are required.

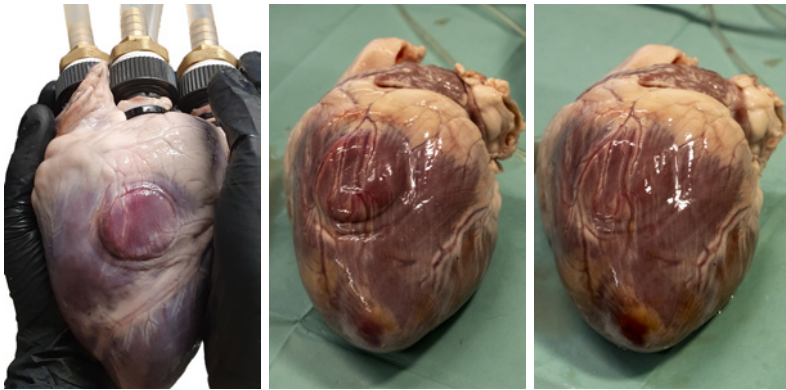


Figure 13.7: Tissue damage on the epicardium after using the device with the vacuum pump on porcine samples. **Left:** picture taken right after using the DCC device on a fresh porcine heart. **Centre:** picture taken right after using the DCC device on an older porcine heart. **Right:** picture taken 15 minutes after using the device on the older porcine heart.

Chapter 14

Conclusions and outlook

In this study, we investigated the capabilities of a novel direct cardiac compression (DCC) device to replace open-chest cardiac compression.

The device showcases a unique approach for VADs, focusing on resuscitative thoracotomy applications. Other devices exist to support patients via epicardial compression, but our design is unique as it uses mechanical actuation through a patch system. The approach of using these devices in emergency cases rather than as *bridge-to-transplant* might serve the further development of similar designs. The system of patches, actuation device, and vacuum pump shows promising results for a proof-of-concept study.

With a series of experiments, we were able to show promising initial results with our DCC device. Our design not only shows a new path for product development, but we also managed to collect data that is not usually published for such devices. Recording pressure and flow rates for both ventricles and creating reasonable statistics with the recorded data is of interest to the field, yet these results are rarely published.

The device not only provides a cardiac output within the range of 1.5 L/min, but it also achieves this with safer pressure levels and greater consistency compared to traditional manual cardiac compression. Our DCC device holds the potential to improve the currently low success rate associated with open-chest cardiac compression during resuscitative thoracotomy. By supporting essential life functions and relieving medical staff from the burden of manual massage, it allows them to focus on delivering care where it is most needed.

We have managed to improve some of the flaws of the previous designs by Kummer et al. [90]. These include the ballooning of the apex, the transmission cable cutting into the epicardium, and the rotation of the patches during compression. The current design is safe, but there is potential for improvements. As we have found during our experiments on cardiac massage, using our fingertips to compress the LV and at the same time softly compressing the RV with the balls of the hands leads to better flow and pressure distribution between the LV and the RV. Adapting the design of the patch system to mimic this kind of compression can lead to improvements

in both cardiac output and peak pressure values.

Further tests are required using porcine heart samples to test the performance of the device. As the device showed promising results on silicone hearts, we expect to be able to reproduce these on porcine heart samples. These samples require special preparation to be able to record flow and pressure data on our test bench. So far, we have not managed to acquire watertight porcine samples with functioning heart valves. With an improved system, following tests on *ex vivo* porcine samples, a cadaver experiment should provide new insights into the performance of the DCC device.

Biocompatibility and the compactness of the device were not the focus of the research but will be essential for *in vivo* tests and future patients.

Based on our experience with the MCL, it is highly recommended to build a completely new test bench for future experiments. The industrial check valves have opening pressures that are too high, forcing the user to use high preload values. The ball valve is also not a precise tool for adjusting the resistance of the system. The Audiowell sensors are sensitive to air bubbles, which caused many problems during the experiments as it not only introduced noise to the data but also created random offsets in the flow rate data. Tilting the platform of the test bench would help with driving the air bubbles to the compliance chambers. The compliance chambers are not precisely adjustable as they use balls to remove air from the chambers, which is not a precise adjustment.

Part V

Ex Vivo Testing

Chapter 15

Introduction

Based on the findings of Part IV, we decided to continue with testing the device on *ex vivo* models. Despite the lack of success in our in-house *ex vivo* tests, we were able to prepare for external tests and improve our design based on the findings from the *in vitro* results and the interaction with the tissue in the preliminary *ex vivo* tests.

Our *in vitro* test showed promising results, with the output of the DCC patch device being comparable to that of OCCC. However, many factors can influence the performance of *in vitro* studies, and they cannot provide the full picture for real-life applications. The tissue mechanics of the heart and the mechanical properties are different for an *ex vivo* heart and an *in vitro* model [121]. The tissue damage of the heart cannot be accurately assessed in an *in vitro* model, and an *ex vivo* heart can provide a better prediction of the clinical outcome. The attachment of the device to the heart is also different, as the wet epicardium is more slippery than the dry silicone surface of the *in vitro* model. We noticed backflow and low pressures in the ventricles during our in-house *ex vivo* tests, assuming that these results were due to valve failure. Also, we must consider the limitations of our MCL setup. Achieving similar results in a different MCL is a good indication of the actual performance of the device.

We tested our device in the *ex vivo* MCL of the *LifeTec Group*, Eindhoven, Netherlands. Their expertise has served Kummer et al. in their previous studies [90]. Their MCL is a well-established test bench for *ex vivo* studies and has been used in many studies [122]. The MCL is an open-loop system, which allows for the control of preload and afterload values for both the left and right heart circulations individually. Their setup includes cameras recording valve functions, allowing us to assess their functions during the tests.

We prepared for the test with the same patches, but a new manual actuator was designed for this study. We wanted to test the DCC device against OCCC and to compare the performance of the two methods. The goal was to show that the DCC patch device can match the output of a trained

surgeon (Prof. Stefanos Demertzis) in an *ex vivo* setting. The results for OCCC in the *in vitro* study [1] were performed by me (Kristóf Sárosi) based on a consultation with Prof. Demertzis. This is an improvement on the methodology, as the device should be compared to OCCC performed by a trained surgeon, not a researcher.

We wanted to compare the results with different hearts as, unlike in the *in vitro* study, different *ex vivo* heart models had different properties. We selected a porcine and an ovine heart for the study. The hearts were prepared by the *LifeTec Group*, and they were stored in a freezer until the day of the experiment. The porcine heart was 14% larger than the ovine heart.

We tested different amplitudes, rates of actuation, systolic fractions, and orientations of the patches. With the manual actuation, we could test the DCC device at different rates of actuation and systolic fractions. We could also assist the surgeon in performing OCCC at different rates of actuation. We also prepared temporary attachments to modify the design of the patches to test different hypotheses regarding patch design.

We found that the DCC patch device could match the performance of OCCC in an *ex vivo* setting, supporting our finding in the *in vitro* study. The DCC patch device was able to generate sufficient cardiac output and mean arterial pressure to sustain the life of a potential cardiac arrest patient. Our output was comparable to results in the literature; however, our tests were done *ex vivo*, while other studies are reported *in vivo*.

This part of the thesis builds on the previously published work on the *ex vivo* testing of the BiVAD system:

K. Sarosi et al. “Direct Cardiac Compression Device for Short-Term Mechanical Circulatory Support”. In: *ASAIO Journal* (June 2025). ISSN: 1058-2916. DOI: <https://doi.org/10.1097/MAT.0000000000002493>

Chapter 16

Methods

This proof-of-concept study aims to showcase the performance of the DCC patch device. We compare the performance of the DCC patch device to OCCC to study whether it could replace OCCC in clinical practice. Other use cases are discussed in Section 17.5. Our study was conducted in an *ex vivo* MCL on a single ovine cardiac arrest model in the *Cardiac BioSimulator* [122] at the facilities of the *LifeTec Group*, Eindhoven.

The *ex vivo* test bench is set up as an open MCL with the ability to control preload and afterload values for both the left and right heart circulations individually. Pressure values are recorded directly at the aorta, pulmonary artery (PA), left atrium (LA), and right atrium (RA). The ultrasonic flow meters were attached to Tygon tubes downstream of flexible silicone tube sections. The results presented in this study were achieved with a preload of 16 mmHg.

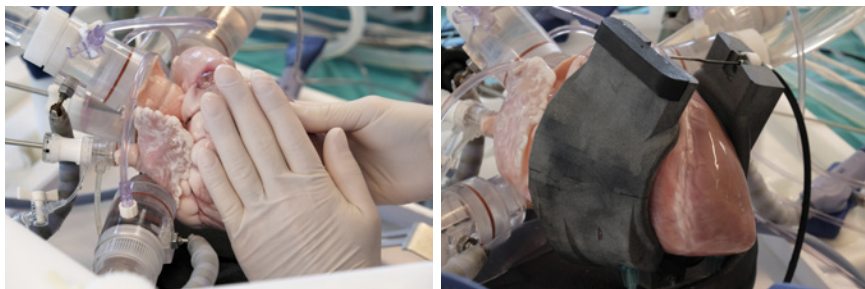


Figure 16.1: The two MCS modes compared. **Left:** OCCC performed on an ovine model in an *ex vivo* MCL. The surgeon uses a two-handed compression technique. **Right:** DCC patch device. The patches are in direct contact with the heart and are actuated through the Bowden cable.

16.1 DCC Device

The DCC patch device consists of two patches, shaped like the cups of a hand, a driveline, and a vacuum line. Figure 16.1 shows the DCC patch device on an *ex vivo* ovine model. The patches are in direct contact with the epicardium, compressing directly on the ventricles. The vacuum is required to keep the device attached to the surface of the heart through the suction cups in the centre of each patch. A magnetic locking mechanism between the two patches allows for quick deployment in emergency scenarios.

To actuate the DCC device, we had two options: motorised actuation or manual actuation. The manual actuator (see Figure 8.2) was built specifically for this study to serve as a backup in case the servo motors failed. During the experiments, the manual actuator proved to be able to deliver significantly higher output than the servo motors.

Further details regarding the DCC patches and the auxiliary parts can be found in Chapters 7 and 8.

16.2 Open-Chest Cardiac Compression

It must be noted that there is no standard technique for OCCC, it is unique to the surgeon. The cardiac surgeon was instructed to perform OCCC just as in the operating room (OR) (see Figure 16.1). With our experimental setup, the surgeon had more direct and real-time information and feedback on the performance of the OCCC than in an OR, as pressure, flow rate data, and videos of the heart valves were shown live on a screen. This might have already resulted in an improved performance of OCCC compared to performing OCCC in an emergency scenario in an OR.

16.3 Study Design

Using the DCC patch device enabled us to tune certain parameters, such as the orientation of the device, the initial distance between the patches, the rate of actuation, and the systolic fraction. Despite having a fully automated actuation system, we wanted to share the results with manual operation as well to demonstrate the full potential of the DCC device. The power output from motorised actuation was deemed insufficient compared to manual actuation. The operator of the manual actuator was instructed to push the lever mechanism at a given rate of actuation with a given speed

of contraction (slow/quick). An audio signal at the desired 100 bpm rate of actuation helped the operator of the DCC patch device to perform regular compression. It must be mentioned that for the proof-of-concept study, the surgeon had no audio signal because they do not have any in the OR in a real emergency scenario. However, we did pace the surgeon in parts of the study to see how it affected the performance of OCCC.

As for this proof-of-concept study, using the DCC device in an *ex vivo* setup for the first time, we adjusted the settings of the patches in a trial-and-error manner. This study does not include a detailed description of this process, only the most important findings and results using the settings as a conclusion of the initial trials. Also, we do not share the results on a porcine sample, as it was not successful for either OCCC or DCC support. In total, 38 tests were conducted in 2 hours using the same ovine model. These tests included OCCC studies, parameter studies using the DCC patch device, and studies with a modified DCC patch design (not included in this study). Along with pressure and flow rate measurements, we also recorded videos of the tricuspid and mitral valves and the experiments themselves. Pressure and flow rate data were recorded for approximately 30 seconds at 1000 Hz for each test. This generated enough data points to create statistically significant measures of performance.

16.4 Data Sampling and Analysis

Post-processing includes filtering the high-frequency noise in the signal, segmenting and averaging the data over cardiac cycles, and calculating the standard deviation in time-series, mean and peak pressure, SV, CO, and rate of actuation. We used Welch's t-test to compare the cases of OCCC and the DCC device, as the ANOVA's homoscedasticity assumption was not met. We used one-way ANOVA for certain sensor data in the parameter studies, as both the Shapiro-Wilk test and the Levene test indicated that we met the conditions for using one-way ANOVA. In any other case, we used Welch's t-test.

Chapter 17

Results and discussion

In this chapter, we present raw data from the pressure sensors and flow meters and statistically evaluated measures of performance such as average cardiac cycle, consistency of output, SV, CO, and peak and mean pressure in the aorta and the PA. The results are compared between OCCC and the DCC patch device with manual and motorised actuation. Using manual actuation, we find that the DCC patch device performs similarly to OCCC, providing the required CO and pressure levels for MCS while maintaining safe operation. The performance of the device and related results are presented in the following paragraphs.

17.1 Motorised Actuation

Initially, we tested the DCC device with motorised actuation with different control parameters, such as amplitude, frequency, systolic fraction, initial contraction, and position on the heart. We compared the performance of the device with that of OCCC and with manual actuation at similar actuation frequencies.

Figure 17.1 shows raw data using OCCC and the DCC patch device with the motorised actuator. Both OCCC and the DCC patches demonstrate significant pressure level differences between the atria and the outflow tracts. This indicates that the flow is driven by the compression, not by the preload of the system.

In the following, we share results from the motorised and manual actuation and compare them to OCCC. These results can be used to assess the performance of the motorised actuation and to point out any limitations regarding power output.

17.1.1 Expected Output

We segmented and averaged the collected data over every cardiac cycle to investigate the expected output and consistency. The results of the averaging

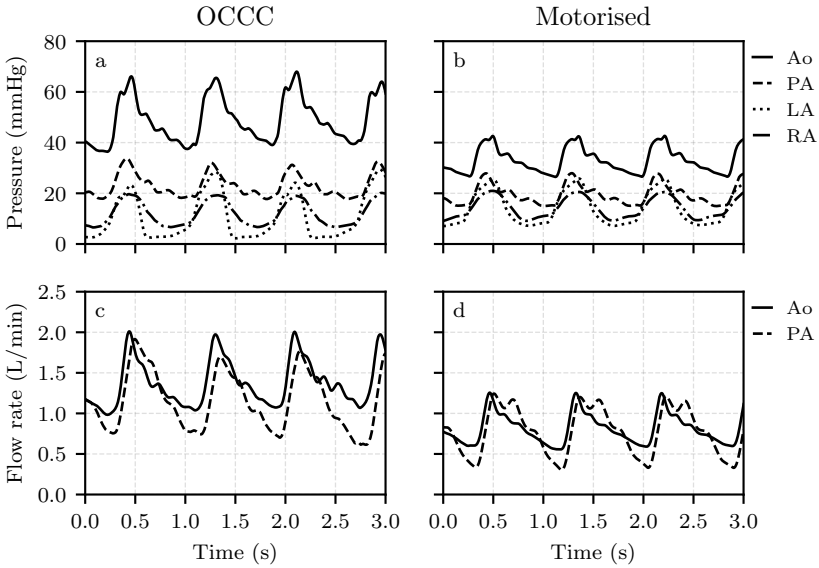


Figure 17.1: Raw data using OCCC (left) and the motorised DCC patch device (right) in the CA model. Nomenclature: Ao - Aorta, PA - Pulmonary artery, LA - Left atrium, RA - Right atrium.

are shown in Figure 17.2. The difference in the length of the cardiac cycles is due to the different rates of actuation (see Figure 17.3). The averaging shows that the consistency of the output with the device was significantly better than with OCCC, but this could be due to the audio pacing. Both OCCC and the manual actuation show higher pressure and flow rate values than the motorised actuation.

17.1.2 Stroke Volume and Rate of Actuation

Both OCCC and the DCC patch device with manual actuation were able to provide around 15 mL of SV despite the different rates of actuation. Figure 17.3 shows SV and rate of actuation data for the three modes of actuation. The stroke volume is significantly lower with the motorised actuation than with OCCC or manual actuation. SV is an important

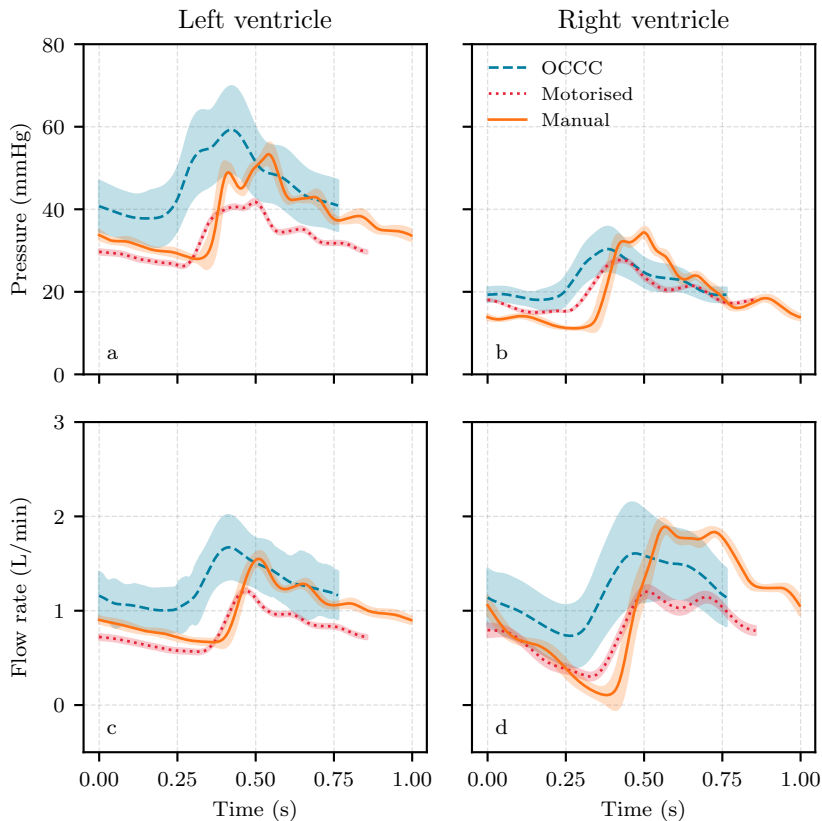


Figure 17.2: Mean flow rate and pressure in the left and the right ventricles. The lines show the mean over all the heart cycles, and the shaded bands show the standard deviation of the collected data over all heart cycles. Nomenclature: OCCC - open-chest cardiac compression, Motorised - using the device with the motorised actuator, Manual - using the device with the manual actuator.

measure as it can be bounded by the design of the patches and reduced by insufficient filling pressure or time.

The surgeon was not aided by an audio signal, and the mean rate of actuation of the OCCC was 80 bpm. The motorised actuator was set to

70 bpm as it was the maximum rate of actuation while keeping the 1:2 systolic fraction at the maximum amplitude. The manual actuation was paced at 60 bpm or 100 bpm. It is important to note that the motorised actuation could not reach 100 bpm with the maximum amplitude and 1:2 systolic fraction. As shown in Figure 17.3, the rate of actuation can vary greatly for OCCC. Long-term OCCC especially suffers from this phenomenon as the surgeon gets fatigued by the extensive pumping action. Given that the total cardiac output is given by the stroke volume and the rate of actuation, the performance can be better measured by CO.

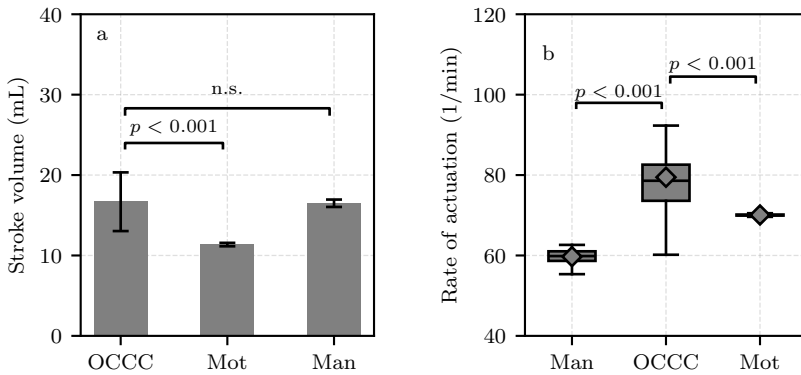


Figure 17.3: Stroke volume (left) and actuation frequency (right) for the three modes of actuation. The means are represented by the rhomboids and the horizontal line represents the median. Nomenclature: OCCC - open-chest cardiac compression, Motorised - using the device with the motorised actuator, Manual - using the device with the manual actuator.

17.1.3 Cardiac Output and Mean Pressures

Figure 17.4 depicts CO with mean arterial pressure (MAP) and mean pulmonary arterial pressure (MPAP) as the main indicators of performance. OCCC can provide significantly higher CO, MAP, and MPAP values than the DCC patch device. However, the manual actuation with a shorter systolic fraction shows better performance than the motorised actuation, despite being operated at a lower rate of actuation. Given that CO is the product of SV and the rate of actuation, the output from the DCC patch device is

expected to improve with actuating the device manually at a higher rate of actuation.

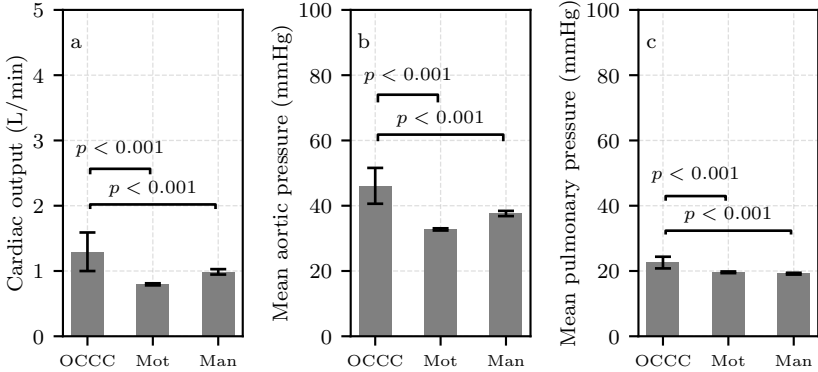


Figure 17.4: Cardiac output and mean pressures of OCCC and DCC support. Nomenclature: OCCC - open-chest cardiac compression, Motorised - using the device with the motorised actuator, Manual - using the device with the manual actuator.

With the technical limitations of the motorised actuation, we decided to continue the study with manual actuation. It is important to note that the results from the motorised actuation are already comparable to results in the literature [78, 86] and show the potential of the DCC patch device. However, to show the full potential of the device, we share the results from manual actuation. The manual actuation was not limited by delivering the expected amplitude and systolic fraction at a higher rate of actuation. The results from the manual actuation are presented in the following section.

17.2 Manual Actuation

We performed studies with manual actuation at 100 bpm with short systolic fractions. The manual actuation was paced by an audio signal, which allowed the operator to maintain a consistent rate of actuation. The results from the manual actuation are presented in the following subsections. We refer to the results with the manual actuation of the DCC device simply as the results with the DCC device as the manual actuation shows the full potential of the device.

17.2.1 Raw Data

Figure 17.5 shows raw data using OCCC and the DCC patch device. Both OCCC and the DCC patches demonstrate significant pressure level differences between the atria and the outflow tracts. This indicates that the flow is driven by the compression, not by the preload of the system. Comparing the results to Figure 17.1, the manual actuation shows significantly higher levels of pressure and flow rate in both ventricles.

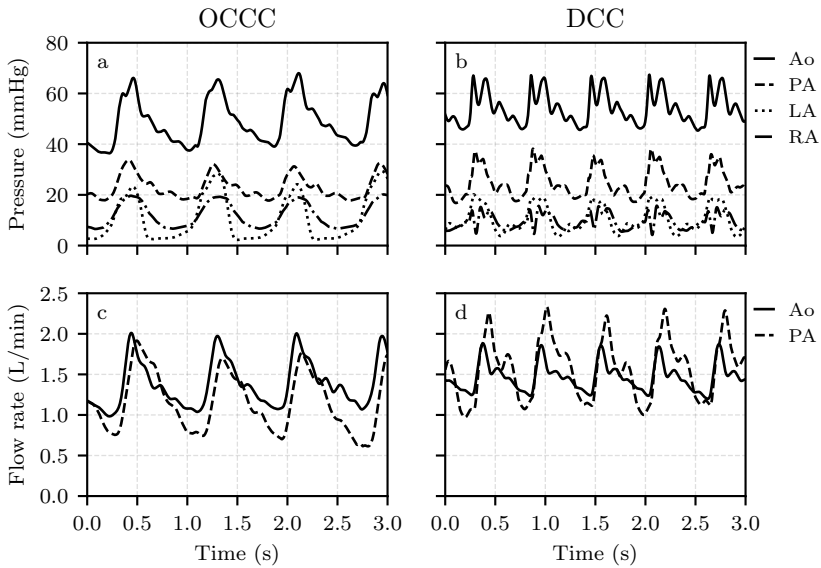


Figure 17.5: Raw data using OCCC (left) and the DCC patch device with the manual actuator (right) in the CA ovine model. Nomenclature: Ao - Aorta, PA - Pulmonary artery, LA - Left atrium, RA - Right atrium.

17.2.2 Expected Output

Figure 17.6 shows similar values in pressure and flow rate for both OCCC and the DCC device, but the consistencies of the two actuation modes are different. Our DCC patch device shows a smaller standard deviation, that

is, a more consistent mode of actuation. The length of the cardiac cycles is different for the two modes of actuation as the DCC patch device was operated at 100 bpm, while OCCC was performed at 80 bpm (see Figure 17.7).

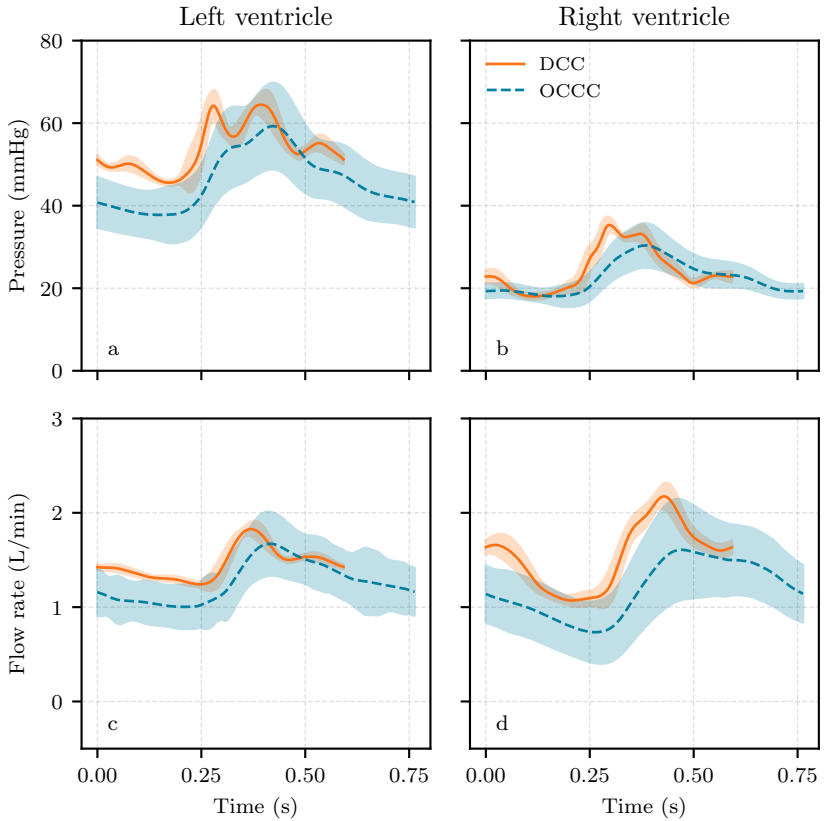


Figure 17.6: Mean flow rate and pressure in the left and the right ventricles. The lines show the mean over all the heart cycles, and the shaded bands show the standard deviation of the collected data over all heart cycles.

17.2.3 Stroke Volume and Rate of Actuation

Even with the higher rate of actuation, the DCC device shows a similar SV to the results at 60 bpm (see Figure 17.3), around 15 mL. Figure 17.7 shows the SV and rate of actuation for both OCCC and the DCC patch device. The DCC device was actuated at a higher rate of actuation than OCCC, and given that their SV is significantly different but still in the same range, we expect the device to perform similarly to OCCC in terms of CO.

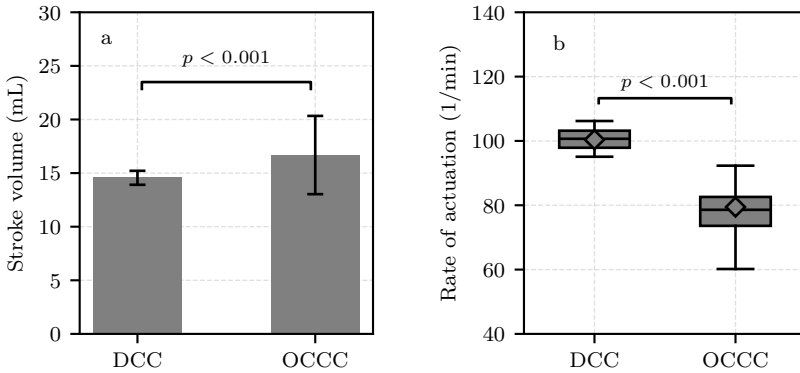


Figure 17.7: Stroke volume (left) and actuation frequency (right) for the two modes of actuation. The means are represented by the rhomboids and the horizontal line represents the median.

17.2.4 Cardiac Output and Mean Pressures

Figure 17.8 shows the output of the DCC device and OCCC. The device shows significantly higher CO, MAP, and MPAP values. Both modes of actuation deliver around 1.5 L/min CO on average. In terms of mean pressure, the DCC patch device delivers significantly higher pressure levels compared to OCCC but still in the safe region for hypertension [123, 124, 28, 125] (references show the following limits: MPAP > 25 mmHg, Aortic pressure > 140/90 mmHg).

The results show that the performance of the DCC patch device is comparable to OCCC. The DCC patch device can provide the required CO and pressure levels for MCS while maintaining safe operation.

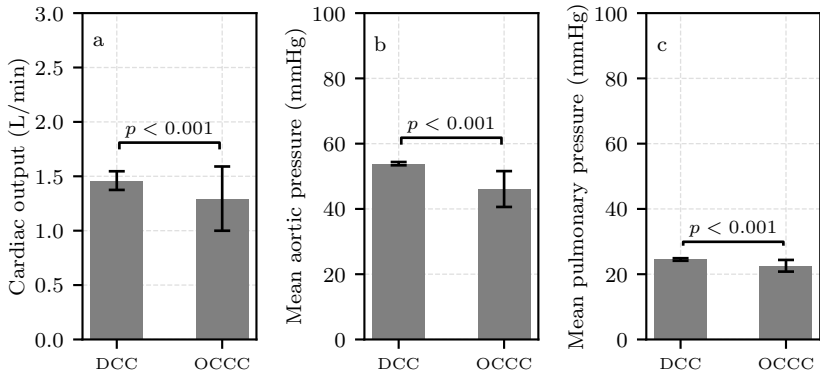


Figure 17.8: Cardiac output and mean pressures of OCCC and DCC support.

17.3 Parameter Study

To obtain the results shown in this chapter, we performed preliminary tests on different actuation frequencies, systole lengths, and patch orientations. We share the most important findings regarding the rate of actuation and using different systolic fractions. Figure 17.9 shows that a higher rate of actuation leads to significantly higher CO, MAP, and MPAP values. Regarding the systolic fraction, CO and MAP are significantly higher with a smaller systolic fraction, while MPAP is less affected.

17.4 Pacing the Surgeon

We intentionally rejected pacing the surgeon as it is not representative of real-world scenarios. However, we did perform studies with pacing the surgeon on different MCL settings, sharing preliminary results below. Figure 17.10 shows results with and without pacing OCCC. Both were performed on the ovine heart at 16 mmHg preload, but the paced OCCC results were taken at lower afterload values. The results show a significant increase in CO, MAP, and MPAP with pacing. However, the level of MPAP indicates the risk of pulmonary hypertension. The study has to be repeated, focusing on the effects of pacing and rate of actuation separately.

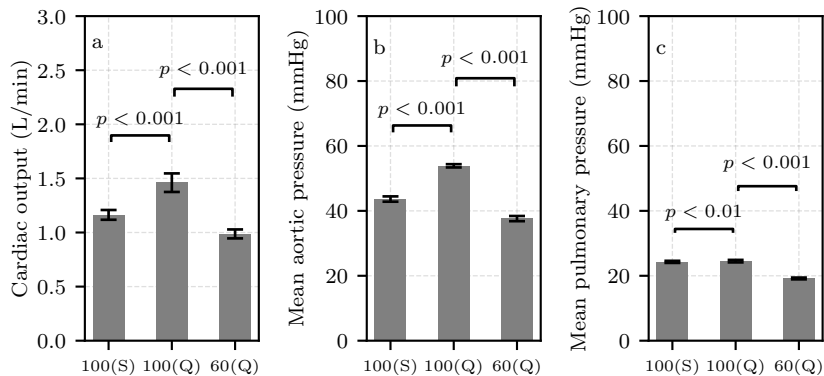


Figure 17.9: Comparing results with different rate of actuation and systolic fraction using the DCC device. The numbers (100 / 60) stand for the rate of actuation in bpm, and the letters (S / Q) stand for the speed of systolic contraction (slow / quick).

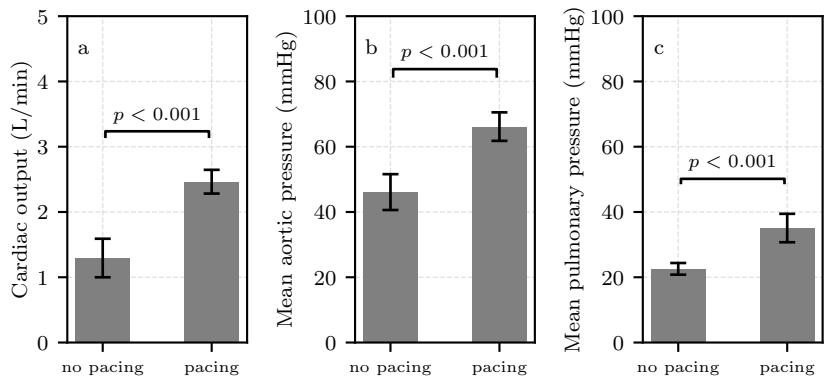


Figure 17.10: The effect of using audio pacing at 100 bpm during OCCC.

17.5 Discussion

The DCC patch device was designed to suit the needs of emergency scenarios in open-chest surgeries or following thoracotomy. Valve regurgitation was monitored by cameras capturing the mitral and tricuspid valves. Valve functions were maintained during device operation. Regarding hypertension, we evaluated mean and peak pressures in the aorta and the pulmonary artery. Using the manual actuation of the device, we did not eliminate the human factor and thus the variability of the output. The operator of the device was not a medical professional, unlike the cardiac surgeon performing OCCC. Using simple audio pacing, we managed to decrease the variability in the output of the device. Issues concerning long-term use, such as increased filling pressure and tissue-device adhesion, are not discussed in this study.

Assessing hypertension is crucial for both the systemic and pulmonary circulations. The pulmonary circulation is especially prone and sensitive to hypertension as the wall of the RV is much thinner than that of the LV. According to the literature, peak and mean pressure in both the LV and the RV are good indicators of hypertension [123, 124, 126]. The pressure values are below the hypertension levels for both the aorta and the pulmonary artery.

Sensitivity to the position of the device could be of interest to surgeons. Should they be concerned about positioning the device perfectly in an emergency? As the two patches of the DCC device are different in size and curvature, this information could be crucial for surgeons. There was indeed a significant difference regarding the different positions of the DCC device on the heart. The intended orientation has a lower output on the pulmonary side than with different orientations. To avoid pulmonary hypertension, we used the device in the intended position. With this, we sacrificed performance for safety. Hypertension is an important factor in assessing the safety of the device. As shown in Figure 17.8, the aortic and PA pressure levels are in the safe range using the DCC patch device.

Reaching high pressures on the systemic side is crucial for organ perfusion. Some studies show improved performance in organ perfusion with pulsatile DCC and MCS devices over continuous ones [23, 45], but the broader literature shows mixed results. Our device can create different pressure increments for the LV and the RV, which is crucial for organ perfusion while avoiding RV hypertension. Organ perfusion is crucial for the long-term survival of patients.

Another important aspect of DCC device feasibility is the interaction of

the tissue with the device. We used our DCC patch device on the ovine model for 2 hours (including pauses). In this time frame, the DCC patch device caused visible damage to the epicardium (see Figure 17.11). We have to add that this study was a proof-of-concept study only for the performance of the device, and not for biocompatibility and long-term use. The surface of the device was not treated after it was manufactured with 3D printing, leaving a rough contact surface for the epicardium.

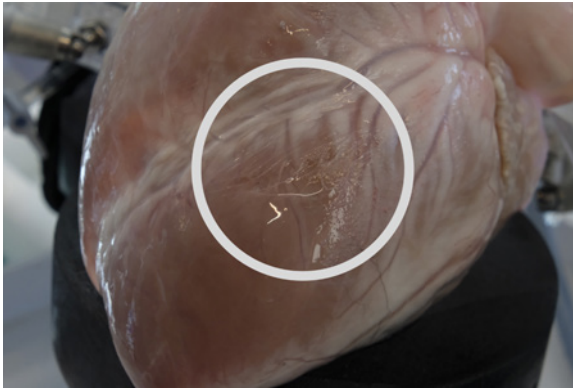


Figure 17.11: Damage to the epicardium after 2 hours of DCC support.

As we pointed out in Chapter 15, most of the recent DCC devices aimed to replace VADs. VAD treatments have improved greatly in the past decades with third-generation devices, reaching more than 80% survival in the first year [37]. Accessibility is still an issue with VADs, requiring improvement, but we see the application of our device in different treatments. Anstadt et al. [23] made a comprehensive review of DCC devices (for them: direct mechanical ventricular actuation (DMVA) devices) and applications. Anstadt's device was a breakthrough in the field of DCC, achieving astonishing results in animal studies, but their human studies did not succeed. We aim to concentrate on different applications, with a focus on short-term MCS. This study was meant to show possibilities for resuscitation in CA, but prolonged support in fibrillation, use in *in vivo* organ preservation, and support in hypothermia could also be explored. According to the publications of the *CorInnova* device, DCC devices might be able to replace ECMO devices in certain scenarios [78]. We aim to use our device as short-term support in open-chest surgeries or following thoracotomy.

Reviewing the studies published for the *CorInnova* DCC device, we can compare our results with one of the most established DCC projects of recent times. In their latest publication on a caprine CA model for paediatric patients, they generated 0.4 L/min CO and a 6 mmHg increase in both MAP and MPAP [78]. Their study on the ovine HF model shows similar results [86]. For comparison, our DCC patch device delivered 1.5 L/min CO, a 36 mmHg increase in MAP, and a 7 mmHg increase in MPAP. We must mention that the *CorInnova* studies were conducted *in vivo*, while our study uses a single *ex vivo* ovine model. However, we believe that the results from the *CorInnova* publications serve as reasonable references for our study.

One of the biggest flaws of previous DCC devices is that, unlike in physiological conditions or during OCCC, the generated pressure increment for the LV and the RV are similar. Initially, in our measurements, the pressure measured at all four measurement points is equal. Any pressure difference between measurement points at the same time instance is created by OCCC or the device. The results in Figures 17.6 and 17.8 show the pressure increment differences created by the DCC device.

This work presents an *ex vivo* study on a single ovine CA model. We used an open MCL provided by the *LifeTec Group*. This MCL allows users to independently control preload and afterload for the LV and the RV. To improve on this study, a closed MCL, a cadaver, and later an *in vivo* model should be used. Translating the results from an *ex vivo* model to an *in vivo* one is not straightforward but could prove to be a valuable reference. Unlike the *CorInnova* device, our DCC patch device cannot be used in a minimally invasive procedure. The DCC patch device requires thoracic surgery or an open chest for deployment. Our design is currently not biocompatible, as this and our previous study [1] assess only the output created by the DCC patches.

The ovine heart was 430 g at the beginning of the study, and it was 444 g at the end of the study. For reference, the average male heart weighs 331 g, and the average female heart weighs 245 g [127].

Considering the results in this chapter, the DCC patch device performed similarly to OCCC carried out by a trained surgeon. Figures 17.5 and 17.6 both show similar flow rates and pressures for the DCC device to OCCC. Regarding the dampened flow rate values, both the tricuspid and mitral valves are closed after systole. As there are no signs of regurgitation, the only contributing factor to this effect is the compliant tubing upstream of the flow meters. This limitation is given by the *Cardiac BioSimulator's* design.

This proof-of-concept study shows the potential of providing life-sustaining support with the DCC patch device for patients in emergencies. The device should be placed on the heart by a surgeon, but it could be operated electronically or manually by other members of the medical team. Replacing OCCC would relieve surgeons from the burden of supporting the circulation of the patient manually, allowing them to carry out other tasks in the OR, stabilising the patient for *bridge-to-decision*. The DCC patch device could also be used bedside in intensive care units for emergencies where OR readmission time could put the patient's life at risk.

Chapter 18

Conclusions and outlook

In this study, we present a novel, fully mechanical Direct Cardiac Compression (DCC) patch device developed at ETH Zürich. The device consists of two patches, similar to clamshells or human hands, directly attached to the surface of the heart. The DCC device was designed for emergency scenarios where quick deployment is crucial. Our aim is to provide short-term circulatory support, avoid contact with the bloodstream, reduce costs, and relieve surgeons from manual cardiac compression. The results show that our device can outperform a trained surgeon by generating matching flows and significantly higher MAP and MPAP, sufficient to sustain the life of a cardiac arrest patient while further treatment options are considered.

The natural continuation of this work is to test the DCC patch device in animal studies. Animal studies could provide more information on the performance of the DCC device, as it should be tested in a natural circulation. The compliance of the heart also changes post-mortem; therefore, testing the device on a recently deceased or live animal should be closer to the expected outcome in practice. Testing the device in an animal is also important for assessing tissue damage, especially to the coronary arteries, as their perfusion cannot be properly assessed in cadaver or *in vitro* tests. The next step is to improve the design based on the feedback from the *ex vivo* tests and to prepare for animal studies. Previous porcine cadaver studies can be found in the doctoral thesis of Thomas Kummer [90].

We drew many conclusions from our study. The DCC patch device is able to generate sufficient flow and pressure values to sustain the life of a cardiac arrest patient. We used the manual actuation mode of the DCC device, as it proved to be better than the motorised actuation. This means that the motorised actuation system needs to be improved beyond the work of this thesis before being used in animal studies. In the meantime, the manual actuation mode can be used for testing the device. Based on the feedback of the operator (Prof. Patrick Jenny), we need improvements to the manual actuator before conducting animal studies. The actuation should be performed by foot rather than by hand, as hitting the lever to achieve

small systolic fractions was painful and cannot be carried out for hours.

The patches damaged the epicardium, and we assume it was because of the rough surface and the high curvature of the RV patch. Reducing the curvature of the patches could help mitigate this issue. However, the size and curvature of the patches play an important role in controlling the pressure increment in the ventricles. The new design should find a balance between the two approaches.

We found that the vacuum was not needed to keep the device on the surface of the heart. Another use case for the vacuum is to lower the filling pressure by pulling the ventricles at diastole. This aspect should be further investigated before removing the vacuum line from the design. Also, with a smoother surface, the vacuum line might be essential in keeping the device in place.

Having tested the DCC patches on two hearts, with the porcine heart being 490 g compared to the 430 g of the ovine heart, we ran into issues as the porcine heart was too large for our patches. The tissue damage was more prominent, and the output from both the device and the surgeon was lower than in the ovine model. To quickly adjust to different sizes of hearts, the current size adjustment method is not sufficient, as mingling with a small screw like an M3 nut in the chest is not practical, especially not in an emergency while wearing gloves. The new design considerations were already developed into a concept by Annika Hämmer during her semester project (see Figure 9.1) [128].

When tilting the patches in the chest cavity, the movement of the Bowden cable could be restricted, resulting in normal forces on the connecting points of the patches. This could lead to the opening of the magnetic lock or the ejection of the other anchor point of the cable on the LV patch. This design feature should also be reconsidered in the new design.

Part VI

Device Optimization in Lumped Parameter Models

Chapter 19

Introduction

Developing and testing devices is an expensive and time-consuming process. In many engineering applications, computers play an essential role not only in designing but also in testing devices in virtual environments. In the medical field, however, this is more complex. While designing, simulating mechanical properties, and flows in the device itself are all possible with current technology, simulating the interaction of the device with the human body is still a challenge. The broad range of scales in the vasculature and the biological, chemical, and physical complexity of the circulatory system require highly complex models that are unfeasible to simulate. Most current simulation frameworks use less complex models, such as lumped parameter models (LPMs) to simulate macro models, or highly complex models, such as finite element method (FEM) models, to simulate only a part of the entire system (e.g. the heart or the aorta). Using the latter approach, unfortunately, does not allow for practical applications in device development, as these models are computationally expensive. LPMs, on the other hand, are computationally efficient but currently cannot account for the complexity of the interaction of the heart and a mechanical circulatory support (MCS) device.

We faced the same dilemma during our project. We inherited a complex FEM model of the heart from the work of Kummer et al. [88], which was coupled with an LPM network model. This framework can simulate the interaction of a direct cardiac compression (DCC) device with the heart. However, the computational cost of this model is too high to be used in the process of device development. Therefore, we had a motivation to merge the capabilities of the FEM model and the LPM. Kummer's work used a coupling between the FEM model and the LPM network based on the volume of the ventricles. The relation between volume and pressure is described by the compliance or the elastance of the heart. Simulating the behaviour of the heart in the LPM with an elastance model is a common approach in the literature. On the other hand, these models do not account for MCS devices.

Another motivation for this part of the project was our experience in the *in vitro* and *ex vivo* experiments. The motorised actuator provides precise control of amplitude, frequency, and systolic fraction of the compression. Also, the position of the DCC device on the heart could affect the outcome of the experiments. Testing all the combinations of control parameters of the actuator and the design parameters of the patches is not feasible in the experimental environment. Therefore, we wanted to create a framework that allows us to test these parameters in an *in silico* environment and to use the results to optimise the design of the DCC device and the set of parameters to be tested during the experiments.

The simple construction of the DCC device leaves only a few parameters to be optimised regarding the BiVAD system. The vacuum pump and the actuator can be considered outside the scope of this optimisation as they can only be limited by power. The patches, however, are in direct contact with the heart, and their design features can affect the outcomes. This could be the size and shape of the contact surface, the flexibility of the material, and the placement on the heart.

Bridging the gap between simulations and experiments is essential in the rapid development of MCS devices [91]. Experiments could be improved by pre-testing sets of parameters regarding the actuation and the placement of the device. Given that we were able to run only 38 tests in our *ex vivo* experiment, reported in Chapter 17, a tool to pre-select the best possible actuation frequency, amplitude, systolic fraction, and placement would have been extremely valuable. This is even more important for animal and clinical tests, as their complexity and costs are higher.

Higher-order models usually work with a combination of different scales, called multi-scale methods. Similar to the work of Kummer et al. [88], 3D FEM simulations are coupled with a 0D network model to simulate the behaviour of the rest of the vasculature, while the FEM code handles solely the deformation of the ventricles. Other approaches exist with breaking down the circulation into different scales. The largest scales, such as the large vessels, are handled by 3D, the arterial tree by 1D CFD simulations, and the rest of the circulation by 0D LPMs [129].

Most of these models only handle pre-set boundary conditions and cannot handle MCS devices. Their purpose is to build a stronger understanding of the vasculature, and they do not aim to support device development. Other works do concentrate on MCS and their effects, including the use of LVADs [130] and DCC devices [88]. These, however, are too complex to be used in device development.

Most of the optimisation tools and *in silico* environments are built on LPMs. They are efficient and easy to handle. Some even use the knowledge acquired from FEM simulations and include it in the LPM [131]. But most of them use only a 0D network model, using an electric analogue network for simulating the circulation [132, 133]. Similar to most higher-order models, these are tools to simulate only the behaviour of the circulation, not the interaction with MCS devices.

However, in recent years, the demand for tools to include MCS devices and still be efficient enough to help the development of new devices has emerged. *In silico* environments were developed to build a better understanding of rotary blood pumps [134], DCC devices [105], and ECMO [135].

As most of these devices are used in patients with pathologies or dysfunctions, their accurate simulation is also essential in predicting outcomes. These conditions can include heart failure [105], valve stenosis [136] and regurgitation [101], and atrial fibrillation [106]. Only simulating the combination of pathologies and MCS can help in reducing the number of experiments needed to develop a device and speeding up the development and approval process.

Recent studies show the need for such a model in the development of DCC devices [105]. DCC devices have to meet a complex set of requirements to be successful in clinical applications, and our model aims to test these requirements in an *in silico* environment. This model allows us to tune our circulation parameters according to patients' conditions for personalised treatment, test different device settings, test the functionality with different pathologies, and design new DCC devices based only on very little a priori knowledge. Using a simple computational framework, designers can test their device parameters such as the amplitude, frequency, systolic fraction, timing, placement, and design parameters in an *in silico* setting. With our model, we can speed up development and potentially reduce the number of animal experiments required to introduce DCC devices to clinical practice. Our goal is to break the century-long deadlock on DCC device applications and to build a better understanding of the compliant nature of the heart.

Chapter 20

Methods

Current methods fail to describe the interaction of the heart and DCC devices without computationally expensive higher-order models, and standard compliance models cannot account for ventricular interdependence. To replace the current standard of multi-scale coupled models, LPMs require new methods to replace higher-order models, such as coupling the LPM circulation with an FEM model. In this section, we discuss our extension to these models, from the idea through derivations to defining model parameters.

20.1 Network Model

LPMs are 0D network models that simulate the pressure and flow rates in both the systemic and pulmonary circulations. The basis for our LPM was taken from the work of Kummer et al. [88]. We wanted to create an LPM that could produce physiological results without the need to couple it with the FEM digital twin model from Kummer's framework. Current similar closed-loop network models are coupled with either a single ventricle [137] or both ventricles [88]. The network consists of resistances and capacitors for most of the nodes, four-element Windkessel elements with inductance for the aorta and the pulmonary artery, and diodes representing heart valves. Figure 20.1 shows the LPM network model. The coupling between the FEM model and the LPM network was replaced by a compliance model, similar to literature standards [131]. The DCC actuation is described by $\omega(t)$, a time-varying input function of device displacement, and e , a coefficient describing heart-device interaction. The list of elements and their parameter values can be found in Appendix B.2.

The following relationship must be satisfied at every node per Kirchhoff's law in an electrical analogue network:

$$\sum_j q_{i-j} = 0, \quad (20.1)$$

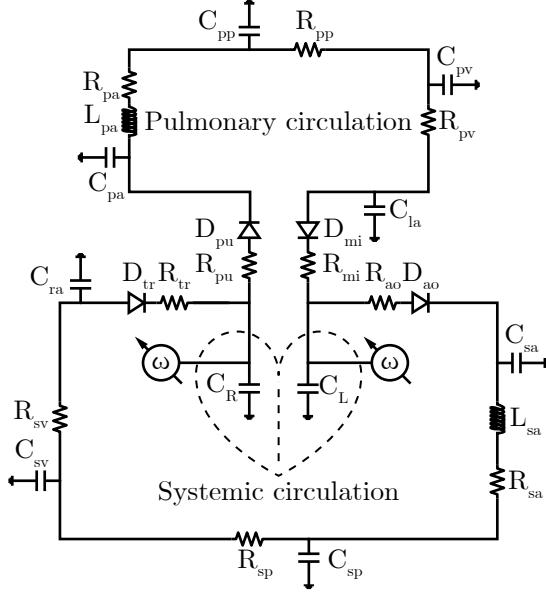


Figure 20.1: The lumped parameter model of our framework. We replaced Kummer’s coupling with a compliance model and the input from the DCC device, $\omega(t)$. The electrical analogue consists of resistance, capacitance, and inductance elements. The nodes are defined at the capacitance elements.

where q_{i-j} is the net flow rate from the i -th node to the j -th node.

The different elements between the nodes describe different relations between pressure and flow rate. The resistances, R_i , describe the viscous losses in the system and define the relation between pressure and flow rate as follows:

$$q_{i-j} = \frac{p_i - p_j}{R}, \quad (20.2)$$

where p_i and p_j are the pressures at the i -th and j -th nodes, respectively. Resistance R is the resistor between the two nodes.

The capacitances, C_i , define the following relation:

$$q_{i-\infty} = C \frac{dp_i}{dt}, \quad (20.3)$$

where $q_{i-\infty}$ is the flow rate to the capacitor of the node, accumulating the volume. C is the capacitance of the capacitor, describing the compliance, the attribute to accumulate volume in a node. And p_i is the pressure at the node.

The inductances, L_i , model the inertia of blood in bigger vessels such as the aorta and the pulmonary artery. It describes the rate of change of flow rate as follows:

$$\frac{dq_{i-j}}{dt} = \frac{p_i - p_j}{L}, \quad (20.4)$$

where q_{i-j} is the flow rate through the inductor between the i -th and j -th nodes, and p_i and p_j are the pressures at the i -th and j -th nodes, respectively. L is the inductance of the inductor.

The inductance elements are combined with resistors between two nodes; therefore, the following relationship holds:

$$\frac{dq_{i-j}}{dt} = \frac{p_i - p_j}{L} - \frac{R}{L} q_{i-j}. \quad (20.5)$$

Equation (20.5) is integrated separately in the solver, leading to a decoupled system of ODEs for the inductance elements. The updated inductive flows are then used in the next time step in the vector $\mathbf{Q}_L$.

The diodes, D_i , describe the function of the valves, defining the flow rate as follows:

$$q_{i-j} = D \frac{p_i - p_j}{R}, \quad (20.6)$$

where q_{i-j} is the flow rate through the diode between the i -th and j -th nodes, and p_i and p_j are the pressures at the i -th and j -th nodes, respectively. D is the state of the diode, representing the heart valves. If the diode is open, $D = 1$, and $D = 0$ when it is closed. Different models use different characteristics for the diodes (see Section 20.3).

Using these relations, the flow rate can be calculated at every node. We still need a model for the heart and the DCC device to simulate the interaction of the two without using the FEM model. Section 20.2 introduces the heart model defining the volume in the ventricles. Using this and the network model, we can calculate the flow rate and the pressure at every node. The core of the model is a system of ODEs that describes the rate of change of

pressures in various nodes. The system can be expressed in matrix form as:

$$\mathbf{C} \frac{d\mathbf{p}}{dt} = \left(\mathbf{A} - \frac{d\mathbf{C}}{dt} \right) \mathbf{p} + \mathbf{Q}_L + \mathbf{S}, \quad (20.7)$$

where:

- $\mathbf{p}$ is the vector of pressures.
- $\mathbf{C}$ is the matrix of compliances.
- $\mathbf{A}$ is the matrix of resistances and diodes.
- $\mathbf{Q}_L$ is the vector of flows from inductive elements.
- $\mathbf{S}$ is the source term of DCC device actuation.

You can find the model for the entire circulation in Appendix B.1.

20.2 Heart Model

If we can define a model describing the volume in the ventricles, we can define the rate of change and therefore the flow rates and the pressure at every node. In this model, we need to integrate the DCC device actuation to simulate the interaction of the heart and the device.

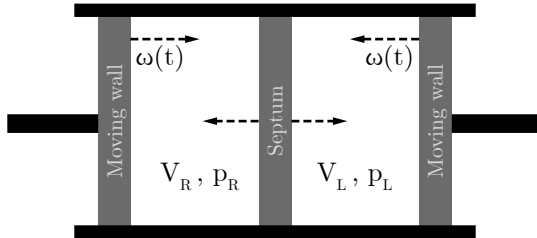


Figure 20.2: Our model was based on the analogue of a piston with three piston heads. The chambers in the piston represent the ventricles, the moving walls the tissue wall deformed by device compression, and the middle piston head the interventricular septum. Note that the medical convention for the sides is flipped as it is defined from the perspective of the patient.

To replace the FEM coupling with a model, we used a simple mechanical analogue. This analogue is depicted in Figure 20.2. We assume that the heart is a bounded volume with two pressurised chambers, divided by a moving wall. Similar to a piston with two moving piston heads moving towards each other and a freely moving piston head dividing the piston into two chambers, we have the LV, RV, and the interventricular septum as the freely moving wall, moving according to the pressure distribution between the two ventricles. Based on this analogue, we can make the following important assumption:

$$V_i = f(p_L, p_R, \omega(t)). \quad (20.8)$$

We assume that the volume in a ventricle depends on the pressure in both ventricles, as depending on the pressure distribution between the two chambers, the freely moving piston head can move and change the volume in the ventricles; and the external compression, $\omega(t)$, as with a moving piston head, the volume changes in the chambers as the piston heads move.

Adding to this assumption, based on models in the literature, we can add that in the case of a beating (self-actuating) heart, the time-varying compliance can also change the volume in a ventricle [131]. With this added assumption, we have the final model:

$$V_i = f(p_L, p_R, C_{iL}(t), C_{iR}(t), \omega(t)). \quad (20.9)$$

Using these assumptions, we could integrate the actuation from the DCC device and the self-actuation of the heart. This should enable us to run DCC simulations for different pathologies, such as CA and HF.

As the solver simulates the rate of change of our solution vector $\mathbf{u}$, we take the total time derivative of Equation (20.9) to integrate it into our model. Using the chain rule, we can define the rate of change of volume in a ventricle as follows:

$$\frac{dV_i}{dt} = \frac{\partial V_i}{\partial p_L} \frac{dp_L}{dt} + \frac{\partial V_i}{\partial p_R} \frac{dp_R}{dt} + \frac{\partial V_i}{\partial C_{iL}} \frac{dC_{iL}}{dt} + \frac{\partial V_i}{\partial C_{iR}} \frac{dC_{iR}}{dt} + \frac{\partial V_i}{\partial \omega} \frac{d\omega}{dt}. \quad (20.10)$$

The rate of change of V , p , C , and ω are either part of the solution vector $\mathbf{u}$ or direct input parameters of the system. The partial derivatives, on the other hand, are unknown coefficients that need to be defined.

20.2.1 Coefficients

Taking the general definition of compliance, we know the following relationship:

$$c = \frac{\partial V}{\partial p}. \quad (20.11)$$

The change in volume with respect to the change in pressure in an elastic vessel is the compliance. Using this knowledge, we can state that the first two unknown coefficients, $\frac{\partial V_i}{\partial p_L}$ and $\frac{\partial V_i}{\partial p_R}$, are compliances, the first being the compliance of the ventricle and the second one being the co-compliance of that same ventricle [131].

The variable $\frac{dC_i}{dt}$ is specific to the changes in compliance, only valid for active heart (i.e. heart failure) simulations. The associated coefficients $\frac{\partial V_i}{\partial C_i}$ are model-specific, depending on the used time-varying elastance model.

As we based our modelling assumptions regarding self-actuation on the literature, it is important to transfer the assumptions in the literature to our model regarding this coefficient. We take assumptions from the well-established time-varying compliance model of Suga et al. and build on their concept. Taking their model, we have the following relation between pressure, elastance, and volume:

$$p(t) = E(t)(V(t) - V_0). \quad (20.12)$$

Rearranging it to a relationship for $V(t)$, we have:

$$V(t) = \frac{p(t)}{E(t)} + V_0. \quad (20.13)$$

Taking the widely used assumption that V_0 is constant and using the definition that compliance is the inverse of elastance, we obtain the following relation after taking the partial derivative with respect to the compliance and using the common relation between elastance and compliance:

$$E = \frac{1}{C} \quad (20.14)$$

and thus, using Equation (20.13):

$$\frac{\partial V}{\partial C} = p. \quad (20.15)$$

The only missing coefficient, $\frac{\partial V_i}{\partial \omega}$, describes the heart-device interaction as it shows the ventricular volume's sensitivity to DCC support. We use the notation e as a short-hand for this partial derivative:

$$e_i = \frac{\partial V_i}{\partial \omega}. \quad (20.16)$$

The coefficients in Equations (20.11) and (20.16) can be device-specific, as we acquire them from experiments or simulations using the DCC device. Therefore, we assume that the published coefficient values are specific to our DCC device presented in Part III. You can find more information about the values of the unknown parameters in Appendix B.2.

With this, using Equation (20.10), we get the following system for the two ventricles:

$$\begin{bmatrix} \dot{V}_L \\ \dot{V}_R \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial V_L}{\partial p_L} & \frac{\partial V_L}{\partial p_R} \\ \frac{\partial V_R}{\partial p_L} & \frac{\partial V_R}{\partial p_R} \end{bmatrix}}_{\mathbf{C}} \begin{bmatrix} \dot{p}_L \\ \dot{p}_R \end{bmatrix} + \underbrace{\begin{bmatrix} \dot{C}_{LL} & \dot{C}_{LR} \\ \dot{C}_{RL} & \dot{C}_{RR} \end{bmatrix}}_{\mathbf{C}} \begin{bmatrix} p_L \\ p_R \end{bmatrix} + \underbrace{\begin{bmatrix} \frac{\partial V_L}{\partial \omega} \\ \frac{\partial V_R}{\partial \omega} \end{bmatrix}}_{\mathbf{e}} \dot{\omega}. \quad (20.17)$$

To see how our model aligns with examples in the literature, we compare it to the work of Suga et al. [96], assuming V_0 is constant.

We can redefine Suga's equation for compliance using the elastance-compliance relation in Equation (20.15) as follows:

$$V(t) = C(t)p(t) + V_0. \quad (20.18)$$

If we take the time derivative of the system, we find that our model extends theirs with only the added DCC support. The rate of change of volume in a ventricle using Suga's model is the following:

$$\dot{V} = C\dot{p} + \dot{C}p. \quad (20.19)$$

From Equation (20.17) follows the general description of our model, well aligned with Suga's Equation (20.19):

$$\dot{V} = \mathbf{C}\dot{p} + \dot{\mathbf{C}}p + \mathbf{e}\dot{\omega}. \quad (20.20)$$

With this, we want to show that our model is, in fact, not that different from the most established model in the field, but it extends it while appreciating the original idea. While Suga et al. came to this model based on

empirical observations, we ended up at a similar conclusion using a simple model assumption and mathematical derivation. Just as they described ESPVR and the linear relation to approximate $E(t)$, we also need to close our equations by defining the missing coefficients $\mathbf{C}$ and e . There are two ways to define these coefficients: either through measurements using the DCC device or using an FEM solver with DCC support features. As studying the compliance of the heart systematically in a simulation environment, focusing on the interaction of the LV and the RV, is a novel approach and a topic on its own, it deserves a separate part in this thesis. Therefore, we will not go into details about the parameter definitions in this chapter, but we will describe the methods in Chapter 24.

The results in this part use the knowledge built in Chapter 25 but at the same time rely on standard models in the literature, such as the time-varying elastance model [99].

Active Heart Model

To simulate the beating heart, in this part, we use the literature-standard time-varying elastance model. This model describes the elastance as a time-varying function, $E(t)$, and we can use this model in our model Equation (20.20). We take the two-Hill function as described by Mynard et al. [99]:

$$E(t) = k \left(\frac{g_1}{1 + g_1} \right) \left(\frac{1}{1 + g_2} \right) + E_{min}, \quad (20.21)$$

where

$$g_1 = \left(\frac{t}{\tau_1} \right)^{m_1}, \quad g_2 = \left(\frac{t}{\tau_2} \right)^{m_2}. \quad (20.22)$$

The parameters m_1 and m_2 are rate constants for the contraction and relaxation, respectively, τ_1 and τ_2 are the time constants of the two phases of the cardiac cycle, and k is a scaling factor.

$$k = \frac{E_{max} - E_{min}}{\max \left[\left(\frac{g_1}{1 + g_1} \right) \left(\frac{1}{1 + g_2} \right) \right]}, \quad (20.23)$$

with the denominator being the maximum value over T , the length of the cardiac cycle. This ensures that the maximum value of $E(t)$ is E_{max} .

The parameters of the model can be found in Appendix B. It is important to note that the pressure and flow rate generated by the time-varying

elastance depend on the *RCL* elements in the circulation. Therefore, as we used the *RCL* elements of Kummer et al. [88] and not Mynard's, we had to tune the parameters of the time-varying elastance model to get physiological results in our network.

20.3 Valve Models

We implemented different valve models to see their effects on the results. Initially, we assumed that the main focus should be on the effects of valve models. However, with the work of Gianluca Tocco [138] and Carlos de la Torre Luque [139], we found that the compliance of the heart and the parameter values of the circulation are equally important for the results. Nevertheless, we have implemented different valve models in the LPM framework to see their effects on the results.

In Chapter 21, we show results for healthy valves using Maynard et al.'s [99] valve model, which is a smooth diode model. We also implemented valve models using the work of Kummer et al. [88], using variable resistance [140], and a simple diode model opening to a positive pressure gradient.

20.3.1 Mynard's Smooth Valve Motion Model

The smooth valve motion model behaves similarly to Kummer's opening and closing models, $\frac{dD_u}{dt}$ and $\frac{dD_l}{dt}$. Although, instead of using fixed rate of change parameters γ , the rate of change depends on the pressure difference Δp across the valve. The model is defined as follows [99]:

$$\frac{dD}{dt} = \begin{cases} (1 - D)K_{vo}(\Delta p - \Delta p_{open}) & \text{if } \Delta p > \Delta p_{open}, \\ DK_{vc}(\Delta p - \Delta p_{close}) & \text{if } \Delta p \leq \Delta p_{close}. \end{cases} \quad (20.24)$$

This function guarantees that the valve converges to $D = 1$ when the valve is opening and $D = 0$ when it is closing.

20.3.2 Kummer's Valve Model

Kummer's valve model describes a general model for valves with three submodels describing valve states close to fully open, fully closed, and in transition states.

$$\frac{d^2 D_m}{dt^2} = \lambda \frac{p_i - p_j}{|p_i - p_j|}, \quad (20.25)$$

$$\frac{dD_u}{dt} = \gamma_u(1 - D_u), \quad (20.26)$$

$$\frac{dD_l}{dt} = -\gamma_l D_l. \quad (20.27)$$

At the start of every time step, we check which model to use.

$$D = \begin{cases} D_u & \text{if } D_u < D_m, \\ D_l & \text{if } D_l > D_m, \\ D_m & \text{otherwise.} \end{cases} \quad (20.28)$$

20.3.3 Variable Resistance Model

The previous valve models could be computationally expensive, slowing down the simulations. Therefore, Regazzoni et al. [100] proposed a model that has only two states: open and closed. The model is defined as follows:

$$R(p_i, p_j) = \begin{cases} R_{min} & \text{if } p_i < p_j, \\ R_{max} & \text{if } p_i \geq p_j. \end{cases} \quad (20.29)$$

Note that this model does not define D but rather the resistance of the valve. This basically modifies the respective resistance element in the $\mathbf{A}$ matrix of the LPM network model without the need to define a diode state.

20.3.4 Heaviside Function Model

The simple diode model uses a Heaviside function to define the state of the valve. The model is defined as follows:

$$D = H(p_i - p_j), \quad (20.30)$$

where $H(x)$ is the Heaviside step function, which is defined as:

$$H(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x > 0, \\ 0.5 & \text{if } x = 0. \end{cases} \quad (20.31)$$

20.4 Simulating Pathologies

To simulate pathologies, we used the valve model of Mynard et al. described in Section 20.3.1 and reduced the output of the time-varying elastance model to account for heart failure. Parameter values for simulating pathologies can be found in Appendix B.2.

20.5 DCC Simulations

Other parameters, which are not included in the derivation of the model but are nevertheless equally important for this study, are the parameters of the DCC device. These could be control parameters such as the amplitude, frequency, systolic fraction, and timing of the compression; or design parameters such as the size of the patches, placement on the heart, using active suction, and the shape of the patches. In this study, we include results regarding the control parameters, but further investigation is needed to account for an integrated design optimisation process in our model and code.

As described in Equation (20.20), the DCC actuation is integrated into the system as a source term, $\omega(t)$, which is a time-varying function describing the displacement of the DCC device. Note, that this displacement is translated into ventricular volume change by the coefficients e_L and e_R . This $\omega(t)$ function can be set by the actuator device, with $\sin(t)^2$ being the default function. To test different actuation control parameters, we need a more general function. Therefore, we define the following parametric relation for the actuation:

$$\omega(t) = A \sin^B \left(\frac{t + \phi T}{T} \pi \right), \quad (20.32)$$

where A controls the amplitude, T the frequency, ϕ allows for a phase shift in heart synchronisation, and B can adjust the systolic fraction.

Using this simple model, we can build a framework to study the effects of the control parameters of the DCC device actuator. This tool is useful for preparing for studies to define the parameter combinations to be tested in the experiments.

Combining the power of modelling pathologies and DCC support, we can simulate the interaction of the heart and the DCC device in different pathologies. This is a novel approach in the field, as most studies focus on other MCS modelling in pathologies, but not DCC devices. Also, using a

model like this can show insights beyond DCC support, such as defining guidelines for every compression-based solution, such as OCCC.

Chapter 21

Results and discussion

In this chapter, results are presented for cardiac arrest and heart failure simulations with and without DCC device support. With this, we want to show the capabilities of the framework to simulate the interaction of DCC devices with the heart and the circulatory system. We also present the effects of different device parameters and physiological variance on the expected outcome. The results are presented mostly as pressure-volume (p-V) loops, as it is the standard for clinicians and engineers alike. To understand the relevance of p-V loops, refer to Figure 1.5.

21.1 Active Heart Simulations

To build the foundation of our studies, an active heart model needs to be included in the framework. This allows us to create a baseline for healthy heart results to compare the performance of the DCC device in pathological conditions. The active heart model was implemented as described by Mynard et al. [99]. It allows us to tune its parameters to achieve physiological results in our LPM network. Again, it is important to note that the parameters in LPMs are tuned to provide the needed results in their respective networks. Taking models from the literature while using a different network with different element parameters means that the implemented model has to be tuned to the new network. The model parameters for the healthy heart and the pathologies can be found in Appendix B.2.

21.1.1 Healthy Heart

Our results for the baseline show a typical pressure-volume loop for a healthy heart in Figure 21.1. The pressure and stroke volume values are aligned with the findings in the literature [127]. These results serve as a comparison for assessing how successful DCC support is in different pathologies. The corners on the p-V loop at the end and the start of systole show the transient effects of the smooth valve functions.

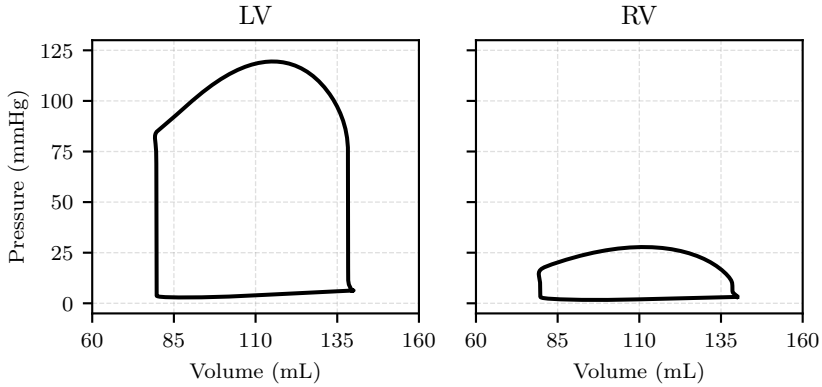


Figure 21.1: The pressure-volume loop of a healthy heart.

21.2 Pathologies

Before testing the capabilities of the framework in simulating DCC device support, we need to create pathological conditions, such as heart failure, cardiac arrest, fibrillation, and valve stenosis and regurgitation. Figure 21.2 shows the pressure-volume loop of reduced cardiac performance, simulating heart failure. The results show a reduced stroke volume and slightly increased end-diastolic pressure, which are typical for heart failure patients.

As we aim to design our DCC device for emergency MCS in cardiac arrest, we also simulate this condition. The pressure-volume loop of a cardiac arrest shows no data, as there is no flow and pressure increase in the heart. Therefore, the baseline condition for cardiac arrest is not indicated in the respective p-V loop figures.

21.3 Simulating DCC Device Support

As the main focus of this part is to demonstrate a simulation framework for simulating the effect of DCC device support and to create an *in silico* testing and optimisation tool, the main results of this chapter are related to the simulated effects of DCC device support in the case of different pathologies.

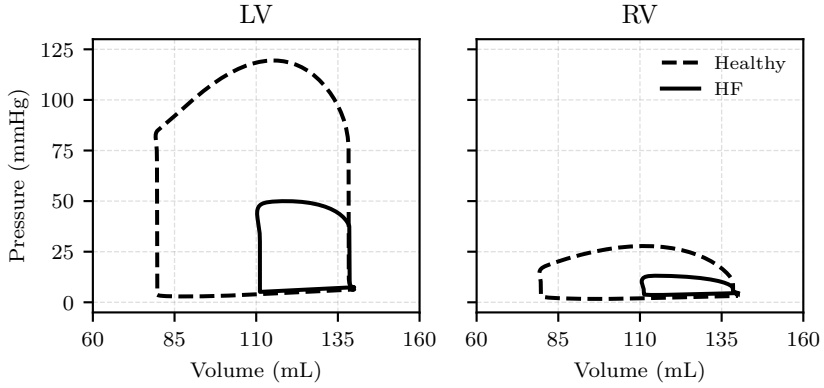


Figure 21.2: The pressure-volume loop of a simulated heart failure without DCC support. Nomenclature: Healthy - reference healthy heart, HF - simulated heart failure.

21.3.1 Cardiac Arrest

The proposed application for our DCC device is short-term support for cardiac arrest patients. Simulating the effects of our device on this patient cohort is essential. Using the initial conditions of Kummer et al. and having no heart activity, we reach the equilibrium pressure of around 5 mmHg in the circulation. Figure 21.3 shows the performance of the DCC device for a patient without a heartbeat. The p-V loop is generated solely by the DCC device. The device parameters were set according to the settings used in Chapter 17, and the heart's passive compliance was set according to the mean values in Chapter 25. We demonstrate that our framework can simulate the effects of DCC device support. Here, the absolute results are not important as no optimisation was performed, but the fact that this model can simulate the effect of DCC device support solely based on the displacement of the device and the coefficient describing the mechanical heart-device interaction.

21.3.2 Heart Failure

In the case of heart failure, the DCC device interacts with a heart that has a natural heartbeat. The pumping action of the DCC device is added to the

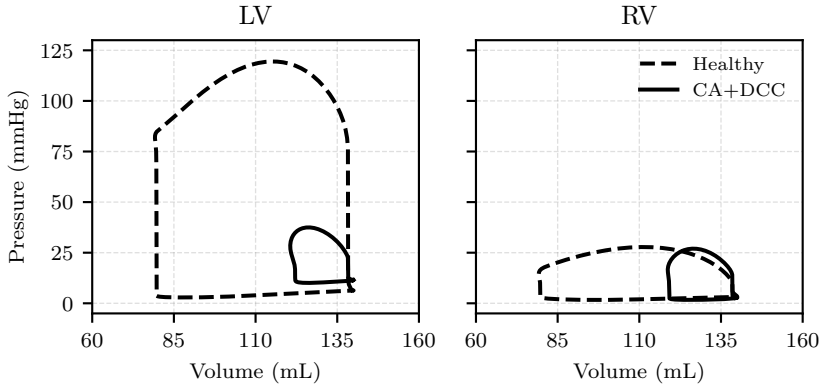


Figure 21.3: The pressure-volume loop of a simulated cardiac arrest with DCC support. Nomenclature: Healthy - reference healthy heart, CA+DCC - simulated cardiac arrest with DCC support.

heart's pumping action, which results in a pressure-volume loop that shows the effects of the DCC device on the heart. The device is in sync with the heart's rhythm. Figure 21.4 shows the p-V loop of a simulated heart failure with DCC support. The DCC device helps to increase the stroke volume and the pressure in the ventricles.

21.3.3 Dual Tachycardia

Another proposed application is the support of patients with atrial and ventricular fibrillation. The DCC device can be used to support the heart in this condition by providing additional pumping action. Figure 21.5 shows the p-V loop of a simulated dual tachycardia (both atrial and ventricular fibrillation) with DCC support. The DCC device is in interaction with a diseased heart that has an elevated beating frequency. This results in almost no pressure increase and stroke volume, while the end-diastolic pressure is elevated. The unnatural shape of the p-V loops in Figure 21.5 shows that the frequency of action for the heart and the device are not the same. The actuation frequency of the DCC device was set to half of the heart's frequency to achieve converging results. Important to note that this is an idealised case, with regular heart beats at 200 bpm. In reality, fibrillation is

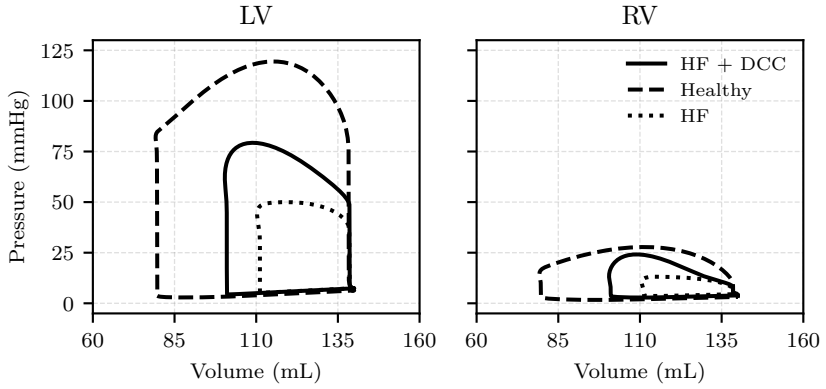


Figure 21.4: The pressure-volume loop of a simulated heart failure with DCC support. Nomenclature: Healthy - reference healthy heart, HF - simulated heart failure, HF+DCC - simulated heart failure with DCC support.

an irregular, elevated rhythm, which would result in more chaotic p-V loops.

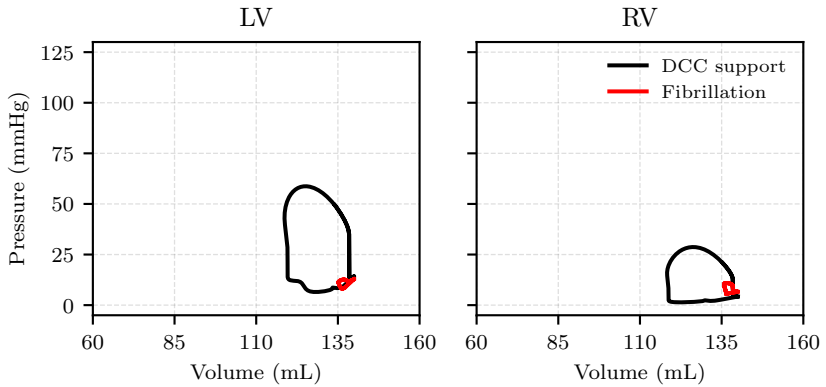


Figure 21.5: The pressure-volume loops of simulated dual tachycardia with DCC support.

21.3.4 Valve Pathologies

Valve pathologies can play a major role in the outcome of DCC device support. As our model cannot account for cardiac remodelling at the current state, we show results with simulated aortic stenosis and tricuspid regurgitation defined by Mynard et al. [99]. The parameters of the heart were set to the healthy values, and the valve parameters were modified according to the pathologies. The results are shown in Figure 21.6 and Figure 21.7.

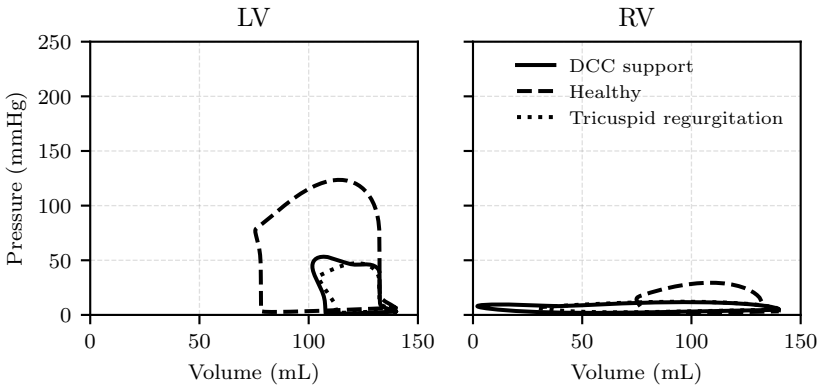


Figure 21.6: The pressure-volume loops of simulated tricuspid regurgitation with DCC support. Regurgitation parameters are set according to literature [99].

21.4 Optimising Actuation

The main motivation behind developing this framework was to create a tool for optimising our DCC device, as the FEM simulations are computationally too expensive to study the effects and settings of the device thoroughly. In the current state of the code, an optimisation algorithm is not implemented, we present a study on the effects of different DCC device control parameters on the pressure-volume loops of cardiac arrest and heart failure.

In our study presented in Chapter 16, we tested different control parameters of the DCC device, such as the actuation frequency, amplitude, and systolic fraction. However, no systematic study was possible in the *ex vivo*

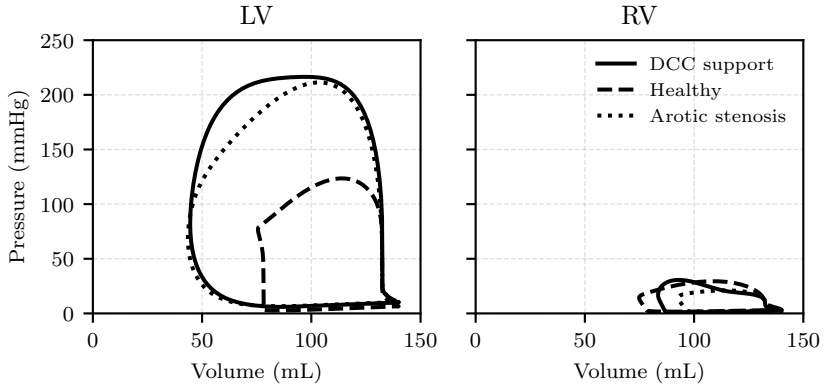


Figure 21.7: The pressure-volume loops of simulated aortic stenosis with DCC support. Stenosis parameters are set according to literature [99].

environment. Figure 21.8 shows the effect of different actuation parameters on the outcomes with cardiac arrest. The results show significant differences in the outcomes, something that we also found in the *ex vivo* experiments.

The more challenging task, however, is to study the effect of the DCC device on heart failure cases. The natural beating of the heart increases the complexity of the task, as the timing of the compression of the DCC device and the asynchrony with the natural heartbeat might have significant effects on the outcome. Figure 21.9 shows the results of our study on the effects of different DCC device control parameters on the outcomes with heart failure. The results show significant differences between the different realisations.

21.5 Discussion

We created a model to account for DCC device compression on the heart, and we created an extensive study on the compliance and the compliant nature of the heart. These studies are reported in Part VII. Nevertheless, there are many other things to discuss in this study that can be observed with our framework.

Unlike examples in the literature [105], we argue that the correct way to study DCC devices in an LPM environment is to use the compliance

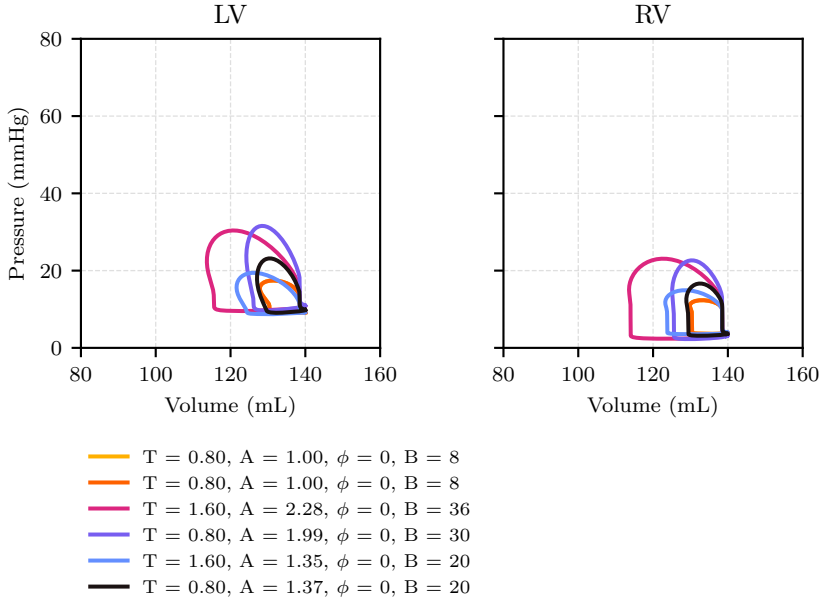


Figure 21.8: The pressure-volume loops of simulated cardiac arrest with different DCC support control parameters. For the parameters, refer to Equation (20.32).

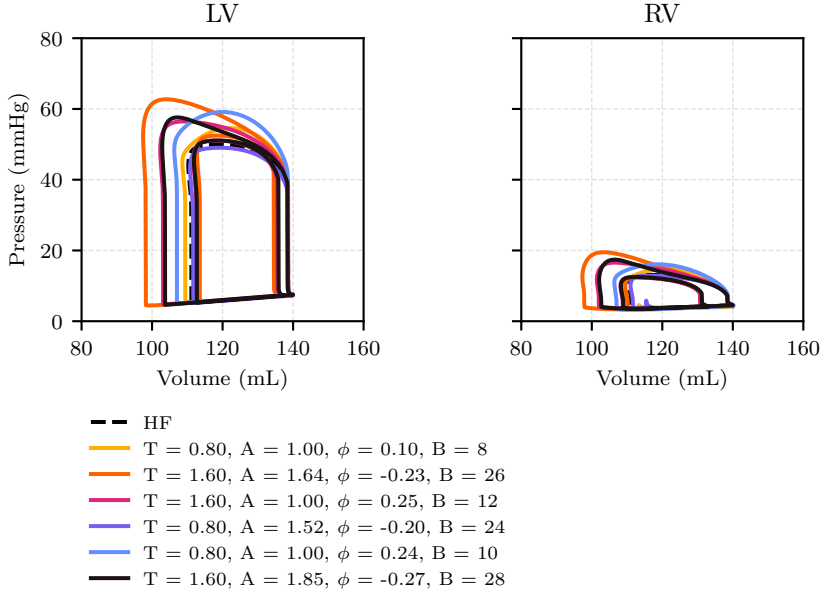


Figure 21.9: The pressure-volume loops of simulated heart failure with different DCC support control parameters. The reference case of heart failure without DCC support is indicated with the dashed lines. For the parameters, refer to Equation (20.32).

and the heart-device interaction knowledge derived from our assumption Equation (20.9) rather than adding the pressure increment from the devices. First of all, to do the latter, one must know the added pressure from the device a priori. But in this case, this framework cannot support DCC device development. The only reason these studies exist is to validate the LPM and not to support the development of devices. Also, we can argue that based on the nature of LPMs, it is trivial that by adding the right amount of pressure in one of the nodes of the network, we can achieve the expected results in a network that was tuned for the case that we are observing.

Our goal, on the other hand, is to provide a tool for DCC device development and to create a better understanding of the interaction between MCS devices and the circulatory system. These devices have been under development for around a century, and we would like to unlock the potential of their design, as they could be useful in many clinical applications [23].

Our framework allows for easily changing parameters either in the circulation, the heart, or the device itself. As we mainly created this tool to help us in device development, we would like to show how it could help us at its current state. Changing physiological parameters helps the user to create patient-specific simulations, falling into the field of human precision medicine, something that is highly sought after at the present time [141]. Changing control parameters of the device helps in planning preclinical or clinical studies during device development. Conducting these studies requires a lot of effort, time, and funds. Preparing for these experiments to maximise their impact by running hundreds, if not thousands, of simulations in the *in silico* environment could be of great help.

At the current state of the code, we do not handle design parameters, such as the geometry of the patches of our DCC device. To include this in an optimisation algorithm, first, we need to check in FEM simulations how each design parameter changes the outcome. After finding these sensitivities, we can implement an approximate protocol for optimising device geometries. Based on these findings, we might be able to provide suggestions on which geometry to use given the patient data.

Chapter 22

Conclusions and outlook

Lumped parameter models are widely used in the field of cardiovascular modelling. However, the inclusion of the effects of MCS devices, such as DCC devices, is limited in the literature. The main challenge in including DCC devices in this framework is that it needs to characterise the interaction of the device with the heart through mechanical properties, not just pressure and flow rate.

We demonstrated the capability of including a DCC device in an LPM environment, and we showed how the framework can be used to simulate the interaction of DCC devices with the heart through compliances. We also presented the effects of different device parameters and physiological variance on the expected outcome. The results show the potential for our framework to be used as a tool for optimising DCC device design and control parameters, which can help in the development of these devices.

Our approach is unique in the simulation of MCS devices in an LPM environment. Other methods use a priori knowledge of the generated pressure by DCC devices to simulate the effects on the heart and the vasculature. Our method, however, uses the compliance of the heart to simulate the interaction of the device with the heart. This approach allows for a more flexible framework, as only limited a priori knowledge (i. e. compliances and the heart-device interaction coefficients) is needed to simulate the effects of DCC devices on the heart.

We built our compliance model with high-fidelity FEM simulations, which allowed us to capture the complex interaction of the DCC device with the heart. We included this data in our computationally efficient LPM framework, which allows us to simulate the effects of DCC devices on the heart in a fast and flexible way. We simulated the effects of our DCC device on different heart pathologies.

The current model uses constant compliances for cardiac arrest simulations, an approach that might over-simplify the complex interaction of the DCC device and the heart. The active heart model simulations use a model from the literature that cannot account for the potential changes in the heart's

mechanical properties during DCC support.

To improve our model, we can include an optimisation process to find the optimal combination of actuation control parameters for different patient cohorts, which can help to further enhance the performance of the DCC device in various clinical scenarios. To optimise the design of the device, design parameters need to be identified, their effects on the outcome need to be characterised, and an optimisation algorithm needs to be implemented. With this, we can improve outcomes with already existing devices and future designs alike.

Part VII

On the Compliance of the Heart

Chapter 23

Introduction

Current elastance models in LPMs are, on the one hand, based on models developed from experiments with isolated ventricles and, on the other hand, struggle to accurately model the interaction of the heart with MCS devices, especially DCC devices. We aim to characterise tissue mechanics in an LPM without the need to couple it with a computationally expensive finite element model (FEM). These tissue mechanics are described by compliance or elastance on a macro scale. This theory is supported by Suga's time-varying elastance model [96], which has proven to be the most influential among the available compliance models. Their work proposes a time-varying elastance model for the systolic phase, while Gaasch developed a model for the diastolic phase [142]. Many have pointed out the limitations of their models since their publications [98, 143], but they are nevertheless still widely used in the literature, even in modern computational models [144, 105].

Recent models can simulate DCC support in LPMs, but they do not use physiological interaction between the device and the biological system [105]. Simply adding the expected pressure increment from the device based on previous studies is not a strong model to use in a device development framework or to build a better understanding of the interaction between the heart and the device. Building a system that can predict the pressure and flow rate response based on device input, such as the displacement of the device, is a more complex task, but it is necessary to create a model that can be used in device development.

While Suga's model was based on empirical observations, we aim to derive a novel model based on the simple piston model, using a simple assumption regarding the ventricular volumes. Our goal is to propose a comprehensive, mathematically sound model for cardiac mechanics on a macro scale with a new compliance model and to introduce parameters in our model to describe the interaction of the tissue with DCC devices. This model could allow us to simulate the interaction of DCC devices with the heart in a computationally efficient way and explore options for patient-specific support.

With that in mind, we propose a new compliance model, which was already introduced in Chapter 20, and we will elaborate on how to define compliance and how to model the heart in this part. The main question that remained unanswered in Equation (20.17) was how to define the compliance and heart-device interaction coefficients.

One option that was mentioned is experiments, using the DCC device from Chapter 7 and the setup from Section 11.1. We performed experiments on a silicone heart and noticed significant differences from the results of FEM simulations. A limitation regarding experiments with our setup is that we can only run cardiac arrest experiments, as the silicone heart cannot beat.

To validate our results on the silicone heart and to compute the compliance of the beating heart, we used the modified code of Kummer et al. [88] to run FEM simulations. We performed many studies with passive and active heart models, with or without DCC support. To compute the heart-device interaction coefficients, DCC support is a must.

Recent studies highlight the differences between patient cohorts and the effects of a lack of data on the population from underrepresented groups [127]. To build a better understanding of the compliant nature of the heart and to improve predictions of DCC support for these patients, we aim to include patient-specific data in our model.

We show that the compliance of the heart is not constant, not even in the case of a passive heart. We demonstrate the importance of co-compliances, highlighting the effects of ventricular interaction. We show differences in the compliance of an active and a passive heart. On the other hand, our results show that the current assumption used in Equation (20.8) might not be sufficient to describe the change in volume in the ventricles. However, our study is a comprehensive work on trying to understand the compliant nature of the heart.

We conclude that the compliance of the heart is a complex phenomenon, and it is not sufficient to use constant or only time-dependent compliances for the entire cardiac cycle. Based on our results, we can state that the compliance of the heart is more complex than what the most common models suggest. We introduce a new model to describe the interaction of the heart with DCC devices. Our results highlight the importance of considering co-compliances in LPMs, and we suggest that future work should focus on refining the model to better capture the dynamics of the heart under various conditions, including patient-specific scenarios.

Chapter 24

Methods

No current elastance or compliance model can predict MCS outcomes by simply defining the interaction of the heart with the device in a physiological way. Increasing the flow rate or the pressure in the nodes in the LPM representing the location where the device interacts with the circulation is a common approach. While this might be useful for turbomachines, such as the *HeartMate 3*, this is not true for DCC devices. They interact with the soft tissue of the heart to provide MCS, requiring a model to translate the motion of the device into pressure and flow increments in the circulation. Current models either do not account for DCC support, or they use preliminary knowledge built on experiments that only serves the validation of the model, but not the generation of new knowledge.

We aim to build a model that can describe the interaction of the heart with DCC devices and to build an understanding of the compliant nature of the heart. We want to build a model that supports our hypothesis in Equation (20.9) to provide data for simulating DCC support. We want to show that the assumptions of current models are inherently limited as they are built on isolated ventricle experiments. However, based on the nature of LPMS, one can come up with an arbitrary model for the heart and can achieve physiological output if they tune the network and parameters accordingly. However, to integrate MCS devices, such as DCC support, into the LPM, we need a model that captures the physiology of the heart.

24.1 Compliance Models

The most commonly used compliance model is the end-systolic pressure-volume relationship (ESPVR) / end-diastolic pressure-volume relationship (EDPVR) model, a combined theory of Suga's [96] and Gaasch's [142]. Even though the original publications did not refer to ESPVR or EDPVR as compliance or elastance, they are widely used as an analogue model [145, 99]. The distinction between these models and actual compliance is very important. Suga et al. conducted their experiments on isolated canine

hearts, finding that the pressure-volume relationship in systole is invariant to the preload in the system. This means that the ESPVR is linear, and regardless of the preload, the end-systolic points of a pressure-volume loop lie on the ESPVR line. Taking this invariance assumption to the entire systolic phase, their hypothesis is that using a simple relation, they can describe the relationship between pressure and volume for every time instance. They describe the systolic phase for a beating heart with the assumption that the pressure can be defined by a linear relationship (the time-varying ESPVR, $E(t)$) with the current ($V(t)$) and the unloaded ventricular volume (V_d). This means that for a given time instance, the compliance is constant at any given preload for an isolated ventricle:

$$p(t) = E(t)(V(t) - V_d). \quad (24.1)$$

Many take ESPVR as an analogue to elastance, especially that it was using the notation $E(t)$. However, the definition of elastance is the change in pressure with respect to the change in volume in an elastic vessel:

$$E = \frac{\partial p}{\partial V}. \quad (24.2)$$

The models of Suga and Gaasch approximate this relation in an isolated ventricular setup, as they derive a direct relation between the pressure in the ventricle and the volume. In a biventricular setting, this relationship is more complex as the pressure in a ventricle can also change with the volume change of the other ventricle. Both Suga's and Gaasch's models are widely used in the literature [145, 99], but there have been reports that these models do not hold for simulations with MCS devices [98, 143].

There exist models with ventricular interaction [131], but they either do not include MCS support or they inherit assumptions from the ESPVR / EDPVR models. Also, no active heart model describes co-compliances as active parameters, assuming that the beating heart functions as two isolated ventricles driving the circulation. To challenge the theory behind the most popular models and to create a mathematically sound model, we build on the previously described assumption in Equation (20.9).

The source terms in active heart models, in most cases, is either the ESPVR / EDPVR model or results from the coupling with higher-order models. Our source terms or input parameters are $\omega(t)$ from the DCC device support and self-actuation, $C_{ij}(t)$, the changing compliance in the ventricles. The following derivation and the proposed model in Equation (20.17) describe

a model for only nodes representing the heart in our LPM. To get the coefficients for the matrix $\mathbf{C}$ and vector $\mathbf{e}$, we need to define the compliance and the heart-device interaction coefficients. We can acquire these coefficients through experiments with the DCC device or through FEM simulations.

24.2 Experiments

In the experimental setup, it is challenging to approximate the partial derivatives in Equation (20.10), namely fixing all variables except for one. Our approach was to isolate a silicone heart, adjust and fix the volume in the ventricles, and compress the heart with the patches in incremental steps in DCC support, while recording the pressure in both the LV and RV in this quasi-static setup. A similar method was described by Kerckhoffs, using FEM simulations [131]. Figure 24.1 shows the experimental setup. Initially, we conducted experiments using the silicone heart, but we found great differences in the obtained compliance and heart-device interaction coefficient values between the FEM and the experimental methods. Therefore, we moved on to reproduce the results using *ex vivo* porcine samples. This method requires watertight sealing of the heart. We could not achieve that on *ex vivo* heart samples; therefore, we decided to use our results from the FEM simulations. Nevertheless, we would like to share our method as it might be a valuable tool to measure these coefficients, given the technical issues are solved.

As the silicone heart is not a biological material, we can only come as close to the properties of the heart as the material allows. Unfortunately, studies on silicone samples show that the material properties of the heart are hard to match [121]. Also, silicone is a passive material, only allowing for cardiac arrest investigations on compliance. As most MCS devices are used on heart failure patients, we need a model to acquire compliance and heart-device interaction coefficients in a beating heart. Therefore, we decided to use the modified FEM simulation framework of Kummer et al. [88] to acquire these coefficients.

24.3 FEM Simulations

For the coupled FEM simulations, we used the extended LifeV [147] FEM code by Kummer et al. [88]; we modified the simulations to gain the compliance and heart-device interaction coefficient values. Another advantage of using

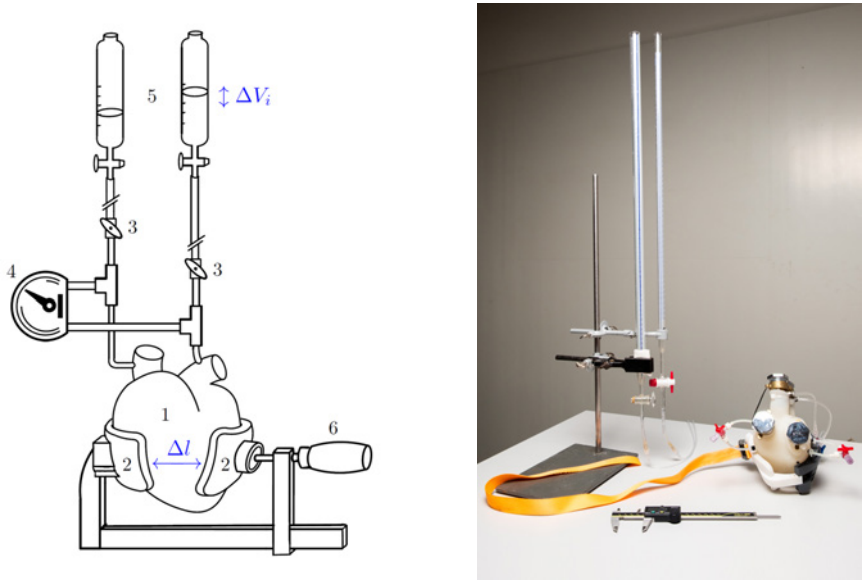


Figure 24.1: The schematics (left) and the experimental setup (right) to acquire compliances and heart-device interaction coefficients. The inflow and outflow tracts of the heart (1) are closed, and the ventricles are connected to burrettes (5). The volume is set in each ventricle with ΔV steps, and then the heart filling valves (3) are closed. The heart is then compressed by the DCC device (2) in Δl steps, using a clamp (6) to allow quasi-static compressions. The pressure is measured with pressure sensors (4) after the pressure has converged to a static value [146].

such a method was that we could test the theory building on Suga's, using a two-Hill periodic function as the elastance model of a beating heart [99]. As these models were built on isolated ventricle experiments, we hoped that we could gain new insights into the compliance of the heart using a biventricular FEM model. Our approach was twofold: computing the compliance and the heart-device interaction coefficient in CA and beating heart simulations.

We modified *LifeV* such that at every time step, we perturbed the pressure in the LV and the RV individually, restored the original state, and then moved the actuation boundary condition forward in time. The flow chart of the perturbation algorithm and an image from the simulations are depicted in Figure 24.2. We use the solution u at time step n and perturb the different parameters of our model, Equation (20.8). We calculate the missing coefficients using finite differences and then reset the pressure boundary

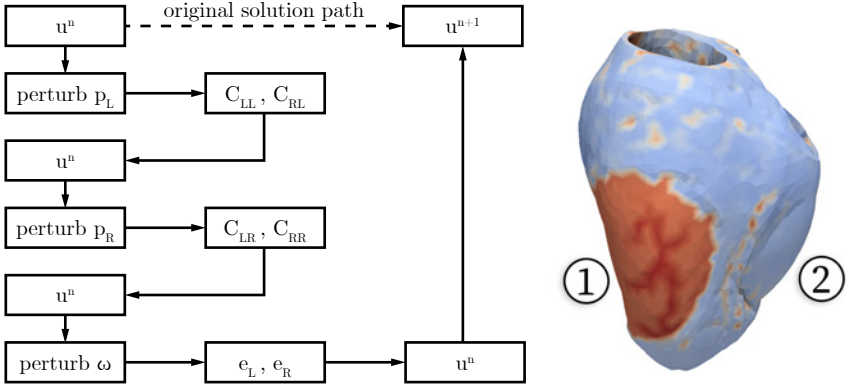


Figure 24.2: Our FEM approach to obtain the missing coefficients to close our model. **Left:** the flow chart of defining the coefficients. **Right:** image showing the FEM simulation. The colouring is based on the local activation levels of the heart. We can set region with reduced cardiac activity (1), and add DCC device support (2).

conditions and the mesh displacement to that of the solution u^n . We continue that until we have acquired all the missing coefficients, reset the solution once again to u^n , and then continue with the FEM simulation and LPM coupling until we reach u^{n+1} .

It must be noted that the range of the heart compliances highly depends on the parameters of the whole circulation, as these parameters are set to achieve physiological results in terms of pressure and flow rates given an assumption of the heart's compliance. In our case, the compliance should be closer to the actual physiological range as it is based on the complex muscle structure simulations in the FEM model rather than a heuristic assumption. The parameters in the circulation were tuned to get physiological results given these compliance properties.

24.3.1 Active Heart Simulations

We used the active heart model by Kummer et al. [88] to compute the compliance and heart-device interaction coefficients in a beating heart. We used the perturbation method described in Figure 24.2 to compute the coefficients using finite differences. We ran studies with healthy and pathologic hearts,

with and without DCC support. Again, to get the heart-device interaction coefficients e_i , the DCC device support must be included in the simulations.

24.3.2 Passive Heart Simulations

For defining the coefficients in CA, we could turn off heart activation and decouple the FEM model from the LPM. There are two strategies: running the DCC simulations, using a passive heart model and perturbing the system in the subroutine at every time step; or setting the pressure manually in each ventricle in a predefined range, creating a random search process, and using the perturbation subroutine at every time step. The former resembles the realistic scenario in CA DCC support simulations, but the explored pressure range is usually small, and it highly depends on the performance of the simulated DCC device. The latter method allows the user to explore the entire physiological pressure range in both the LV and the RV while applying compression with the DCC device boundary condition. To get realistic values in the active heart model, the LPM should be coupled with the FEM model.

24.4 Patient-Specific Models

As these simulations were computationally very expensive, taking about 30,000 CPU hours to run a single heartbeat, training a patient-specific model is not feasible with the current framework. Our approach was to look at the literature to predict the differences and the variance in compliance between different patient groups. Based on this knowledge, we could modify the compliance coefficients in the LPM to simulate not patient-specific scenarios, but rather exploring how much effect compliance and heart-device interaction coefficients have on the expected outcome of DCC support.

Here, we propose a method to define compliance and heart-device interaction coefficients for our novel heart model in our LPM network in Part VI. Our approach can define compliance coefficients in an experimental and a simulation environment. By leveraging the FEM framework, we can compute these coefficients even in a beating heart, something that is not feasible in an experimental setup. Based on this approach, we can explore the compliant nature of the heart in a controlled environment and have an insight into properties we would not have otherwise. We want to use the collected data to build a general compliance model, simulate the effects of compliance,

thus exploring outcomes for different patient cohorts, and to create a better understanding of the interaction between the heart and DCC devices.

The model is integrated into our LPM framework introduced in Chapter 20. The compliance and heart-device interaction coefficients can be defined in the LPM, and the model can be used to simulate DCC support. The model is capable of simulating both CA and HF scenarios, and it can be used to explore the effects of different device parameters on the expected outcome. The model can also be used to simulate patient-specific scenarios by modifying the compliance coefficients and pathologies based on literature data.

Chapter 25

Results and discussion

In this chapter, results and findings related to the compliance of the heart are presented. Understanding how compliance changes with pressure or time, how it changes between individuals, and how it affects the outcome of DCC device support are crucial for our project. We aim to build a new, comprehensive compliance model to improve the current standards in LPMs. We approach this by using experiments and a higher-level FEM model to study compliance and the interaction of the heart and the DCC device.

25.1 Passive Heart Compliance

First, we used our FEM simulations without activation to simulate CA conditions. One of our assumptions regarding the compliance and the heart-device interaction parameter was that they are not constant throughout the compression and that the co-compliances are significant regarding the outcome of the simulations. We tested two methods to acquire the coefficients: experiments on a silicone heart model and a perturbation method on the FEM model of Kummer et al. [88].

25.1.1 Experiments

Using silicone hearts in the experiments for measuring the coefficient leaves us with uncertainty, given the reports on their mechanical properties and their relation to biological tissue [121]. Figure 25.1 shows the results of the experiments. We find that the mean values of each coefficient are close to the value 0, indicating that the heart is very stiff. Also, the range goes from negative to positive values for all of the coefficients, something that we cannot explain at the moment. Given these results, we decided to proceed with FEM simulations as they can reproduce physiological results in the simulation environment.

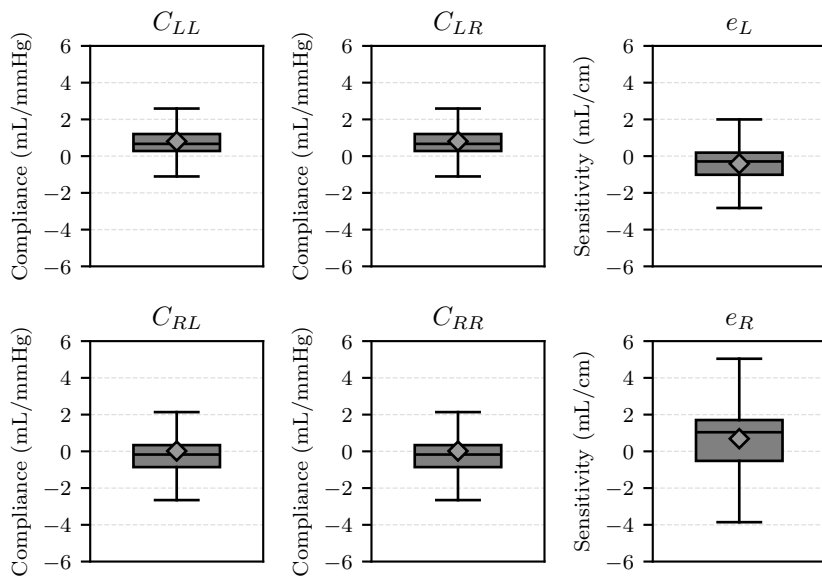


Figure 25.1: Compliance and heart-device coefficients collected by using the experimental setup. The boxplot indicates the interquartile range with the gray box, the median with the line in the box, and the mean value with the rhombus.

25.1.2 FEM Simulations

To study cardiac arrest cases in the FEM framework, we turned off the heart activation in Kummer's model and collected the results regarding the coefficients through the perturbation subroutine (Figure 24.2).

Figure 25.2 shows the boxplot of these coefficients in our simulations. They show that indeed, despite the common assumption in the literature, the compliance of the heart is not constant, and the different coefficients have different variances. The larger the variance of the coefficient, the bigger the modelling error introduced by using only a constant value for the compliances. Now that the variance and the mean values of the coefficients have been presented, we should also have a look at the distribution of these coefficients in their value range. Figures 25.3 and 25.4 show the distribution of the compliance and heart-device interaction coefficients, respectively. Some of the coefficients show a bimodal distribution, indicating that the single mean value might not be sufficient to describe the compliance of the coefficient.

We also found that based on the mean values of these compliances, using the relation in Equation (20.10), the co-compliances should have a significant effect on the final results as their value is in the same order of magnitude as the other compliances. However, we can only draw conclusions on that after running the LPM simulations. Regarding the sensitivity to DCC device actuation, the RV shows a significantly higher sensitivity compared to the LV. This means that for the same increase in displacement, there is a different change in the volume of the ventricles.

25.1.3 Effects of Compliance

After presenting a thorough study on compliance and heart-device interaction, we can present results using this knowledge in the LPM network. First, we present the results of a cardiac arrest heart with different passive compliances for different realisations.

Compliance Variation

As we previously made the assumption that the compliance and heart-device interaction coefficients are constant, we might have introduced a source of error in the evaluation. Also, we previously indicated that as most current models use this approach, they might suffer from the same issue. To study how the compliance and heart-device interaction coefficients affect the outcome of DCC device support in a cardiac arrest case, we ran simulations

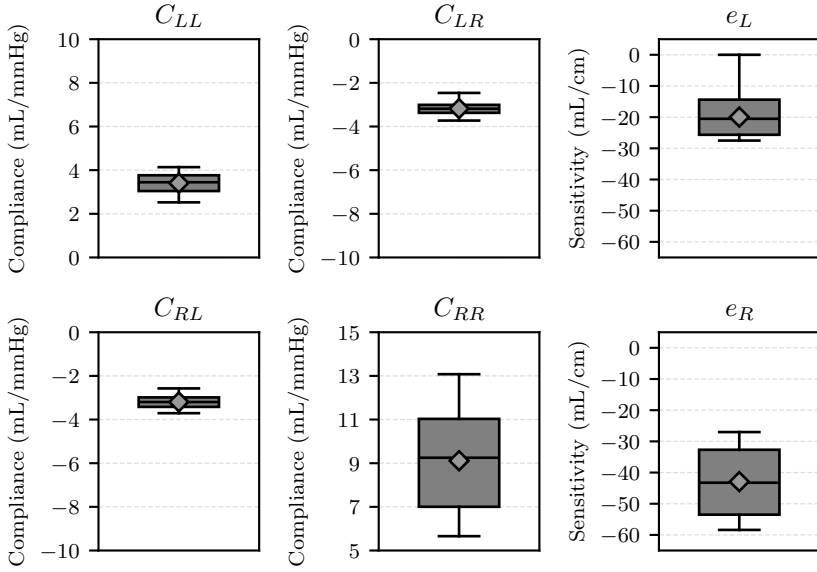


Figure 25.2: Results of the compliance and heart-device interaction coefficients, using the FEM model. The boxplot indicates the interquartile range with the gray box, the median with the line in the box, and the mean value with the rhombus.

with a set of coefficient values within the respective standard deviation ranges. We used the mean values of the coefficients from Figure 25.2 as the baseline (red curve), and then we randomly sampled individual realisations with constant coefficients within the standard deviation ranges. Figures 25.5 and 25.6 show the results of the simulations. The results show that the compliance and heart-device interaction coefficients do not have a significant effect on the outcome of DCC device support in a cardiac arrest case.

Co-compliances

As most current models neglect the effects of co-compliances, except for a few examples [131], we also wanted to study how the co-compliances might affect the outcome of DCC device support. We ran a simulation with the same compliance and heart-device interaction coefficients, but this time

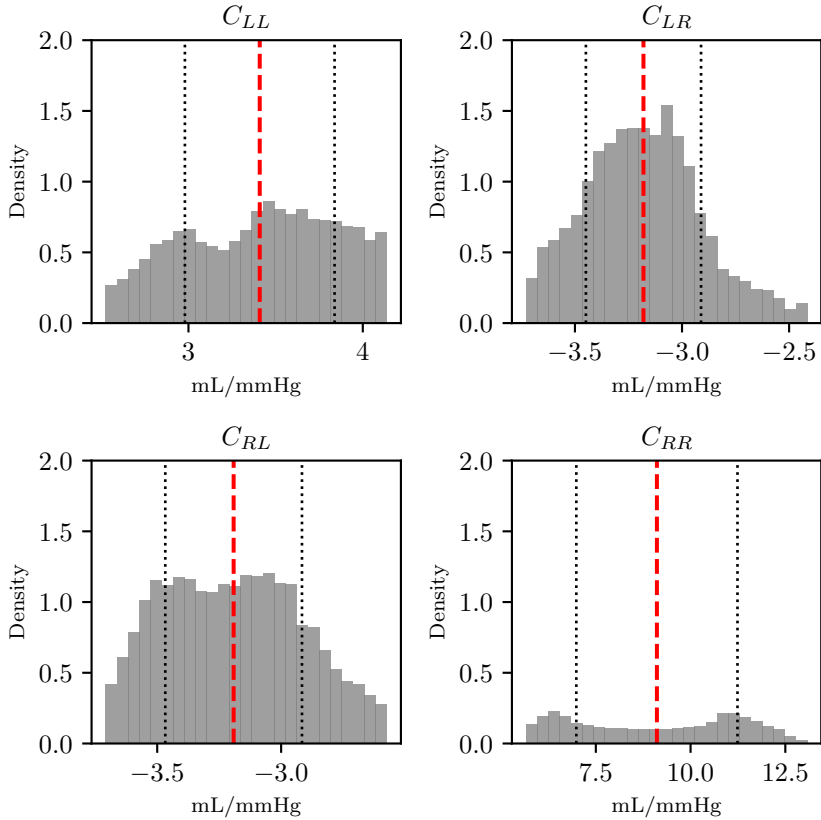


Figure 25.3: The histograms of each compliance coefficient. The red vertical line indicates the mean, and the dashed black lines indicate the standard deviation range of the coefficients.

we set the co-compliances to zero. The results in Figure 25.7 show results with non-zero and zero co-compliances and that the co-compliances have a significant effect on the outcome of DCC device support in a cardiac arrest case. This is an important finding, as it shows that the co-compliances might not be neglected in LPMs.

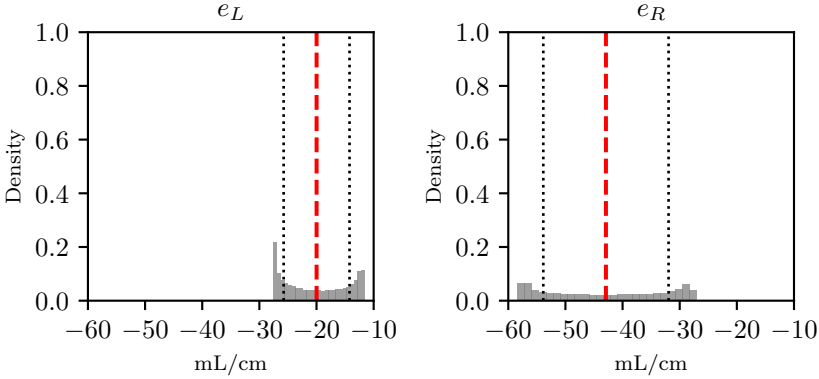


Figure 25.4: The histograms of each heart-device interaction coefficient. The red vertical line indicates the mean, and the dashed black lines indicate the standard deviation range of the coefficients.

25.2 Active Heart Compliance

Next, we investigated how compliance changes with time in a beating heart. As mentioned in Chapter 24, we were concerned about the validity of isolated ventricle models in MCS applications. To study how compliance changes in the beating heart, we set up a case for a healthy heart simulation without DCC device support. One of the reasons for our investigation was to see if the two-Hill periodic solution is a good representation of cardiac compliance. Figure 25.8 shows the compliance of a healthy, beating heart as a function of time from our FEM model. Notice that we were not only able to acquire time-varying compliance models for $C_{LL}(t)$ and $C_{RR}(t)$ but also for the co-compliances $C_{LR}(t)$ and $C_{RL}(t)$, something that has never been defined for active heart models. The data shows that $C_{LL}(t)$, $C_{LR}(t)$, and $C_{RL}(t)$ are taking similar amplitudes, suggesting that co-compliances might be essential for the simulations.

If we want to compare our compliance results with the commonly used two-Hill periodic function, we need to invert the compliance matrix, normalise the elastance coefficients, and then normalise their values. Normalising is also important, as there is no agreement in the literature regarding the range of values compliances can take. In LPMs, the parameters of the circulation

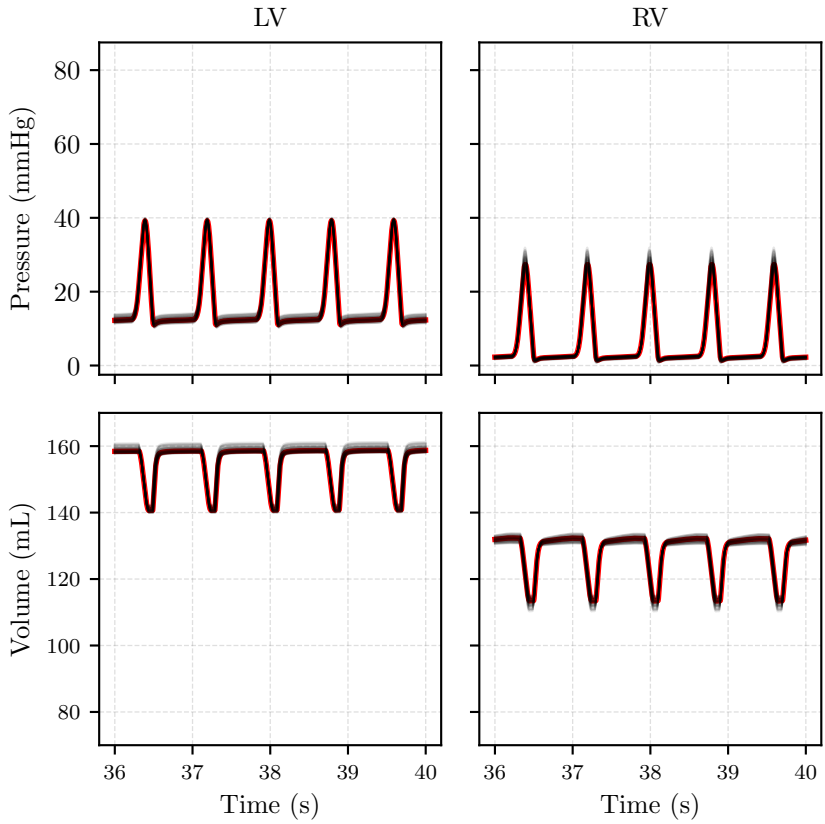


Figure 25.5: Pressure and volume of DCC support in simulated cardiac arrest cases with different compliance and heart-device interaction coefficients sampled from the standard deviation range of each coefficient. The red curve shows the result using the mean values of each coefficient, while the grey curves show the individual realisations. For the mean value and the standard deviation ranges, see Figures 25.3 and 25.4.

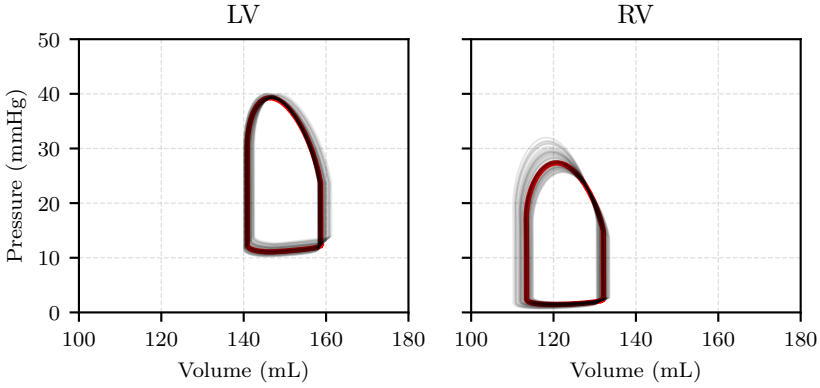


Figure 25.6: Pressure-volume loops of DCC support in simulated cardiac arrest cases with different compliance and heart-device interaction coefficients sampled from the standard deviation range of each coefficient. The red curve shows the result using the mean values of each coefficient, while the grey curves show the individual realisations. For the mean value and the standard deviation ranges, see Figures 25.3 and 25.4.

and the elastance in the time-varying elastance models are tuned to get physiological results, which does not necessarily mean that the parameter values themselves are also physiological. Figure 25.8 also shows the difference between our compliance model and the two-Hill elastance function from the literature [99]. We compared the values of $C_{LL}(t)$ for both methods. We found that our active heart FEM model produces a much shorter systole than Mynard’s elastance model and a similarly rapid drop in elastance during diastole. Interestingly, our FEM model produces the peak elastance not at the end of systole (set to 250 ms) but at a much earlier time (160 ms), while Mynard’s model uses parameters τ_1 and τ_2 to control the systolic fraction, but the peak elastance is set between these two time parameters.

25.3 Active vs Passive Heart Compliance

To further study the range of compliances and their variance, we investigated the active heart model and compared it to the CA model. The reason behind this is that due to the activation in the FEM model, the muscle fibres contract, which could modify the contractility of the heart. Therefore, we

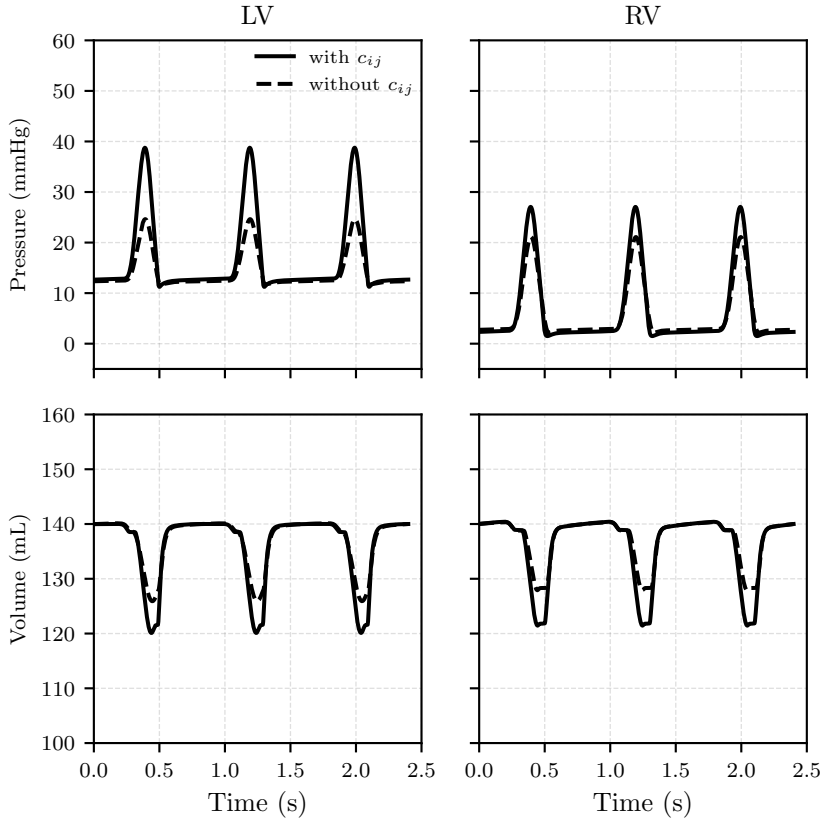


Figure 25.7: Comparing two cardiac arrest cases with simulations using zero and non-zero co-compliances c_{ij} (i.e. c_{LR} and c_{RL}).

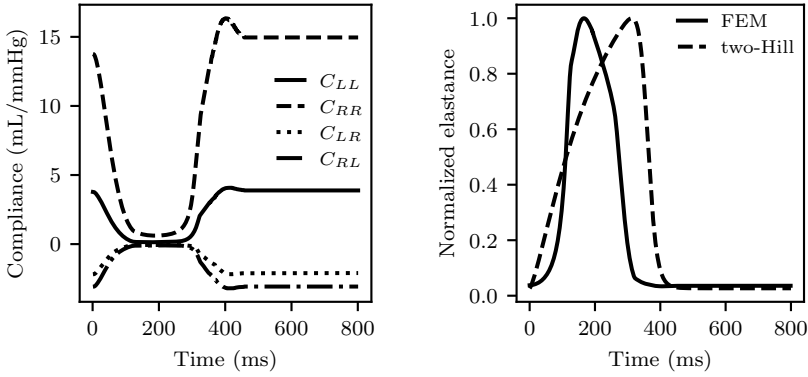


Figure 25.8: Compliance of the active FEM heart model (left), and comparing it to a standard time-varying elastance model [99] (right).

wanted to quantify to what extent the compliance changes with an activated heart.

For the CA cases, we decided to set the pressure in the ventricles manually, as with DCC device support on a CA model, we could not reach the same pressure and volume ranges as the active heart can. We assumed that the compliance might be a function of pressure; therefore, we wanted to make sure that we ran both investigations in the same pressure range. We decided to randomly sample (p_L, p_R) pairs from the domain of $p_L = (0, 120)$ and $p_R = (0, 40)$ and set these as pressure boundary conditions in the FEM model. For this method, we decoupled the FEM from the LPM as no pressure exchange was required between the two models. We sampled 100 ventricular pressure pairs, created an interpolated list with the maximum increment of $|\Delta p| = 0.1$ mmHg, and used this list as preset boundary conditions for every time step.

With the active heart simulation, we kept the coupling between the FEM and the LPM models and extracted the coefficients for one heart cycle. We turned off the DCC support for this study, as with high pressures, the added compression led to unphysiological stresses and diverging simulations. We acquired the coefficients with the same perturbation method as before (see Figure 24.2).

Figure 25.9 shows the difference in compliance between an active heart

and a passive heart simulation. There is a significant difference in the range and the values of the coefficients. Apart from C_{RR} , the heart shows lower compliance levels in absolute values, meaning that the heart is stiffer when it is active, except for the right ventricle. Regarding the significant difference between the mean and median values of the active heart compliances, Figure 25.8 shows further insights. Interestingly, the co-compliances in the CA case sometimes reach positive values (not shown on the boxplot diagrams), something that we did not expect. This means that an increase in the pressure in the other ventricle results in an increased volume in the observed ventricle. Looking at the FEM model in Figure 25.10, due to the increased pressure, the apex elongates and essentially stretches both ventricles. This behaviour is unnatural in terms of tissue mechanics, as with increased pressure, a healthy heart produces apical contraction and not elongation.

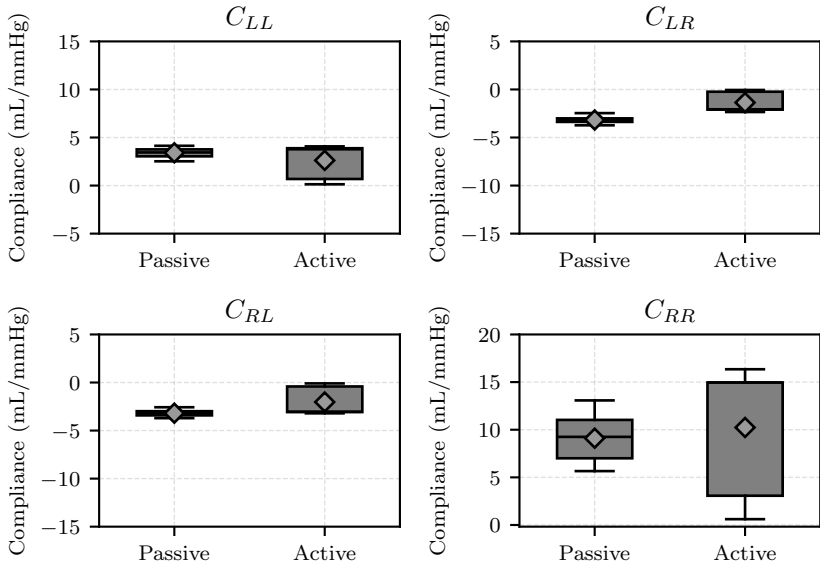


Figure 25.9: The compliance coefficients of the heart models, acquired in the FEM model for a passive (CA) and an active heart. The boxplot indicates the interquartile range with the gray box, the median with the line in the box, and the mean value with the rhombus.

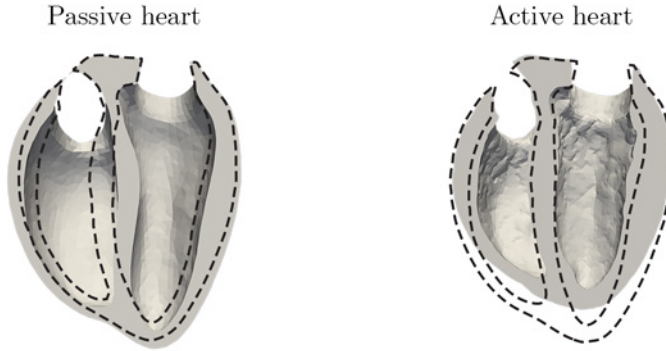


Figure 25.10: The results from FEM simulations for both a passive (cardiac arrest) and an active heart. The dashed lines indicate the shape of the heart at the initial state. The slices show the geometry at the highest pressure in the ventricles for the passive heart and the end of systole for the active heart.

25.4 Discussion

Similar to examples in the literature [131], we use the mean value of heart compliance parameters in a biventricular setting. Just like Kerckhoffs, we did studies on the compliance of the heart and decided to show our results with constant compliances. Building a surrogate model for the complex compliance behaviour is not an easy task. We trained models with polynomial fitting and linear least squares with l_2 regularisation (Ridge), but the small inaccuracies of the surrogate models led to significant and unphysiological changes in the volume of the ventricles. We trained our models to fit the following functions: $C = f(p_L, p_R, \omega)$ and $e = f(p_L, p_R, \omega)$. We believe that by finding an accurate surrogate model, we could provide new insights into the compliant nature of the heart. At the current stage of the research, further studies are needed to evaluate the passive and active compliance models with Kummer's FEM model. For this reason, we used only the constant compliance and heart-device interaction coefficients and the two-Hill time-varying elastance model in Part VI.

Based on these models, we might be able to point out crucial parameters for the circulation. Understanding how compliance works in different loading and physiological conditions, and device placement are crucial for predicting the outcome for a patient. These studies could also provide valuable input for

surgeons on the placement of these devices, especially in emergency scenarios when time is pressing and accurate placement might not be possible. To study this phenomenon, a thorough investigation is needed, including using FEM simulations.

After understanding how compliance changes for an individual, it is also important to understand how these values change in a broader population. We believe that the probabilistic representation of the results should serve engineers and clinicians better than only reporting the expected values. The variance of the outcome could be very different among groups of our population. We used a generic 3D heart geometry for training our models, but to explore the differences of human hearts from the perspective of compliance, we must redo our study with different heart geometries from actual 3D computed tomography (CT) images. As pointed out in [127], differences in the anatomy of underrepresented populations can affect the lives of patients. To learn how different hearts can behave from the perspective of compliance, we must run a thorough examination of multiple heart geometries of multiple groups of our global population.

More and more studies concentrate on the differences between patient groups, highlighting the effects on the treatment of those from underrepresented populations. In [127], the authors point out the differences in male and female heart properties, from expected ventricular volume, size, systolic and diastolic pressures, strain rates, heartbeat frequency, and other parameters. As half of the population is female, it is crucial to understand how these differences affect the compliance of the heart and thus the outcome of DCC or MCS device support. Finding crucial parameters and analysing the sensitivity of the outcome to these parameters can help the survival of patients.

Exploring these sensitivities can create a rapid protocol for handling patients in emergencies and planning long-term therapies with DCC devices. If we can find sensitivities to parameters, we can help clinicians support patients. They could be served with information on the expected outcome and the uncertainty, given the medical history of the patient and their population group.

Chapter 26

Conclusions and Outlook

Compliance or elastance is the link between pressure and volume in the complex function of the heart. Current LPM models, however, use very simple, often linear or constant models to describe the compliance of the heart. We wanted to investigate compliance in more detail and to propose a new, more complex model that is essential for MCS applications.

Our results show a thorough investigation of compliance and a novel method to describe the interaction of a DCC device with the heart through heart-device interaction coefficients, e . Our aim was to replace the computationally expensive FEM model by Kummer et al. [88] to create a framework for device optimisation.

The results of our study indicate that co-compliances (C_{LR} and C_{RL}) are essential for the interaction of the heart with the DCC device, a factor that is often neglected in the literature. Our model is the first LPM that includes DCC device support, and the thorough investigation of compliance was essential for this.

We propose a novel method for describing and capturing the compliance and the heart-device interaction coefficients through FEM computer simulations. We integrated these results into our LPM framework, presented in Part VI.

Our model currently relies on computationally expensive FEM simulations to build a surrogate model. Increasing the complexity of the model might not be sufficient, given the number of FEM simulations that would be required for this task. Currently, our heart model needs further investigation to validate the surrogate model.

To support patient-specific applications and to build a better understanding of the mechanical aspects of the heart, we need to run FEM simulations on CT geometries of actual patients.

Part VIII

Conclusions and Outlook

Chapter 27

Summary and Final Remarks

This thesis has presented the design and development of a novel direct cardiac compression (DCC) device aimed at supporting surgeons in cardiac arrest cases. The project built upon the foundational work of Thomas Kummer, who laid the groundwork for the DCC concept. While the motivation for this work was the development of the device, many challenges were encountered. These led to research into developing a novel lumped parameter model for simulating the interaction of the device with the heart in an efficient computer simulation, developing a new mock circulation loop (MCL), assembling a new data acquisition system, researching suitable silicone materials for *in vitro* experiments, and conducting extensive *in vitro* and *ex vivo* testing.

The design of the DCC device includes the DCC contact patches, the motorised and manual actuators, and the vacuum pump. The patches are designed to compress the left and right ventricles of the heart, with distinct shapes to control the pressure in each ventricle. The device can be deployed quickly in an open-chest emergency scenario, providing immediate support to the heart for short-term applications. It uses only mechanical elements, compared to other DCC devices which are driven by pneumatics or hydraulics.

The *in vitro* and *ex vivo* experiments showed that the device can achieve comparable performance to open-chest cardiac compression, with a cardiac output of 1.5 L/min and a mean arterial pressure of 50 mmHg. Our results show the potential for the device to replace open-chest cardiac compression in emergency scenarios, relieving the medical team from manual compression. This could allow the surgeon to focus on other tasks, such as preparing for surgery or performing other life-saving procedures, while the DCC device supports basic life functions, similar to ECMO support but without direct contact with the blood.

To optimise the design and settings of the device, a novel lumped parameter model (LPM) was developed. The model simulates the interaction of the device with the heart, allowing for efficient computer simulations. The model allows for testing device parameters in an *in silico* environment, which is

essential for optimising the device before *in vitro* or *ex vivo* experiments.

To model the interaction of the device with the heart in the LPM, a novel compliance model was developed for the ventricles. This model allows for the investigation of the compliant nature of the heart and its effects on direct cardiac compression outcomes. This surrogate model was built on finite element method (FEM) simulations and *in vitro* experiments, extending currently available time-varying and constant compliance models.

Chapter 28

Outlook

During this project, which involved device development, experimental instrumentation design, developing simulation frameworks, modelling heart compliance, and conducting experiments, it is only natural that we generated more questions than answers.

Despite promising results, the device is still in its early stages of development. The results presented in this thesis were acquired in *in vitro* and *ex vivo* experiments, and the device has not yet been tested in *in vivo* conditions. The next step is to test the device in a porcine cadaver model, which will provide more insight into its performance and potential limitations.

The current design does not account for biocompatibility, and a plan for providing a sterile device for applications is needed. The patches are not biocompatible, and the current size adjustment mechanism is not feasible for emergency scenarios in an operating room, so a solution for quick adjustment is needed. As we did not focus on the size of the device, it is relatively bulky and requires optimisation for a more compact design. We also noticed tissue damage in the *ex vivo* experiments, especially on the left ventricle, which indicates that the curvature of the right ventricle patch is too great for larger hearts. To further reduce tissue damage, the surface of the patches should be smoothened. To reduce the size of the patches, we are currently working on a new design without vacuum channels, using a net to keep the device in place. The current magnetic locking mechanism is ideal for emergency scenarios, but it can open when lateral forces are exerted on the Bowden cable. An improved locking mechanism is needed to ensure that the device stays in place during operation, even when the patches are tilted in the chest cavity.

The actuation of the device was not the main focus of this project, but it was essential to operate the device. The current manual actuator is not ideal for long-term use, and a new foot pedal design is in development. The pedal could include adjustments to set the amplitude, maximum contraction, and initial position of the patches. This would allow for more precise control of the device during operation, even with manual actuation. Also, a new

motorised actuator could be designed with much improved power output.

For better *in vitro* testing, we propose developing an improved and more compact mock circulation loop (MCL) for future experiments. The current MCL was designed by Kummer et al. [90] for different instrumentation, and although the current design is validated and tested, a new design is needed to improve usability.

To improve the current simulation framework, an optimisation algorithm should be included to find the optimal combination of actuation control parameters for different patient cohorts. Currently, the framework does not allow for optimising the design of the patches. A thorough study is needed to identify key design parameters and their effect on the outcome, and to include these in the optimisation process.

While our model can provide improvements on currently available compliance models, it still fails to capture the full complexity of heart actuation. Future work should focus on refining the model further, potentially incorporating more variables based on anatomical and physiological considerations. To improve the general understanding of the compliant nature of the heart, we need to run FEM simulations on computed tomography (CT) geometries of actual patients. This would allow for patient-specific applications and a better understanding of the general mechanical behaviour of the heart.

Part IX

Appendices

Appendix A

In vitro Testing

This part of the appendix contains material from the following:

K. Sarosi et al. “Demonstration of a Mechanical External Biventricular Assist Device for Resuscitative Thoracotomy”. In: *Cardiovascular Engineering and Technology* (July 2025). ISSN: 1869-408X. DOI: <https://doi.org/10.1007/s13239-025-00793-y>

J. V. Spitzer. “Testing and developing and extraventricular assist device”. Master’s Thesis. ETH Zurich, 2022

A.1 Actuation System

Table A.1: Parts list of the actuation box, listing major parts.

Function	Qty	Name
Digital servo motor	2	ANNIMOS DS5160SSG 180° (60kg·cm)
Servo controller	1	Arduino Nano 5V ATmega328P
Main controller	1	Arduino Uno WiFi Rev2

The design utilises two Arduino boards, namely Uno WiFi Rev2 and Nano ATmega328P. These boards communicate asynchronously through their respective hardware UART ports. The Uno-Nano configuration resembles a cerebrum-cerebellum design, where the Arduino Uno WiFi Rev2 instructs the Arduino Nano ATmega328P to control the servo motors. The servo motors, requiring a 5 V logic level for PWM control, execute specific movements as directed. In real time, the Uno WiFi Rev2 transmits servo positions (analogue input) and set parameters from potentiometers via WiFi UDP to the PC. Simultaneously, it awaits start-stop commands through HTTP requests from the PC. The Uno WiFi Rev2 also updates the 24x4 LCD panel.

Appendix A In vitro Testing

To combine the pulling force generated by both servo motors, a Y-shaped bar assembly, referred to as the fork, has been introduced and illustrated in Figure A.1. The patches are actuated with a $\sin^2(t)$ signal.

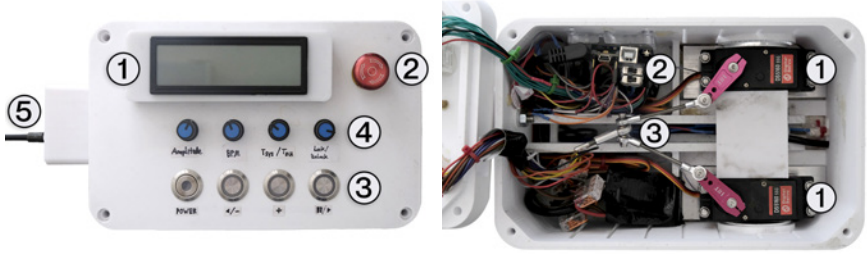


Figure A.1: **Top:** The actuation device box, showing the control panel. The LCD display (1) shows the initial position of the compression, the amplitude, the BPM value, and the time period ratio. The main function buttons (3) set the initial position, and they start and stop operation. The operator can set the amplitude, the BPM, and the $T_{\text{systolic}}/T_{\text{diastolic}}$ period ratio with the knobs (4). The Bowden cable is connected to the actuation box on the left (5). The operation can be stopped immediately by the emergency stop button (2). **Bottom:** The actuation device box's internal elements. The two servo motors (1), the Arduino Uno and Nano (2), the electronics to control the motors, the display, and to set the values with the knobs on the control panel. The Bowden cable connects to the motors via a fork (3).

A.2 Data Acquisition System

Table A.2: Parts list of the data acquisition system, listing major parts.

Function	Qty	Name
In-line ultrasonic flow meter	4	Audiowell HS0016
Evaluation board (Audiowell)	4	TI EVM430-FR6047
Clamp-on ultrasonic flow meter	1	EmTec BioProTT
Evaluation board (EmTec)	1	EmTec DIGIFLOW-EXT1
Flexible tubing	4	Saint-Gobain Tygon E-3603 3/4"
Data processing board	2	Arduino Portenta H7
Connectivity board	2	Arduino Portenta H7 Breakout Board
Analog pressure sensor	4	Honeywell ABPMRNT005PGAA3

The flow and pressure data within the system are obtained through the

use of four digital UART ultrasonic flow meters and four analogue pressure sensors, with simultaneous acquisition being a key requirement.

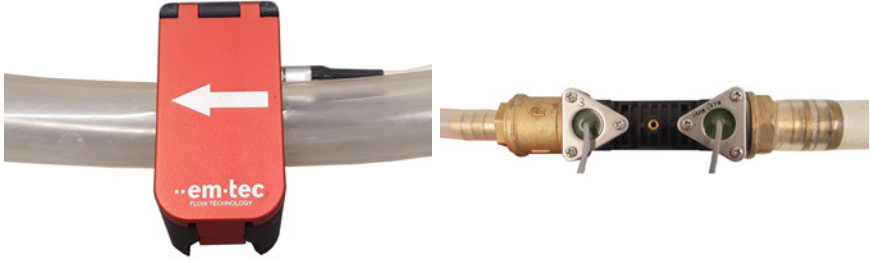


Figure A.2: **Left:** EmTec BioProTT clamped on a Saint-Gobain Tygon E-3603 tube. The signal is transmitted to a EmTec DIGIFLOW-EXT1 board, and then the data is streamed to the PC via WiFi by an Arduino Portenta H7. **Right:** Audiowell HS0016 connected to the test bench. The signal is transmitted to a TI EVM430-FR6047 board, and then the data is streamed to the PC via WiFi by an Arduino Portenta H7.

The EmTec DIGIFLOW-EXT1 utilises an ST Microelectronics 144-pin ST10 chip running at 40 MHz, transmitting flow rate and debug information from the BioProTT Clamp-On Transducer at 57600 baud/s via UART (RS232). On the other hand, the TI EVM430-FR6047 ultrasonic sensing evaluation boards have a 100-pin MSP430-FR6047 chip with separate 6 MHz crystal oscillators, transmitting the flow rate of each Audiowell HS0016 ultrasonic flow sensor at 115200 baud/s via UART.

Analogue pressure sensor data is read out via the internal ADC of the STM32H747 sequentially at a high sampling rate of 3.6 million samples per second. Therefore, the analogue readout is practically instantaneous. However, for digital sensors employing UART or I2C protocols, the Arduino Portenta H7 needs to wait for the arrival of new packets.

The Arduino Portenta H7, equipped with four UART and eight analogue ports, is utilised for data acquisition. To accommodate the five flow sensors (1x EmTec BioProTT + 4x Audiowell HS0016, see Figure A.2), an additional Portenta H7 is dedicated solely to the EmTec sensor. This configuration provides flexibility, allowing the removal of the EmTec sensor if necessary. However, its presence is advantageous as the Audiowell sensors are sensitive to the presence of air bubbles. The EmTec sensor was used to calibrate each Audiowell sensor. Their performance was found to be excellent; therefore, the EmTec BioProTT only remained in the setup to monitor the performance

of the Audiowell sensors.

In summary, two independent Arduino Portenta H7-based systems were constructed for sensor data acquisition:

1. WiFi data transmitter unit for the Audiowell flow sensors and the analogue Honeywell pressure sensors (see Figure 11.2): This unit includes four EVM430-FR6047 boards, four Honeywell ABPMRNT005PGAA3 pressure sensors, and an Arduino Portenta H7.
2. WiFi data transmitter unit for the EmTec flow sensor (see Figure 11.2): This unit is specifically designed for the EmTec sensor, with a dedicated Arduino Portenta H7.

A.3 Silicone Heart

Table A.3: Technical overview of the Dragon SkinTM10 Fast material, which was used for our silicone heart. Data provided by Smooth-on company, the producers of Dragon SkinTM.

Dragon Skin TM 10 Fast	
Mixed Viscosity (ASTM D-2393)	23000 cps
Specific Gravity (ASTM D-1475)	1.07 g/cc
Specific Volume (ASTM D-1475)	25.8 cu. in./lb.
Pot Life (ASTM D-2471)	8 min.
Cure Time	75 min.
Shore A Hardness (ASTM D-2240)	10A
Tensile Strength (ASTM D-412)	475 psi
100% Modulus (ASTM D-412)	22 psi
Elongation at Break % (ASTM D-412)	1000%
Die B Tear Strength (ASTM D-624)	102 pli
Shrinkage (ASTM D-2566)	<0.001 in./in.

Calibration Report

emtec

Product				Serial No.
BioProTT™ Clamp-On Transducer BCT 3/4 x 1/8" (ID12530)				100217
Report for			Overall Result	
Initial Calibration				
Check of Offset Stability (at room temperature)	Error Limit [ml/min]	Start Value [ml/min]	End Value [ml/min]	Result
Flow Drift over 30min at zero flow (0,1 Hz)	< ±15	-4	2	Pass

Calibration information for Check of Flow Accuracy

Table No.	Medium	Temp* [°C]	Tube Material	Max. Flow [ml/min]	AC ^{xx}
1	Water	25	Tygon S3 E-3603 [#170]	30000	No
2	Water/Gly40%	25	Tygon S3 E-3603 [#170]	30000	No
3					
4					
5					
6					
7					

*Medium Temperature; other environment conditions are insignificant for calibration **AC= Acoustic Couplant applied

Check of Flow Accuracy (Definition: Accuracy < ±3% of reading ± 200 ml/min)

Table No.	Flow at [ml/min]	Error Limit [ml/min]	Error Limit [%]	Actual Value [ml/min]	Actual Error [%]	Result
1	1000	770 – 1230	±23,0 %	998	-0,2 %	Pass
	5000	4650 – 5350	±7,0 %	4952	-1,0 %	Pass
	10000	9500 – 10500	±5,0 %	9962	-0,4 %	Pass
2	1000	770 – 1230	±23,0%	1024	2,4 %	Pass
	5000	4650 – 5350	±7,0%	5068	1,4 %	Pass
	10000	9500 – 10500	±5,0%	10075	0,8 %	Pass
3						

Figure A.3: The calibration data sheet of the EmTec BioProTT.

Appendix B

Lumped Parameter Model

B.1 Model Formulation

The model consists of 10 nodes at different locations of the LPM network:

1. Left Ventricle (LV)
2. Systemic Arteries (SA)
3. Systemic Capillaries (SP)
4. Systemic Veins (SV)
5. Right Atrium (RA)
6. Right Ventricle (RV)
7. Pulmonary Arteries (PA)
8. Pulmonary Capillaries (PP)
9. Pulmonary Veins (PV)
10. Left Atrium (LA)

Each node is connected to a compliance element, allowing for volume change at the node as Kirchhoff's law states.

State Vectors and Source Terms

The pressure vector $\mathbf{p}$ contains the pressures for the 10 nodes of the network:

$$\mathbf{p} = [p_{lv}, p_{sa}, p_{sp}, p_{sv}, p_{ra}, p_{rv}, p_{pa}, p_{pp}, p_{pv}, p_{la}]^T. \quad (\text{B.1})$$

The inductive flow vector $\mathbf{Q}_L$ accounts for flow through the inductance elements:

$$\mathbf{Q}_L = [0, -q_{L,SA}, q_{L,SA}, 0, 0, 0, -q_{L,PA}, q_{L,PA}, 0, 0]^T, \quad (\text{B.2})$$

Appendix B Lumped Parameter Model

where $q_{L,SA}$ and $q_{L,PA}$ are calculated using the inductance equation, Equation (20.5). As Equation (20.5) defines the time derivative of the flow, we include it in the solution vector of the ODE system and use the integrated value in the next time step.

The source term $\mathbf{S}(t)$ models the effect of an external device on the ventricles:

$$\mathbf{S}(t) = \left[-e_l \frac{d\omega}{dt}, 0, 0, 0, 0, -e_r \frac{d\omega}{dt}, 0, 0, 0, 0 \right]^T, \quad (\text{B.3})$$

where e_l and e_r are heart-device interaction coefficients and $\frac{d\omega}{dt}$ is the rate of change of device displacement.

Matrix of Resistances and Diodes (A)

The matrix $\mathbf{A}$ represents the conductance between the compartments. It is a 10×10 matrix where the elements A_{ij} represent the resistance and diode elements between node i and j . The matrix is defined as:

$$\mathbf{A} = \begin{pmatrix} -\frac{D_{mi}}{R_{mi}} - \frac{D_{ao}}{R_{ao}} & \frac{D_{ao}}{R_{ao}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{D_{mi}}{R_{mi}} \\ \frac{D_{ao}}{R_{ao}} & -\frac{D_{ao}}{R_{ao}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{R_{SP}} & \frac{1}{R_{SP}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{R_{SP}} & -\frac{1}{R_{SP}} - \frac{1}{R_{SV}} & \frac{1}{R_{SV}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{R_{SV}} & -\frac{1}{R_{SV}} - \frac{D_{te}}{R_{tr}} & \frac{D_{te}}{R_{tr}} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{D_{te}}{R_{tr}} & -\frac{D_{te}}{R_{tr}} - \frac{D_{pu}}{R_{pu}} & \frac{D_{pu}}{R_{pu}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{D_{pu}}{R_{pu}} & -\frac{D_{pu}}{R_{pu}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{R_{PP}} & \frac{1}{R_{PP}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{R_{PP}} & -\frac{1}{R_{PP}} - \frac{1}{R_{PV}} & \frac{1}{R_{PV}} & 0 \\ \frac{D_{mi}}{R_{mi}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{R_{PV}} & -\frac{1}{R_{PV}} - \frac{D_{mi}}{R_{mi}} \end{pmatrix} \quad (\text{B.4})$$

where D_{xx} is the state of a valve and R_{xx} is the resistance.

Compliance Matrix (C)

The compliance matrix **C** is a matrix containing the compliance of each compartment, with off-diagonal terms representing ventricular interaction.

$$\mathbf{C} = \begin{pmatrix} c_{ll} & 0 & 0 & 0 & 0 & c_{lr} & 0 & 0 & 0 & 0 \\ 0 & C_{SA} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & C_{SP} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{SV} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{ra} & 0 & 0 & 0 & 0 & 0 \\ c_{rl} & 0 & 0 & 0 & 0 & c_{rr} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & C_{PA} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & C_{PP} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & C_{PV} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_{la} \end{pmatrix} \quad (\text{B.5})$$

The terms $c_{ll}, c_{lr}, c_{rl}, c_{rr}, c_{ra}, c_{la}$ can be constant (passive heart) or time-varying (active heart).

B.1.1 Active Heart and Compliance

The model can operate in two modes for cardiac compliance:

1. **Passive Heart:** Ventricular and atrial compliances are constant.
2. **Active Heart:** Ventricular compliances are time-varying. The time derivative of the compliance matrix, $\frac{d\mathbf{C}}{dt}$, becomes non-zero. The compliance of the atria is also time-varying, using the same model as the ventricles.

B.2 Simulation Parameters

The following tables list the constant parameters used in the simulation, as defined in the main body of text.

B.2.1 Resistances

Table B.1: Resistance Parameters

Parameter	Value (mmHg s/mL)	Description
R_{SA}	0.06	Systemic arterial
R_{SP}	0.987	Systemic capillary
R_{SV}	0.0113	Systemic venous
R_{PA}	0.023	Pulmonary arterial
R_{PP}	0.0894	Pulmonary capillary
R_{PV}	0.0056	Pulmonary venous
R_{TR}	2.5e-3	Tricuspid valve
R_{MI}	2.5e-3	Mitral valve

B.2.2 Compliances

Table B.2: Passive Compliance Parameters

Parameter	Value (mL/mmHg)	Description
C_{SA}	0.28	Systemic arterial
C_{SP}	3.72	Systemic capillary
C_{SV}	111.11	Systemic venous
C_{RA}	31.25	Right atrial
C_{PA}	0.76	Pulmonary arterial
C_{PP}	5.8	Pulmonary capillary
C_{PV}	25.37	Pulmonary venous
C_{LA}	19.23	Left atrial
C_{LL}	2.905	Passive V_L-p_L
C_{LR}	-1.791	Passive V_L-p_R
C_{RL}	-1.343	Passive V_R-p_L
C_{RR}	4.784	Passive V_R-p_R

B.2.3 Inductances

Table B.3: Inductance Parameters

Parameter	Value (mmHg s ² /mL)	Description
L_{SA}	0.22e-3	Systemic arterial
L_{PA}	0.18e-3	Pulmonary arterial

B.2.4 Valve Dynamics Parameters

Table B.4: Mynard's Valve Model Parameters

Valve	K_{VO} (cm/dynes/s)	K_{VC} (cm/dynes/s)
Aortic	0.06	0.06
Pulmonary	0.10	0.10
Mitral	0.15	0.20
Tricuspid	0.15	0.20

Table B.5: Kummer's Valve Model Parameters

Valve	K (s/mL)	λ (s ²)	γ_l (s)	γ_u (s)
Aortic	3.75e-4	5000	1000	1000
Pulmonary	1.4e-3	2000	1000	1000
Mitral	–	30000	1000	1000
Tricuspid	–	30000	1000	1000

B.2.5 Active Heart Parameters

The valve parameters were set according to the work of Mynard et al. [99].

Table B.6: Active Heart Parameters

Chamber	Case	T (s)	$E_{min}(mmHg/mL)$	$E_{max}(mmHg/mL)$
LV	Normal	0.8	0.08	4.5
	Heart Failure	0.8	0.08	0.75
	Fibrillation	0.3	1.5	3
RV	Normal	0.8	0.04	0.9
	Heart Failure	0.8	0.08	0.15
	Fibrillation	0.3	0.3	0.6
LA	Normal	0.8	0.08	0.17
	Heart Failure	0.8	0.08	0.17
	Fibrillation	0.3	0.085	0.17
RA	Normal	0.8	0.04	0.15
	Heart Failure	0.8	0.04	0.15
	Fibrillation	0.3	0.075	0.15

B.3 Numerical Solution

The system of ODEs is solved using the `solve_ivp` function from SciPy's integration library, with the 'RK45' method, an explicit Runge-Kutta method of order 5. The simulation is run with a time step Δt that is automatically adjusted by the solver.

The state vector $\mathbf{u}$ contains:

- pressures
- inductive flows
- volume
- patch displacement
- active compliances
- Mynard's valve states

The numerical solution is obtained by initialising the system state and then using an iterative ODE solver. The logic is described in Algorithm 1. The core of the algorithm is the function `du_dt`, which is called at each time step by the solver to compute the derivatives of the state vector.

Algorithm 1 Cardiovascular Model Simulation

```

1: while  $t < t_{end}$  do
2:   Update valve states  $D_i$ 
3:   Compute resistance of pressure-dependent valves  $R_{AO}, R_{PU}$ 
4:   Assemble the compliance matrix  $\mathbf{C}(\mathbf{p}, t)$ 
5:   if Active Heart then
6:     Compute  $\mathbf{c}_{ij}$  and  $\dot{\mathbf{c}}_{ij}$  of active nodes
7:   end if
8:   Assemble the resistance matrix  $\mathbf{A}(\mathbf{p}, t)$ 
9:   Assemble the inductive flow vector  $\mathbf{Q}_L$ 
10:  Assemble the source term  $\mathbf{S}(t)$ 
11:  Invert the compliance matrix  $\mathbf{C}(\mathbf{p}, t)$ 
12:  Compute  $\dot{\mathbf{p}} = \mathbf{C}^{-1} \left( \left( \mathbf{A} - \frac{d\mathbf{C}}{dt} \right) \mathbf{p} + \mathbf{Q}_L - \mathbf{S} \right)$ 
13:  Compute  $\dot{\omega}(t)$ 
14:  Compute  $\dot{\mathbf{Q}}_L$ 
15:  Compute  $\dot{\mathbf{D}}$ 
16:  Compute the flows in the network  $\dot{\mathbf{V}} = \mathbf{A}\mathbf{p} + \mathbf{Q}_L$ 
17:  Assemble  $\dot{\mathbf{u}} = [\dot{\mathbf{p}}, \dot{\mathbf{Q}}_L, \dot{\mathbf{V}}, \dot{\omega}(t), \dot{\mathbf{c}}_{ij}, \dot{\mathbf{D}}]^T$ 
18:   $\mathbf{u} \leftarrow \mathbf{u} + \dot{\mathbf{u}}\Delta t$ 
19:   $t \leftarrow t + \Delta t$ 
20: end while

```

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